ON SOME POSITIVITY PROPERTIES OF THE RIEMANN Ξ FUNCTION

SUVRIT SRA*

ABSTRACT. We prove strict log-concavity of the positive functions Θ and Λ with one-sided and bilateral Laplace transforms $1/\xi(\frac{1}{2}+\sqrt{s})$ and $1/\Xi$, respectively, where $\Xi(z)=\xi(\frac{1}{2}+iz)$. For either function, log-concavity is the Pólya frequency property of order two (PF_2). Without assuming the Riemann hypothesis (RH), but using its published verification up to height about $3\cdot 10^{12}$, we further prove that both densities are PF_{k_*} for $k_* = 1\,810\,540\,931\,075$. The proof views Θ as a small signed perturbation of a density that is PF of every order, with an error bound independent of the order, so the numerical check reduces to scalar inequalities. Since Λ is a Gaussian mixture of Θ, it inherits the same order. Conversely, a pole obstruction shows that if either density is PF_k, then every zeta zero lower than the $(k+2)$-nd lies on the critical line. Finally, we prove that if RH holds up to height $T > \frac{1}{2}$, then Ξ satisfies the generalized Laguerre inequalities $L_n[\Xi](x) \geq 0$ of every order n for $|x| \leq T - \frac{1}{2}$, extending the case $n=1$ of Csordas, Ruttan and Varga (1991).

github.com/suvrit/preprints riem-xi v3 2 Nov 2026

1. INTRODUCTION

For a connoisseur of positivity, the Riemann Ξ function offers a tantalizing array of questions. While the hardest of these are equivalent to the Riemann hypothesis (RH) itself, we find that even weak necessary conditions for RH already repay study. The ideas permeating our study are classical: Fourier–Laplace transforms, the Laguerre–Pólya class, Pólya frequency functions, among others. Our goal is a richer understanding of these positivity properties without assuming RH, and ideally with as little information about the zeta zeros as possible.

We use the usual normalization (Titchmarsh, 1986, Section 2.1) for Ξ:

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad c_0 = \xi(\tfrac{1}{2}) > 0, \qquad \Xi(z) = \xi(\tfrac{1}{2}+iz), \tag{1.1}$$

and define the following functions

$$F(z) := \Xi(z)/c_0, \qquad h(z) := \xi(\tfrac{1}{2}+\sqrt{z})/c_0, \qquad B(s) := \log h(s) \quad (s \geq 0). \tag{1.2}$$

Let $\mathcal{Z}_+$ denote the multiset of nontrivial zeta zeros $\rho = \beta + i\gamma$ with $\gamma > 0$, and define

$$H(t) := \sum_{\rho \in \mathcal{Z}_+} e^{-t(\gamma - i(\beta - \frac{1}{2}))^2} \qquad (t > 0).$$

Under RH three positivity properties hold: (a) H is *completely monotone* (i.e., $(-1)^n H^{(n)} \geq 0$ for all $n \geq 0$); (b) by Schoenberg's theorem (Schoenberg, 1951; Gröchenig, 2020), $1/(c_0 h)$ and $1/\Xi$ are the one-sided and bilateral Laplace transforms of Pólya frequency (PF) functions of every order; and (c) Ξ satisfies every generalized Laguerre inequality on $\mathbb{R}$. Conversely, each of these properties implies RH (Proposition A.1; Csordas and Varga, 1990, Theorem 2.9). Without RH, an off-line zero ρ and its reflection $1-\bar{\rho}$ both lie in $\mathcal{Z}_+$ and contribute complex-conjugate terms to H. So H stays real, but these terms oscillate in t and can in principle make it negative. We ask what parts of (a)–(c) follow from the available bounds on those oscillations.

Our main results are the following. Items marked * overlap in part with earlier or parallel work; the precise relation is described in Section 5 and in the discussion of the corresponding result.

Key words and phrases. Riemann Xi; Pólya frequency; log-concavity; generalized Laguerre; complete monotonicity.

* TU Munich, Dept. of Mathematics, School of CIT; Garching, Germany. Email: s.sra@tum.de.

(1)* Let Θ on $(0,\infty)$ and Λ on $\mathbb{R}$ be the positive densities with one-sided and bilateral Laplace transforms $1/(c_0 h)$ and $1/\Xi$, respectively. They exist without RH. More generally, for every $\alpha > 0$, $h^{-\alpha}$ is the Laplace transform of a positive self-decomposable law, and $[c_0/\xi(\frac{1}{2}+\|u\|)]^{\alpha}$ is strictly positive definite on $\mathbb{R}^d$ for every d (Theorem 1.1). The existence question of Roynette and Yor (2005) was answered by Polson (2026b, v24, Theorem A); the proof here uses zeros only through height 100 and adds the regularity of the densities. This result answers a positive-definiteness question raised by Gröchenig (2020).

(2) The densities Θ and Λ are strictly log-concave; the proof uses zeros only through height 100. Using higher zero verification through height $3\,000\,175\,332\,800$, we prove $\Theta, \Lambda \in \mathrm{PF}_{k_*}$ for $k_* = 1\,810\,540\,931\,075$ (Theorem 1.2). For each k, the $k \times k$ minors formed by $\Theta(x-y)$ are also positive when all differences are sufficiently small and positive (Proposition C.25).

(3)* If RH holds up to height $T > \frac{1}{2}$, then Ξ satisfies the generalized Laguerre inequalities $L_n[\Xi](x) \geq 0$ of every order n for $|x| \leq T - \frac{1}{2}$, where $L_n[\Xi](x)$ is the coefficient of y^{2n} in $|\Xi(x+iy)|^2$ (Theorem 1.3). The case $n = 1$ goes back to Csordas et al. (1991).

The inputs to our results are the critical strip, explicit zero counting, and published zero verification. Our conclusions do not extend the verified height, and they are merely weak necessary conditions. The global Laguerre inequality $L_1[\Xi] \geq 0$ (Csordas, 1998, Problem 2) remains open, and in our view, it is a much stronger necessary condition than those studied in this paper. Below, in §§1.1–1.3 we present the formal statements of our main results.

1.1. **Construction of the reciprocal laws Θ and Λ.** Assuming RH, the densities Θ and Λ exist and are PF_∞ by Schoenberg's theorem (Schoenberg, 1951, Theorems 1 and 2), specialized to Ξ in Gröchenig (2020, Theorems 3–5). But without RH, at least their existence follows easily. Using the notation from (1.1) and (1.2), from Hadamard's product for h over the zeros (Lemma 2.1) we obtain $B'(s) = \int_0^\infty e^{-st} H(t)\,\mathrm{d}t$. So $H > 0$ makes B a *Bernstein function*, nonnegative with completely monotone derivative, and $h^{-\alpha} = e^{-\alpha B}$ the Laplace transform of a probability law.

Theorem 1.1 (Reciprocal xi and probability laws)**.** *For $t > 0$, $H(t) > 0$ and $H'(t) < 0$. The function B is Bernstein, with the representation*

$$B(s) = \int_0^\infty (1 - e^{-st}) \tfrac{H(t)}{t}\,\mathrm{d}t. \tag{1.3}$$

Moreover, for every $\alpha > 0$ there is a positive self-decomposable random variable S_α with $\mathbb{E}e^{-sS_\alpha} = h(s)^{-\alpha}$. It has a C^∞ density positive on $(0,\infty)$. For every $d \geq 1$,

$$h(\|u\|^2)^{-\alpha} = c_0^{\alpha}\,\xi(\tfrac{1}{2}+\|u\|)^{-\alpha} \tag{1.4}$$

is strictly positive definite on $\mathbb{R}^d$ and is the characteristic function of an infinitely divisible law. Its Gaussian-mixture density is positive on $\mathbb{R}^d$ and extends to an entire function on $\mathbb{C}^d$.

1.2. **Log-concavity and finite PF orders.** As noted, RH implies Θ and Λ are PF_∞ functions; without RH, we can still obtain lower PF orders. A function f is PF_m if $\det[f(x_i - y_j)]_{i,j=1}^k \geq 0$ for every $1 \leq k \leq m$ and every pair of increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$. For a nonnegative function with interval support, log-concavity is equivalent to PF_2.

Theorem 1.2 (Log-concavity and PF_k)**.** *The densities satisfy $(\log\Theta)''(x) < 0$ for $x > 0$ and $(\log\Lambda)''(x) < 0$ for $x \in \mathbb{R}$. Put $k_* = 1\,810\,540\,931\,075$. The extension of Θ by zero to $(-\infty, 0]$ belongs to PF_{k_*}. For every $1 \leq k \leq k_*$ and increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$,*

$$\det[\Theta(x_i - y_j)]_{i,j=1}^k \geq 0, \tag{1.5}$$

with strict inequality if and only if $x_i > y_i$ for every i. The density Λ belongs to PF_{k_*} with strict minors: for every $1 \leq k \leq k_*$ and increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$,

$$\det[\Lambda(x_i - y_j)]_{i,j=1}^k > 0.$$

The large finite order uses the published verified height; its scalar criterion is Theorem 3.4. Appendix C.10 establishes that k_* is the maximum of that explicitly specified sufficient criterion.

1.3. **Generalized Laguerre inequalities for Ξ.** Under RH, Ξ satisfies the generalized Laguerre inequalities on $\mathbb{R}$; we prove that it satisfies all of them on an interval, which is once again merely a weak necessary check. For a real entire function f, define the (generalized) Laguerre transform

$$L_n[f](x) := \frac{1}{(2n)!}\sum_{j=0}^{2n}(-1)^{j+n}\binom{2n}{j}f^{(j)}(x)f^{(2n-j)}(x), \qquad n \in \mathbb{N}_0. \tag{1.6}$$

These are the coefficients of y^{2n} in $|f(x+iy)|^2$; in particular, $L_0[f] = f^2$ and $L_1[f] = (f')^2 - ff''$. Satisfying the generalized Laguerre inequality $L_n[f] \geq 0$ for each $n \geq 0$ implies that f is in the Laguerre–Pólya class, so for instance the truth of $L_n[\Xi] \geq 0$ would imply RH. Csordas et al. (1991) proved $L_1[\Xi] \geq 0$ on an interval using known zero verification (Csordas, 2015, Remark 4.8); we extend that result to show that $L_n[\Xi] \geq 0$ on that interval.

Theorem 1.3 (Generalized Laguerre on an interval)**.** *Let $T > 1/2$, and suppose all nontrivial zeta zeros $\beta + i\gamma$ with height $0 < \gamma \leq T$ satisfy $\beta = 1/2$. Then,*

$$L_n[\Xi](x) \geq 0 \qquad (n = 0, 1, 2, \ldots, \quad |x| \leq T - \tfrac{1}{2}).$$

Moreover, $L_n[\Xi](x) > 0$ unless x is a zero of Ξ of multiplicity exceeding n.

By Platt and Trudgian (2021, Theorem 1) the critical line hypothesis holds for $T = T_* := 3 \cdot 10^{12}$, so every $L_n[\Xi] \geq 0$ on $|x| \leq T_* - \frac{1}{2}$. For $n = 1$, this use of verified zeros goes back to Csordas et al. (1991, Corollary to Lemma 4), who assume that the zeros are simple; see also Csordas et al. (1992, p. 398), Csordas (1998, remark after Problem 2) and Csordas (2015, Remark 4.8).

2. Proof of Theorem 1.1: the reciprocal laws

This section proves Theorem 1.1. The construction needs two analytic facts: the canonical product must yield a convergent Laplace integral, and H must stay positive and decreasing. We establish the convergence first, then compare the possible negative tail with one retained real zero. Sections 2.3 and 2.4 then construct the laws and the densities Θ and Λ. Endpoint asymptotics, used for the regularity of the laws, are proved in Appendix B.

2.1. **Analytic and numerical inputs.** We use the classical analytic continuation, functional equation, zero symmetries, and critical strip of ζ (Titchmarsh, 1986, Chapter II). In particular $\delta_\rho := \beta - \frac{1}{2}$ satisfies $|\delta_\rho| < 1/2$, ξ has order one, and the nontrivial zeros are its only zeros. An elementary theta representation, recalled in Appendix B, also gives $c_0 > 0$.

Let $N(Y)$ count upper-half-plane zeros of ordinate at most Y, with multiplicity, and write

$$M(Y) = \frac{Y}{2\pi}\left(\log\frac{Y}{2\pi} - 1\right), \qquad M(0) = 0.$$

We use the following consequence of Bellotti and Wong (2025, Theorem 1.1):

$$|N(Y) - M(Y)| \leq \tfrac{1}{8}\log(Y+2) + 4 \qquad (Y \geq 0). \tag{2.1}$$

For all $U \geq 100$, (2.1) implies

$$N(Y) \leq Y\log Y \quad (Y \geq U), \qquad \sum_{\gamma > U}\gamma^{-2} \leq \frac{2(\log U + 1)}{U} < 1. \tag{2.2}$$

The sum estimate follows by Stieltjes integration by parts, dropping the nonpositive boundary term.

Put $U_{\mathrm{PT}} = 3\,000\,175\,332\,800$, the full published verified height. We also use the round value $T_* = 3 \cdot 10^{12}$. The proved finite zero inputs are

$$\gamma \leq U_{\mathrm{PT}} \implies \delta_\rho = 0, \qquad 14 < \gamma_1 < 15, \tag{2.3}$$

where γ_1 is the first positive zero ordinate. The first assertion follows from Platt and Trudgian (2021, Theorem 1), which verifies height 3 000 175 332 800. The first-zero enclosure is contained in rigorously isolated zero data (Platt, 2017; The LMFDB Collaboration, 2026). Only the portion through height 100 is needed for Theorem 1.1 and the log-concavity in Theorem 1.2. Table 1 lists the numerical inputs of each result; the certificates are in Appendix C.

For $\rho \in \mathcal{Z}_+$ put $\lambda_\rho := (\gamma - i\delta_\rho)^2$, so that $H(t) = \sum_{\rho\in\mathcal{Z}_+} e^{-t\lambda_\rho}$.

Lemma 2.1 (Product and convergence)**.** *The function h is entire of order $1/2$, and*

$$h(z) = \prod_{\rho\in\mathcal{Z}_+} (1 + z/\lambda_\rho). \tag{2.4}$$

The product converges locally uniformly. If $a_\rho = \operatorname{Re}\lambda_\rho$, then $a_\rho > 0$, $\sum a_\rho^{-1} < \infty$, and $h(s) > 0$ for $s \geq 0$. The series for H and its derivatives converge locally absolutely on $\operatorname{Re} t > 0$.

Proof. Evenness turns the order-one Xi function into an order-$1/2$ entire function under squaring. The pair $\rho, 1-\rho$ yields one squared zero; exactly one member is in $\mathcal{Z}_+$. Hadamard factorization at order less than one therefore yields (2.4), with constant one. For large ordinates $a_\rho = \gamma^2 - \delta_\rho^2 \geq \gamma^2 - 1/4$, and zero counting gives reciprocal-square summability. At low ordinates (2.3) makes $a_\rho = \gamma^2 > 0$. Conjugate factors in (2.4) combine into positive squared moduli on the positive axis; real factors are positive. On a compact subset of $\operatorname{Re} t > 0$, the Gaussian decay $e^{-c\gamma^2+C\gamma}$ absorbs every fixed power of γ, proving the last claim. □

2.2. **Positivity of H and $-H'$.** A high zero can oscillate only after its exponential weight has become small. The estimate below makes this delay quantitative for both H and $-H'$. Keeping a single low real zero gives a common positive term against which all negative contributions of the tail can be measured. The same comparison is the large-time step in Polson (2026b, v24, Proposition 6.5) and Polson (2026a, v9, proof of Theorem 22); the estimate below makes it valid for every $t > 0$.

Put $q = \gamma_1^2 \in (196, 225)$ and

$$a_\rho = \gamma^2 - \delta_\rho^2, \qquad b_\rho = 2\gamma\delta_\rho, \qquad \lambda_\rho = a_\rho - ib_\rho.$$

Then $|b_\rho| \leq \gamma$ and $|\lambda_\rho| = \gamma^2 + \delta_\rho^2$.

Lemma 2.2 (One retained real zero)**.** *Suppose $100 \leq U \leq U_{\mathrm{PT}}$, and define*

$$H_{>U}(t) = \sum_{\gamma>U} e^{-t\lambda_\rho}, \quad K_U(t) = e^{-qt} + H_{>U}(t), \quad \varepsilon_U = U^4 e^{-U}.$$

For every $t > 0$,

$$K_U(t) \geq (1-\varepsilon_U)e^{-qt} > 0, \qquad -K_U'(t) \geq (1-\varepsilon_U)qe^{-qt} > 0.$$

In particular $H > 0$ and $-H' > 0$.

Proof. The real summands of H and $-H'$ are respectively

$$e^{-a_\rho t}\cos(b_\rho t), \qquad e^{-a_\rho t}[a_\rho\cos(b_\rho t) + b_\rho \sin(b_\rho t)].$$

Both are nonnegative when $|b_\rho|t \leq \pi/2$. A negative contribution requires $t > \pi/(2\gamma)$. For $\gamma \geq 100$, $a_\rho - q \geq \gamma^2 - 1/4 - 225 \geq 3\gamma^2/4$. When the corresponding contribution is negative, we have

$$e^{-(a_\rho - q)t} \leq e^{-3\pi\gamma/8} \leq e^{-\gamma}.$$

Its negative part is consequently at most $e^{-qt}e^{-\gamma}$ in the first sum, and at most $qe^{-qt}\gamma^2 e^{-\gamma}$ in the second, using $|\lambda_\rho| \leq q\gamma^2$. By (2.2) and monotonicity of y^2e^{-y} and y^4e^{-y} for $y \geq 100$,

$$\sum_{\gamma>U} e^{-\gamma} < U^2 e^{-U}, \qquad \sum_{\gamma>U} \gamma^2 e^{-\gamma} < U^4 e^{-U}.$$

Subtract from the retained terms e^{-qt} and qe^{-qt}. Finally $H - K_U$ is a finite sum of positive real exponentials, proving the assertions about H. □

2.3. **The Bernstein representation and the laws.** We first derive the Lévy representation of B from the canonical product. Termwise it is Frullani's integral $\log(1 + s/\lambda) = \int_0^\infty (1 - e^{-st})e^{-\lambda t}\,\mathrm{d}t/t$; for a positive real rate λ this is the Lévy representation of a gamma law. When every $\lambda_\rho > 0$, summation gives Thorin's representation of a generalized gamma convolution (Bondesson, 1992, Chapter 3). Without RH the exponents λ_ρ may be complex, and the positivity of the resulting Lévy density $H(t)/t$ is the content of Lemma 2.2. For the xi function the identity is written in Polson (2026b, v24, Theorem 5.3). Lemma 2.4 supplies smooth positive densities and negative moments. Gaussian subordination gives the radial characteristic functions, and the growth of B gives their entire inverse Fourier transforms.

The integral characterization of Bernstein functions and the complete Bernstein subclass are described in Schilling et al. (2012, Chapters 3 and 6); here a Poisson random measure constructs the laws directly.

Proposition 2.3 (Bernstein representation)**.** *Equation* (1.3) *holds, and for* $s \geq 0$, $n \geq 0$,

$$(-1)^n B^{(n+1)}(s) = \int_0^\infty t^n e^{-st} H(t)\,\mathrm{d}t > 0.$$

Moreover $\int_0^\infty |H(t)|\,\mathrm{d}t < \infty$.

Proof. Logarithmic differentiation of (2.4) yields $B'(s) = \sum_\rho (s + \lambda_\rho)^{-1}$. Interchanging with the elementary Laplace integral is justified by

$$\sum_\rho \int_0^\infty |e^{-(s+\lambda_\rho)t}|\,\mathrm{d}t = \sum_\rho (s + a_\rho)^{-1} < \infty.$$

Thus $B'(s) = \int e^{-st} H(t)\,\mathrm{d}t$ and $\int |H| \leq \sum a_\rho^{-1} < \infty$. Integrate in s, using $B(0) = 0$, to obtain (1.3). The positivity follows from Lemma 2.2; differentiation is justified by integrability at zero and exponential decay at infinity. The Lévy condition follows from $\int (1 \wedge t)H(t)\,\mathrm{d}t/t < \infty$. □

For $\alpha > 0$ take a Poisson random measure on $(0, \infty)$ of intensity $\alpha H(t)\,\mathrm{d}t/t$, and let S_α be the sum of its points. It is finite almost surely because its expectation is $\alpha \int H(t)\,\mathrm{d}t < \infty$. The Laplace functional proves

$$\mathbb{E}e^{-sS_\alpha} = e^{-\alpha B(s)} = h(s)^{-\alpha}. \tag{2.5}$$

Independent sums add the parameter α, so these laws form a convolution semigroup. For $0 < c < 1$,

$$B(s) - B(cs) = \int_0^\infty (1 - e^{-st}) \frac{H(t) - H(t/c)}{t}\,\mathrm{d}t. \tag{2.6}$$

Since H is decreasing, the Lévy density $(H(t) - H(t/c))/t$ in (2.6) is nonnegative. Equation (2.6) therefore constructs an independent $R_c \geq 0$ with $S_\alpha \overset{d}{=} cS'_\alpha + R_c$, proving self-decomposability. This is the classical criterion: a law on $[0, \infty)$ whose Lévy density has the form $k(t)/t$ with k nonincreasing is self-decomposable (Sato, 1999, Corollary 15.11). For the xi function see also Polson (2026b, v24, Theorem 5.6(ii)).

Lemma 2.4 (Density, support, and moments)**.** *For every* $\alpha > 0$, S_α *has a* $C^\infty(\mathbb{R})$ *density* θ_α, *zero on* $(-\infty, 0]$ *and positive on* $(0, \infty)$. *Its Laplace transform is holomorphic on* $\operatorname{Re} s > -q$. *It has exponential moments of all orders* $r < q$, *no exponential moment of order* $r \geq q$, *and finite negative moments of all orders.*

The proof extracts gamma convolution factors of arbitrarily high order from the Lévy measure. This gives smoothness and full positive support; the remaining moment estimates are in Appendix B.4.

The exponential-moment threshold is also in Polson (2026b, v24, Corollary 6.8(iii)) and Polson (2026a, v9, Theorem 22(iii)).

Let G_d be standard Gaussian and independent of S_α. The vector $X_{\alpha,d} = \sqrt{2S_\alpha}G_d$ has characteristic function (1.4) and density

$$p_{\alpha,d}(x) = \int_0^\infty (4\pi u)^{-d/2} e^{-\|x\|^2/(4u)} \theta_\alpha(u)\,\mathrm{d}u > 0.$$

Its infinite divisibility follows by adding the parameter α, and its self-decomposability follows from (2.6) with c^2 instead of c. This is the usual Gaussian-subordination mechanism (Schoenberg, 1938); see also Roynette and Yor (2005, Proposition I.8.1). For the xi function with $\alpha = d = 1$ it is Polson (2026b, v24, Theorem A). The Gaussian Gram matrix on distinct points is positive definite for each $u > 0$: its quadratic form is the integral of the squared modulus of a nonzero exponential polynomial against a positive Gaussian density. Integration against θ_α proves strictness for (1.4). By (B.5), its characteristic function is integrable against $e^{A\|t\|}$ for every $A > 0$. The inverse Fourier integral therefore defines an entire extension of $p_{\alpha,d}$ to $\mathbb{C}^d$. This proves Theorem 1.1.

Theorem 1.1 also removes the hypothesis from Roynette and Yor (2005, eq. (V.36)), which they prove under RH: for every $\alpha > \frac{1}{2}$ there is a random variable $\widetilde{H}_\alpha \geq 0$ with

$$\mathbb{E}e^{-\lambda^2 \widetilde{H}_\alpha/2} = \frac{\xi(\alpha)}{\xi(\alpha+\lambda)} \qquad (\lambda \geq 0). \tag{2.7}$$

Indeed, $\widetilde{H} = 2S_1$ satisfies $\mathbb{E}e^{-\beta^2\widetilde{H}/2} = c_0/\xi(\frac{1}{2}+\beta)$ for $\beta \geq 0$, and their lemma (Roynette and Yor, 2005, Lemme V.3.4), applied to $\widetilde{H}$ with shift $\alpha - \frac{1}{2} > 0$, gives (2.7). For $\alpha \geq 1$, $\widetilde{H}_\alpha/2$ can be taken to be a generalized gamma convolution (Polson, 2026a, v9, Theorem 18). The question (V.35) of Roynette and Yor, which asks for the additional factor $\exp(\lambda\xi'(\alpha)/\xi(\alpha))$ on the right of (2.7), is not settled by this argument; see Polson (2026a, v9, Remark 20).

2.4. **The densities Θ and Λ.** In the notation of Gröchenig (2020, Theorem 5), $\Xi(z) = \Xi_1(-z^2)$ with $\Xi_1 = c_0 h$. Set

$$\Theta(u) = \frac{\theta_1(u)}{c_0}, \qquad \Lambda(x) = \frac{p_{1,1}(x)}{c_0} = \frac{1}{c_0}\int_0^\infty \frac{e^{-x^2/(4u)}}{\sqrt{4\pi u}}\theta_1(u)\,\mathrm{d}u. \tag{2.8}$$

Both integrate to $1/c_0$ on their respective domains.

Corollary 2.5 (Positive versions of the two Laplace representations)**.** *The density Θ is smooth, supported in $[0,\infty)$, and strictly positive on $(0,\infty)$. The density Λ is even, strictly positive on $\mathbb{R}$, and extends to an entire function. They satisfy*

$$\begin{aligned} \frac{1}{c_0 h(s)} &= \int_0^\infty e^{-su}\Theta(u)\,\mathrm{d}u, && \operatorname{Re} s > -\gamma_1^2, \\ \frac{1}{\Xi(z)} &= \int_{\mathbb{R}} e^{-zx}\Lambda(x)\,\mathrm{d}x, && |\operatorname{Re} z| < \gamma_1. \end{aligned} \tag{2.9}$$

In particular, for arbitrary $c_j \in \mathbb{C}$ and $t_j \in \mathbb{R}$,

$$\sum_{j,k} \frac{c_j\overline{c_k}}{\xi(\frac{1}{2}+t_j-t_k)} = \int_{\mathbb{R}} \left|\sum_j c_j e^{it_j x}\right|^2 \Lambda(x)\,\mathrm{d}x \geq 0. \tag{2.10}$$

The inequality is strict for distinct t_j and a nonzero coefficient vector.

Proof. The first transform follows from (2.5) and Lemma 2.4; equality on the full half-plane follows by holomorphic continuation. For real $|z| < \gamma_1$, conditioning the Gaussian mixture gives

$$\int_{\mathbb{R}} e^{-zx}\Lambda(x)\,\mathrm{d}x = c_0^{-1}\mathbb{E}e^{z^2 S_1} = \frac{1}{c_0 h(-z^2)}.$$

The exponential moments of the resulting symmetric variable give holomorphy of the bilateral transform throughout the strip, so the identity theorem proves (2.9) there. Take $z = it$ and use the functional equation and evenness to obtain the Fourier transform $1/\xi(\frac{1}{2}+t)$. This proves (2.10). Strictness follows because $\Lambda > 0$ and a nonzero exponential polynomial with distinct frequencies cannot vanish identically. □

Consequently the exact inverse Fourier transform specified in Gröchenig (2020, Theorem 4) is

$$\frac{1}{2\pi}\int_{\mathbb{R}} \frac{e^{-itx}}{\xi(\frac{1}{2}+t)}\,\mathrm{d}t = \Lambda(x) > 0.$$

The Gaussian mixture in (2.8) gives a positive representation of this inverse transform and proves the positive-definiteness property discussed after Theorem 5 of Gröchenig (2020). Konstantopoulos et al. (2024, Section 4.1) ask whether F belongs to their subclass $\mathcal{D}_P$ of van Dantzig functions. By Theorem 1.1, F has the properties that Konstantopoulos et al. (2024, Theorem 4.1) derive for members of $\mathcal{D}_P$: $\log F(it) = B(t^2)$, and the Lévy density ℓ of B satisfies $(t\ell(t))' = H'(t) < 0$, so B lies in their class $\mathcal{B}_J$. These properties therefore cannot exclude F from $\mathcal{D}_P$. The same observation is made in Polson (2026b, v24, Section 1.2), and Polson (2026b, v24, Section 13) further restricts the possible representations.

3. Proof of Theorem 1.2: Log-concavity and finite PF orders

Log-concavity follows from exponential tilting and the low verified zeros. The higher orders follow by comparing an exact real exponential product with a small signed correction. Rank comparison under exponential tilting and a moving polynomial basis give a global bound that reduces all required determinant signs to scalar inequalities. A separate analytic theorem transfers the same finite order from Θ to Λ.

3.1. Log-concavity from the low verified zeros.

The one-sided density has a boundary at zero, where a small additive error need not preserve logarithmic curvature. Exponential tilting avoids that difficulty. A second low real zero keeps the tilted residual self-decomposable; its unimodality and its shape before the mode then control the required primitive on the whole positive axis. The same tilting argument also gives bilateral log-concavity. Only the zeros through height 100 are used in this subsection.

Log-concavity is additional information: the self-decomposable law Gamma(α, b), with $0 < \alpha < 1$ and $b > 0$, has density $g(x) = Cx^{\alpha-1}e^{-bx}$ and $(\log g)''(x) = (1-\alpha)/x^2 > 0$. Here the work is to verify that the required exponential tilts of the laws of S_1 and $X_{1,1}$ remain self-decomposable, so that classical shape theorems apply. These reciprocal densities also differ from the Riemann Fourier kernel Φ, whose PF_2 property is recorded in Csordas (1998, Corollary 2.11), and from the xi Taylor coefficient sequence studied by Katkova (2007).

We use two facts from the theory of self-decomposable laws. Every such law on $\mathbb{R}$ is unimodal (Yamazato, 1978). A drift-free positive self-decomposable law with a smooth positive density has *Yamazato's property*: its density is log-concave before its mode and nonincreasing after it. This is Yamazato's result, as recalled by Watanabe (1992, pp. 50–51), whose Theorem 1.2 extends it to unimodal one-sided Lévy processes without drift. For a variable V with a finite exponential moment at r, its *exponential tilt* is the probability law $e^{rx}\Pr(V \in \mathrm{d}x)/\mathbb{E}e^{rV}$. Tilting a positive infinitely divisible law with Lévy density $k(t)/t$ replaces $k(t)$ by $e^{rt}k(t)$.

The next elementary lemma is a localized strict version of the classical passage from a log-concave density to its distribution function; compare Bagnoli and Bergstrom (2005, Theorem 1 and Corollary 1). They credit Prékopa for the integration principle and Dierker for the elementary proof. We give the precise version needed here.

Lemma 3.1 (Log-concavity of a primitive)**.** *Let $g > 0$ be C^1 on $(0,\infty)$ and locally integrable at zero. Suppose that, whenever $g'(x) > 0$, $\log g$ is concave on $(0,x)$. Then $G(x) = \int_0^x g(u)\,\mathrm{d}u$ satisfies $(\log G)''(x) < 0$ for every $x > 0$. In particular this holds for a positive smooth density with Yamazato's property, or for a positive log-concave function.*

Proof. If $g'(x) \le 0$, then $G(x)g'(x) - g(x)^2 < 0$. Otherwise put $r = g'(x)/g(x) > 0$. The tangent inequality for $\log g$ gives

$$G(x) \le g(x)\int_0^x e^{r(u-x)}\,\mathrm{d}u = g(x)\frac{1-e^{-rx}}{r} < \frac{g(x)}{r}.$$

Again $Gg' - g^2 < 0$, which is exactly the asserted curvature sign. □

Proposition 3.2 (The one-sided reciprocal density)**.** *The one-sided density satisfies*

$$(\log\Theta)''(x) < 0 \qquad (x > 0).$$

Its extension by zero to $(-\infty, 0]$ belongs to PF_2.

Proof. Put $q = \gamma_1^2 \in (196, 225)$ and let m_1 be the multiplicity of the first ordinate. Choose a real ordinate $\eta \in (21, 22)$; its existence is certified in Appendix C.19. We claim that

$$J(t) = e^{qt}(H(t) - m_1 e^{-qt}) > 0, \qquad J'(t) < 0 \quad (t > 0). \tag{3.1}$$

To see this, retain the term at η and compare it with $e^{qt}H_{>100}(t)$. Put $\tau = \eta^2 - q > 216$. The shifted tail exponents are $\lambda_\rho - q$. Both their real contributions and negative derivatives are nonnegative when $|b_\rho|t \le \pi/2$. A negative contribution therefore requires $t > \pi/(2\gamma)$, while

$$(a_\rho - q) - \tau = a_\rho - \eta^2 \ge 3\gamma^2/4, \qquad |\lambda_\rho - q| \le 2\gamma^2 \le \tau\gamma^2.$$

The proof of Lemma 2.2 now applies verbatim, giving

$$e^{-\tau t} + e^{qt}H_{>100}(t) \ge (1-\varepsilon_{100})e^{-\tau t}, \quad -\frac{\mathrm{d}}{\mathrm{d}t}(e^{-\tau t} + e^{qt}H_{>100}(t)) \ge (1-\varepsilon_{100})\tau e^{-\tau t}.$$

The other terms in J are positive real exponentials with positive rates, which proves (3.1).

Subtracting all m_1 first-zero factors gives $S_1 \overset{d}{=} \mathrm{Gamma}(m_1, q) + V$, with independence and with V having Lévy density $(H(t) - m_1 e^{-qt})/t$. All remaining rates have real part greater than q, so $M := \mathbb{E}e^{qV} < \infty$. The tilted law $V^{(q)}$ has zero drift and Lévy density $J(t)/t$, hence is self-decomposable (a tilt in the opposite direction appears in Polson, 2026a, v9, Theorem 22(ii)). The gamma-extraction argument of Appendix B.4 gives it a smooth positive density: removing finitely many factors and tilting preserve the small-time divergence of J and its positive exponential lower bound.

Let F_q be its distribution function. Yamazato's property and Lemma 3.1 give $(\log F_q)'' < 0$. The density of $\mathrm{Exp}(q) + V$ is

$$f(x) = qMe^{-qx}F_q(x), \qquad x > 0,$$

so it has the same strict logarithmic curvature. If $m_1 > 1$, convolve with the other $m_1 - 1$ exponential factors. Each step replaces f by $qe^{-qx}\int_0^x e^{qu}f(u)\,\mathrm{d}u$; since $e^{qu}f(u)$ is log-concave, Lemma 3.1 preserves strict curvature. This proves the claim for $\theta_1 = c_0\Theta$, without assuming simplicity. A log-concave function on its interval support, extended by zero, is PF_2. The support can force some ordered minors to vanish. □

Proposition 3.3 (The bilateral reciprocal density)**.** *The density Λ satisfies $(\log\Lambda)''(x) < 0$ on $\mathbb{R}$. Every ordered PF_2 minor is strictly positive.*

Proof. Gaussian subordination (Sato, 1999, Theorem 30.1) gives the Lévy density of $X_{1,1}$ as $A(|x|)/|x|$ (for the Xi function, cf. Polson, 2026b, v24, Section 2 and Proposition 12.1), where

$$A(x) = \sum_{\rho\in\mathcal{Z}_+} e^{-\gamma x}\cos(\delta_\rho x), \qquad x > 0.$$

Indeed, the Gaussian integral applied to each term of H gives $e^{-(\gamma-i\delta_\rho)|x|}/|x|$. Absolute convergence for $x \neq 0$ justifies the interchange: the integrals of the absolute values sum to $|x|^{-1}\sum_\rho e^{-\sqrt{a_\rho}|x|} < \infty$. Conjugation gives the displayed real sum. Put $a = \gamma_1 < 15$. For every $|s| < a$, both $e^{sx}A(x)$ and $e^{-sx}A(x)$ are positive and decreasing. Retain the same ordinate $\eta \in (21, 22)$. For either sign, the tail summands and their negative derivatives are nonnegative for $x \leq \pi$, because $|\delta_\rho| \leq 1/2$. A negative term therefore has $x > \pi$ and

$$e^{-(\gamma-\eta)x} \leq e^{-\pi(\gamma-22)}, \qquad |\gamma \mp s - i\delta_\rho| \leq 2\gamma \leq (\eta \mp s)\gamma^2.$$

The reciprocal-square tail bound makes the negative tail contributions in either estimate, relative to the retained term, at most

$$\varepsilon = 100^4 e^{-78\pi} < 1.$$

All remaining verified terms have positive coefficients and rates $\gamma \mp s > 0$. The tilt exists because $\mathbb{E}e^{sX_{1,1}} = \mathbb{E}e^{s^2 S_1} < \infty$ for $|s| < a$. Its Lévy density is $e^{sx}A(|x|)/|x|$, so the two monotonicity statements show that it is self-decomposable (Sato, 1999, Theorem 15.10), and hence unimodal.

Choose η as the retained ordinate in Theorem C.20. This extracts a $\mathrm{Lap}(a)$ factor, with a remaining law having positive density on the whole line. If $p = c_0\Lambda$, convolution with that factor gives $|p'(x)| < ap(x)$ for every x: the positive contributions from each side of x make the triangle inequality strict. Were $(\log p)''(x_0) > 0$, the tilt with $s = -p'(x_0)/p(x_0) \in (-a, a)$ would have a strict local minimum at x_0, contradicting unimodality. Therefore $\log p$ is concave. If it were affine on an interval, analyticity would force $p(x) = Ce^{bx}$ everywhere, which cannot be a probability density. Thus it is strictly concave, and all PF_2 minors are positive. For the stronger pointwise curvature assertion, apply Theorem C.10 with $m = 2$ to $\theta = c_0\Theta$, using Proposition 3.2 and the regularity verified after that theorem. It gives $\mathscr{D}_2(c_0\Lambda; x) > 0$, hence $(\log\Lambda)''(x) < 0$. This step needs Theorem C.10: strict concavity allows $(\log\Lambda)''$ to vanish at isolated points, as $-x^4$ does at 0, whereas the theorem writes $\mathscr{D}_2(c_0\Lambda; x)$ as a positive multiple of a matrix density positive at xI. □

These arguments establish order two without the high verified height. They do not by themselves give order three. For example, if $0 < \alpha < 1$ and $b > q$, the q-tilt of a $\mathrm{Gamma}(\alpha, b)$ residual is self-decomposable. Yet the density f of $\mathrm{Exp}(q) + \mathrm{Gamma}(\alpha, b)$ satisfies $f(x) = Cx^\alpha(1 + O(x))$ near zero, with a convergent power series in the last factor, and hence

$$-\det[f^{(i+j)}(x)]_{i,j=0}^2 \sim 2C^3\alpha^2(\alpha-1)x^{3\alpha-6} < 0.$$

Thus this density fails PF_3. Higher orders need further information. The next subsection supplies it through an exact signed correction. Appendices C.18 and C.20 keep direct certificates for Θ at order four and Λ at order 256 as independent checks.

3.2. **High finite orders from a signed perturbation.** Order two used only the zeros through height 100; the higher orders use the full verified height. Replacing each unverified squared rate $\lambda_\rho = (\gamma - i\delta_\rho)^2$ by the real rate γ^2 writes the law of S_1 exactly as a real exponential product, which is PF_∞, convolved with a signed correction (Lemma C.4). The correction is small in weighted total variation, but that alone does not preserve small determinants. The key estimate is that, in a basis that moves with the evaluation point, a shift by u enlarges the Hankel matrix of derivatives of the selected product by at most the factor e^{Qu}, where Q bounds the selected rates uniformly in the order (Theorem C.9). Its proof rests on an exact Grassmannian integral and an order comparison for exponentially tilted corner arrays. All determinant signs then reduce to scalar inequalities.

For a verified height $100 \leq U \leq U_{\mathrm{PT}}$ and value $0 \leq a < U^2 - 1/4$, we use the following:

$$\mathcal{R}_U = \frac{\log(U/(2\pi)) + 1}{2\pi U} + \frac{\log(U+2)/4 + 8 + 1/16}{U^2},$$

$$c_U(a) = 1 - \frac{a + 1/4}{U^2}, \qquad \epsilon_U(a) = \exp\left\{\frac{\mathcal{R}_U}{4}\left(1 + \frac{c_U(a) + 2}{c_U(a)^2}\right)\right\} - 1. \tag{3.2}$$

Also put $C(T) = M(T) - \log(T+2)/8 - 4$, the lower bound for $N(T)$ in (2.1).

Theorem 3.4 (Scalar criterion for both reciprocal densities)**.** *Let m, n be positive integers with $n \geq 2m$ and $\frac{1}{2}(n-2m+1) \geq \sqrt{(m-1)(n-m+1)}$ (for instance $n \geq 8m$), and let $100 \leq T \leq U \leq U_{\mathrm{PT}}$. Put $Q = T^2$. Suppose $C(T) \geq n$, $Q < U^2 - \frac{1}{4}$, and $\epsilon_U(Q) < 1$. Then, the zero extension of Θ belongs to PF_m, with ordered minors strictly positive exactly when $x_i > y_i$ for every i. The density Λ has strictly positive ordered minors through order m on the entire real line.*

Proof. The first inequality selects n verified exponential factors with rates at most Q, counted with multiplicity; let f be their product. Lemma C.4 writes the law of S_1 as the real exponential product convolved with a signed measure $\mathcal{M}_U$, and by the last two inequalities $\mathcal{M}_U - \delta_0$ has weighted variation at most $\epsilon_U(Q) < 1$ with weight e^{Qu}. Theorem C.9, whose proof rests on Lemma C.6 and Theorem C.8, then gives the minors of $f * \mathcal{M}_U$ on all of $(0, \infty)$. Appendix C.8 passes to Θ through the support boundary and the remaining infinite convolution. The Gaussian-mixture Theorem C.10, whose regularity hypotheses hold for the reciprocal laws, then gives Λ at the same order. □

Proof of Theorem 1.2. Propositions 3.2 and 3.3 prove the strict curvature assertions using only zeros through height 100. For the higher order, we take the exact profile (C.28) at the full published height U_{PT}. The directed scalar inequalities in (C.29) verify every hypothesis of Theorem 3.4, including the exact factor budget. The uniform error is below 0.999999764856, against threshold one. The theorem gives both global PF assertions, including their different strictness conditions. □

The order-k_* result also has an operator interpretation. Let $v(g)$ be the largest number of sign changes of g along increasing finite sequences, with zero values discarded. For every bounded real Borel function g, the finite-order variation-diminishing theorem (Karlin, 1968, Chapter 5, Theorem 3.1) gives

$$v(\Lambda * g) \leq v(g) \qquad \text{if } v(g) \leq k_* - 1.$$

The convolution converges absolutely because Λ is integrable. The bound is the finite-order counterpart of Gröchenig (2020, Theorem 6).

There is also a necessary constraint on large orders. Theorem C.11 proves that the first nonreal poles obstruct PF after a precise number of preceding real pole occurrences, including ties and multiplicities. By Corollary C.12, for $f \in \{\Theta, \Lambda\}$,

$$f \in \mathrm{PF}_k, \quad N(Y) \leq k+1 \quad \Longrightarrow \quad \text{RH through height } Y.$$

For instance, if $\mathrm{PF}_{10^{24}}$ held, it would imply RH through height 10^{23}.

4. Proof of Theorem 1.3: Laguerre inequalities on an interval

This section proves Theorem 1.3 by expressing each L_n as a coefficient of a product with nonnegative coefficients.

4.1. Coefficients of the squared modulus.

The definition (1.6) also applies to real C^{2n} functions. For example,

$$L_2[f] = \tfrac{1}{4}(f'')^2 - \tfrac{1}{3}f'f''' + \tfrac{1}{12}ff^{(4)}.$$

Lemma 4.1 (Generating and multiplication identities)**.** *For a real entire f and real x, y,*

$$|f(x+iy)|^2 = f(x+iy)f(x-iy) = \sum_{n=0}^{\infty} L_n[f](x)y^{2n}.$$

For real entire f, g,

$$L_n[fg](x) = \sum_{k=0}^{n} L_k[f](x)L_{n-k}[g](x). \tag{4.1}$$

Local uniform convergence of real entire functions $f_j \to f$ implies $L_n[f_j] \to L_n[f]$ locally uniformly, for each fixed n.

Proof. Multiply the two Taylor series at x. Odd powers of y cancel, and the coefficient of y^{2n} is $\sum_{j=0}^{2n} i^j(-i)^{2n-j} f^{(j)}(x) f^{(2n-j)}(x)/(j!(2n-j)!)$. Since $i^j(-i)^{2n-j} = (-1)^{j+n}$, this is (1.6). Absolute convergence on compact sets justifies the multiplication. Multiplying the generating identities for f and g proves (4.1). Cauchy's integral formula implies convergence of each fixed derivative under local uniform convergence; the last statement follows from the finite sum in (1.6). □

4.2. **The zero-exclusion principle.** The Xi function satisfies the hypothesis of the next lemma by Lemma 2.1.

Lemma 4.2 (All-order zero-exclusion principle)**.** *Let $f \not\equiv 0$ be a real entire function with a representation*

$$f(z) = Cz^d e^{-a_0 z^2 + b_0 z} \prod_{\zeta \in \mathcal{Z}_f^*} E_1(z/\zeta), \qquad E_1(w) = (1-w)e^w, \tag{4.2}$$

where $C \in \mathbb{R} \setminus \{0\}$, $d \in \mathbb{N}_0$, $a_0 \geq 0$, $b_0 \in \mathbb{R}$, and $\mathcal{Z}_f^*$ is the multiset of nonzero zeros, invariant under complex conjugation, with $\sum_{\zeta \in \mathcal{Z}_f^*} |\zeta|^{-2} < \infty$. The product is grouped into real zeros and conjugate pairs. Let $I \subset \mathbb{R}$ be an interval such that

$$|x - \operatorname{Re}\zeta| \geq |\operatorname{Im}\zeta| \quad (x \in I,\ \zeta \in \mathcal{Z}_f^* \setminus \mathbb{R}).$$

Then $L_n[f](x) \geq 0$ for every $n \in \mathbb{N}_0$ and every $x \in I$.

In particular, if every nonreal zero $\zeta = a + ib$ satisfies $|a| > T$ and $|b| \leq \Delta$, with $T > \Delta \geq 0$, then

$$L_n[f](x) \geq 0 \qquad (n \in \mathbb{N}_0,\ |x| \leq T - \Delta).$$

Likewise, if $|\operatorname{Im}\zeta| \leq \Delta$ and no nonreal zero has real part in $[x_-, x_+]$, where $x_+ - x_- > 2\Delta$, the conclusion holds on $[x_- + \Delta, x_+ - \Delta]$.

Proof. Fix $x \in I$ and regard $Y = y^2$ as a formal variable. For a real zero $r \neq 0$,

$$E_1((x+iy)/r)E_1((x-iy)/r) = \frac{e^{2x/r}}{r^2}((x-r)^2 + Y),$$

which has nonnegative coefficients in Y. The factor z^d similarly contributes $(x^2+Y)^d$.

For a conjugate pair $\zeta = a+ib, \overline{\zeta} = a-ib$, $b \neq 0$, set $Q_{a,b}(z) = (z-a)^2 + b^2$. Their canonical-product factors multiply to

$$E_1(z/\zeta)E_1(z/\overline{\zeta}) = \frac{Q_{a,b}(z)}{a^2+b^2} \exp\left(\frac{2az}{a^2+b^2}\right).$$

With $u = x - a$, direct multiplication gives

$$Q_{a,b}(x+iy)Q_{a,b}(x-iy) = (u^2+b^2)^2 + 2(u^2-b^2)Y + Y^2.$$

Its coefficients are nonnegative by $|u| \geq |b|$. The exponential and normalizing factors contribute the positive, Y-independent factor $e^{4ax/(a^2+b^2)}/(a^2+b^2)^2$.

Finally, the leading factor in (4.2) contributes

$$C^2 e^{-2a_0x^2 + 2b_0x} e^{2a_0 Y},$$

whose series in Y has nonnegative coefficients because $a_0 \geq 0$. Every finite grouped product therefore has nonnegative coefficients in Y, by ordinary series multiplication. The canonical products converge locally uniformly, and hence so do their products at $x+iy$ and $x-iy$ on compact complex y-disks. Cauchy's coefficient formula shows that every coefficient converges to the corresponding coefficient of $f(x+iy)f(x-iy)$. Those limiting coefficients are $L_n[f](x)$. This proves the first assertion without division by $f(x)$ and so also at real zeros, including multiple zeros.

For $|x| \le T - \Delta$, $|x - a| \ge |a| - |x| > \Delta \ge |b|$, proving the symmetric-interval assertion. For the final assertion, $a < x_-$ or $a > x_+$, so every $x \in [x_- + \Delta, x_+ - \Delta]$ satisfies $|x - a| > \Delta \ge |b|$. □

Proof of Theorem 1.3. A zeta zero $\rho = \beta + i\gamma$ corresponds to the Xi zero $z_\rho = \gamma - i(\beta - \frac{1}{2})$. Thus the critical strip gives $|\operatorname{Im} z_\rho| < 1/2$, and the hypothesis gives $|\operatorname{Re} z_\rho| > T$ for every nonreal Xi zero. Apply Lemma 4.2 with this T and $\Delta = 1/2$. For the Xi function one can equivalently use the polynomial truncations $c_0 \prod_{\gamma \le Y}(1 - z^2/\lambda_\rho)$ directly: real zeros occur as pairs and nonreal zeros as conjugation-symmetric quartets.

To obtain strictness, fix x in the stated interval and write $\Xi(z) = (z - x)^{m_x} g(z)$, where $g(x) \neq 0$. All nonreal zero pairs satisfy the strict inequality $|x - \operatorname{Re}\zeta| > |\operatorname{Im}\zeta|$. Thus each remaining real-zero factor has two positive coefficients in Y, and each nonreal pair has three. There are infinitely many remaining factors, by zero counting. For any $k \ge 0$, choose a finite product P whose polynomial in Y has degree at least k; its kth coefficient is positive. The remaining factor g/P satisfies Lemma 4.2 and is nonzero at x, so $L_k[g](x) \ge L_k[P](x)(g/P)(x)^2 > 0$. Finally $|\Xi(x + iy)|^2 = y^{2m_x}|g(x + iy)|^2$ gives exactly the stated equality cases, and positive normalization preserves them for F. □

5. Related work

5.1. **The reciprocal law.** The reciprocal law was asked for, and constructed under RH, by Roynette and Yor (2005, eq. (0.29) and Théorème V.3.2). Polson's work connects reciprocal entire functions, Hadamard products, and Thorin representations (Polson, 2018, Section 2.1). In Polson (2026b, v24, Theorems A and 5.3), whose T is our S_1, the Lévy density $H(t)/t$, self-decomposability, and Gaussian-mixture construction coincide with ours. Polson (2026a, v9, Theorem 22) also gives the self-decomposable law. The two proofs use the explicit formula near zero and, at later times, verification through heights 1000 and 200, respectively. Our proof extends their large-time comparison with one low zero to every $t > 0$ (Section 2.2), so it needs neither the explicit formula nor verification beyond height 100. Positive powers and arbitrary dimension follow directly from infinite divisibility and Gaussian subordination; the smoothness and positivity of the densities, the negative moments of S_α, and the entire extension are not stated there.[1] Both papers also deduce unimodality of S_1 from Yamazato's theorem (Polson, 2026b, v24, Corollary 6.8(iv); Polson, 2026a, v9, Theorem 22(ii)), which Theorem 1.2 strengthens to log-concavity.

5.2. **Pólya frequency orders.** Gröchenig's Xi specialization of Schoenberg's transform theorem (Schoenberg, 1951; Gröchenig, 2020) identifies $\Theta, \Lambda \in \mathrm{PF}_\infty$ with RH. For Λ this equivalence is already in Roynette and Yor (2005, Théorème V.3.2); see also Konstantopoulos et al. (2024, Corollary 2.4). Theorem 1.2 establishes finite PF orders for these densities. Its log-concavity argument combines Yamazato's one-sided shape property (Yamazato, 1978; Watanabe, 1992) with estimates for exponential tilts. For higher orders, we separate a real exponential product from a finite signed correction, and control the latter in normalized derivative Grams. The classical orbital integral and the uniform corner-array law (Olshanski, 2013) identify the finite-exponential determinants; we prove the comparison of their separately tilted laws at successive ranks and the resulting moving-basis bound. Classical Wronskian and composition arguments (Karlin and Studden, 1966; Karlin, 1968) give Θ's support-dependent strictness. The Hermitian derivative-ensemble identities of Förster et al. (2021), combined with a conditional Gaussian matrix, give the same-order forward implication from Θ to Λ.

[1] Theorem 1.1, the equivalence of RH with (ii)–(vi) in Proposition A.1, and the observation on Konstantopoulos et al. (2024, Theorem 4.1) in Section 2.4, were obtained independently of version 24 of Polson (2026b). They are in the first version of this manuscript, recorded in commit `eae8e7e0154b` of the author's repository on 11 September 2026 at 13:45 UTC; version 24 was submitted to arXiv at 16:43 UTC that day.

5.3. **Laguerre inequalities.** Csordas et al. (1991, Lemma 4) prove the Laguerre inequality on intervals away from the nonreal zeros. Let $f(z) = Ce^{-az^2+bz}\prod_\zeta(1-z/\zeta)e^{z/\zeta}$ be a real entire function with $C \neq 0$, $a \geq 0$, and every nonreal zero in the strip $|\operatorname{Im} z| < 1$. If all zeros with real part in an interval $[x_-, x_+]$, where $x_+ - x_- > 2$, are real and simple, then $L_1[f] > 0$ on $[x_- + 1, x_+ - 1]$; see also Csordas (2015, Proposition 2.2). Their proof uses the second logarithmic derivative. Our coefficient argument is the classical product argument for Csordas and Varga (1990, Theorem 2.9), applied to conjugate zero pairs; it gives every L_n on the same interval (Lemma 4.2) and, for Ξ, determines equality from zero multiplicity.

5.4. **Equivalent forms of RH.** The equivalence of RH with complete monotonicity of H is a special case of Zhang (2023, v8, Lemma 3). Polson's RH equivalences (Polson, 2026b, v24, Proposition 3.2 and Theorem 5.6; Polson, 2026a, v9, Theorem 26) identify complete monotonicity of H with the generalized gamma convolution property of S_1; Proposition A.1 also treats H/t^α for every finite $\alpha > 0$. Polson (2026b, v24, Theorem 7.1) shows moreover that Hankel positivity of every order at a single time, with both Hankel blocks of the Stieltjes moment problem, is equivalent to RH; Polson (2026c, v25, Section 8.3) discusses the unshifted test.

5.5. **The xi function in probability.** The xi function itself also occurs as a probability transform. Biane et al. (2001) survey the probability laws whose transforms are ξ and the Jacobi theta function. Nakamura (2015, Theorems 1.1–1.2) shows that $\xi(\sigma - it)/\xi(\sigma)$ is a characteristic function of t for every real σ, and formulates an RH equivalence through *pretended infinite divisibility* for every $1/2 < \sigma < 1$. Nakamura's signed Lévy–Khintchine-type representations omit the usual integrability condition on the total variation of the signed measure. The quasi-infinitely divisible class studied by Lindner et al. (2018) retains that condition. Nakamura and Suzuki (2023, Theorems 1.1–1.2) construct from the explicit formula a function that is the characteristic function of an infinitely divisible law if and only if RH holds. Under RH its Lévy measure is $\sum_\gamma m_\gamma \gamma^{-2}\delta_{-\gamma}$, where γ ranges over distinct signed zero ordinates and m_γ is the multiplicity. For the reciprocal transform treated here, $1/h$ is infinitely divisible unconditionally, with the positive Lévy density $H(t)/t$; RH is instead the statement that this density is completely monotone (Proposition A.1). The normalized pair $(F(t), 1/F(it))$ is a van Dantzig pair (Konstantopoulos et al., 2024).

AI USAGE

The results in this paper were developed through extensive use of an AI assistant (OpenAI's GPT Pro). None of the claims were obtained zero-shot: they emerged through sustained back-and-forth discussion, including failed attempts, counterexamples, audits, and repairs. The reciprocal-density revisions, verification programs, and attribution review also used OpenAI Codex. Anthropic's Claude (Opus 5.5 and Fable 5.1) was used extensively to assist with the writing. Assistant checks and program reruns are not external peer review or formal verification.

ACKNOWLEDGEMENTS

The author (SS) thanks the Alexander von Humboldt (AvH) Foundation for their generous support via a Humboldt Professorship in AI.

REFERENCES

Mark Bagnoli and Theodore C. Bergstrom. Log-concave probability and its applications. *Economic Theory*, 26(2):445–469, 2005. doi: 10.1007/s00199-004-0514-4.

Chiara Bellotti and Peng-Jie Wong. Improved estimates for the argument and zero-counting function of the Riemann zeta-function, 2025. With an appendix by Andrew Fiori. Preprint, arXiv:2412.15470v2, submitted 7 July 2025. Constants as in v2. Published in *Mathematics of Computation* (online 14 November 2025), doi: 10.1090/mcom/4133.

Philippe Biane, Jim Pitman, and Marc Yor. Probability laws related to the Jacobi theta and Riemann zeta functions, and Brownian excursions. *Bulletin of the American Mathematical Society (New Series)*, 38(4):435–465, 2001. doi: 10.1090/S0273-0979-01-00912-0.

Lennart Bondesson. *Generalized Gamma Convolutions and Related Classes of Distributions and Densities*, volume 76 of *Lecture Notes in Statistics*. Springer, New York, 1992. doi: 10.1007/978-1-4612-2948-3.

George Csordas. Convexity and the Riemann ξ-function. *Glasnik Matematički*, 33(53)(1):37–50, 1998. URL https://web.math.pmf.unizg.hr/glasnik/33.1/33105.pdf.

George Csordas. Fourier transforms of positive definite kernels and the Riemann ξ-function. *Computational Methods and Function Theory*, 15(3):373–391, 2015. doi: 10.1007/s40315-014-0105-8.

George Csordas and Richard S. Varga. Necessary and sufficient conditions and the Riemann hypothesis. *Advances in Applied Mathematics*, 11(3):328–357, 1990. doi: 10.1016/0196-8858(90)90013-O.

George Csordas, Arden Ruttan, and Richard S. Varga. The Laguerre inequalities with applications to a problem associated with the Riemann hypothesis. *Numerical Algorithms*, 1(2):305–329, 1991. doi: 10.1007/BF02142328.

George Csordas, Wayne Smith, and Richard S. Varga. Level sets of real entire functions and the Laguerre inequalities. *Analysis*, 12(3–4):377–402, 1992. doi: 10.1524/anly.1992.12.34.377.

H. E. Daniels. Saddlepoint approximations in statistics. *The Annals of Mathematical Statistics*, 25(4):631–650, 1954. doi: 10.1214/aoms/1177728652.

Yanik-Pascal Förster, Mario Kieburg, and Holger Kösters. Polynomial ensembles and Pólya frequency functions. *Journal of Theoretical Probability*, 34:1917–1950, 2021. doi: 10.1007/s10959-020-01030-z. Theorem numbering agrees with arXiv:1710.08794v4.

Karlheinz Gröchenig. Schoenberg's theory of totally positive functions and the Riemann zeta function, 2020. Preprint, arXiv:2007.12889v1; theorem numbering as in v1. Published in *Sampling, Approximation, and Signal Analysis* (S. D. Casey, M. M. Dodson, P. J. S. G. Ferreira, and A. Zayed, eds.), Applied and Numerical Harmonic Analysis, Birkhäuser, Cham, 2023, 193–210, doi: 10.1007/978-3-031-41130-4_9.

Fredrik Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633.

Samuel Karlin. *Total Positivity, Vol. I.* Stanford University Press, Stanford, CA, 1968.

Samuel Karlin and William J. Studden. *Tchebycheff Systems: With Applications in Analysis and Statistics*. Interscience Publishers, New York, 1966.

Olga M. Katkova. Multiple positivity and the Riemann zeta-function. *Computational Methods and Function Theory*, 7(1):13–31, 2007. doi: 10.1007/BF03321628.

Takis Konstantopoulos, Pierre Patie, and Rohan Sarkar. A new class of solutions to the van Dantzig problem, the Lee–Yang property, and the Riemann hypothesis. *Annales de l'Institut Fourier*, 74:377–421, 2024. doi: 10.5802/aif.3600.

Alexander Lindner, Lei Pan, and Ken-iti Sato. On quasi-infinitely divisible distributions. *Transactions of the American Mathematical Society*, 370(12):8483–8520, 2018. doi: 10.1090/tran/7249.

Takashi Nakamura. A complete Riemann zeta distribution and the Riemann hypothesis. *Bernoulli*, 21(1):604–617, 2015. doi: 10.3150/13-BEJ581.

Takashi Nakamura and Masatoshi Suzuki. On infinitely divisible distributions related to the Riemann hypothesis. *Statistics & Probability Letters*, 201:109889, 2023. doi: 10.1016/j.spl.2023.109889.

NIST. NIST Digital Library of Mathematical Functions. https://dlmf.nist.gov/, Release 1.2.8 of 2026-09-15, 2026. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F.

Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

Grigori Olshanski. Projections of orbital measures, Gelfand–Tsetlin polytopes, and splines. *Journal of Lie Theory*, 23(4):1011–1022, 2013. doi: 10.5802/jolt.762.

David J. Platt. Isolating some non-trivial zeros of zeta. *Mathematics of Computation*, 86(307): 2449–2467, 2017. doi: 10.1090/mcom/3198.

David J. Platt and Tim Trudgian. The Riemann hypothesis is true up to $3 \cdot 10^{12}$. *Bulletin of the London Mathematical Society*, 53(3):792–797, 2021. doi: 10.1112/blms.12460.

Nicholas G. Polson. On Hilbert's 8th problem, 2017. Preprint, arXiv:1708.02653v1, submitted 7 August 2017. Theorem numbers refer to v1.

Nicholas G. Polson. van Dantzig pairs, Wald couples and Hadamard factorisation, 2018. Preprint, arXiv:1804.10043v1, submitted 19 April 2018. Section numbers refer to v1.

Nicholas G. Polson. Riemann, Thorin, van Dantzig pairs, Wald couples and Hadamard factorisation, 2026a. Preprint, arXiv:1804.10043v9, submitted 17 September 2026. Theorem numbers refer to v9.

Nicholas G. Polson. On Hilbert's 8th problem, 2026b. Preprint, arXiv:1708.02653v24, submitted 11 September 2026. Theorem numbers refer to v24; the next version, v25 (19 September 2026), is a substantially different paper.

Nicholas G. Polson. On Hilbert's 8th problem, 2026c. Preprint, arXiv:1708.02653v25, submitted 19 September 2026; a substantially different paper from v24. Section numbers refer to v25.

Bernard Roynette and Marc Yor. Couples de Wald indéfiniment divisibles. Exemples liés à la fonction gamma d'Euler et à la fonction zeta de Riemann. *Annales de l'Institut Fourier*, 55(4): 1219–1283, 2005. doi: 10.5802/aif.2125.

Ken-iti Sato. *Lévy Processes and Infinitely Divisible Distributions*, volume 68 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1999.

René L. Schilling, Renming Song, and Zoran Vondraček. *Bernstein Functions: Theory and Applications*, volume 37 of *De Gruyter Studies in Mathematics*. De Gruyter, Berlin, 2nd edition, 2012. doi: 10.1515/9783110269338.

I. J. Schoenberg. Metric spaces and completely monotone functions. *Annals of Mathematics*, 39(4): 811–841, 1938. doi: 10.2307/1968466.

I. J. Schoenberg. On Pólya frequency functions. I. The totally positive functions and their Laplace transforms. *Journal d'Analyse Mathématique*, 1:331–374, 1951. doi: 10.1007/BF02790092.

The LMFDB Collaboration. The L-functions and modular forms database: Riemann zeta function and zeros of $\zeta(s)$. `https://www.lmfdb.org/L/Riemann/`, `https://www.lmfdb.org/zeros/zeta/`, 2026. Rigorously computed zero data and provenance; accessed 10 September 2026.

E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Clarendon Press, Oxford, 2nd edition, 1986. Revised by D. R. Heath-Brown.

Toshiro Watanabe. On Yamazato's property of unimodal one-sided Lévy processes. *Kodai Mathematical Journal*, 15(1):50–64, 1992. doi: 10.2996/kmj/1138039526.

D. V. Widder. Necessary and sufficient conditions for the representation of a function by a doubly infinite Laplace integral. *Bulletin of the American Mathematical Society*, 40(4):321–326, 1934. doi: 10.1090/S0002-9904-1934-05862-2.

Makoto Yamazato. Unimodality of infinitely divisible distribution functions of class L. *The Annals of Probability*, 6(4):523–531, 1978. doi: 10.1214/aop/1176995474.

Ruiming Zhang. On certain genus 0 entire functions, 2023. Preprint, arXiv:2206.05104v8, submitted 24 December 2023. Lemma numbering refers to v8.

Appendix contents

Appendix A. Equivalent forms of the Riemann hypothesis 16
Appendix B. Estimates for H and the reciprocal laws 17

Appendix C. Higher Pólya frequency orders 19

Appendix A. Equivalent forms of the Riemann hypothesis

Extended to every order, the positivity properties of Sections 2.3 and 3 are each equivalent to RH. Schoenberg's characterization relates PF_∞ to real zeros of an entire reciprocal transform (Schoenberg, 1951). Its Xi specialization (Roynette and Yor, 2005, Théorème V.3.2; Gröchenig, 2020, Theorems 3–5) combines with the theory of complete Bernstein functions to give the following equivalences. A Bernstein function B is *complete*, written $B \in \mathrm{CBF}$, when its Lévy density is completely monotone. The equivalence of (i) and (ii), and the continuation to the slit plane used in the proof, are also in Polson (2026b, v24, Propositions 3.2–3.3 and Theorem 5.6(iii)) and Polson (2026a, v9, Lemma 8 and Theorem 26); the continuation argument goes back to Polson (2017).

Proposition A.1 (Equivalent forms of the Riemann hypothesis)**.** *The following assertions are equivalent:*

(i) *RH;*
(ii) *H is completely monotone;*
(iii) *$H(t)/t$ is completely monotone;*
(iv) *for one (equivalently, every) finite $\alpha > 0$, $H(t)/t^\alpha$ is completely monotone;*
(v) *$[H(u_i + u_j)]$ is positive semidefinite in every finite size for all $u_i > 0$;*
(vi) *$B \in \mathrm{CBF}$;*
(vii) *$\Theta \in \mathrm{PF}_\infty$;*
(viii) *$\Lambda \in \mathrm{PF}_\infty$.*

Under these equivalent conditions the Hankel kernels of H and H/t^α are strictly totally positive in every order for every $\alpha > 0$.

The central implication is analytic. Complete monotonicity gives a positive Laplace representation, which extends h'/h or $\log h$ to the plane slit along the negative real axis. Such an extension excludes every zero of h off that axis, and hence every zeta zero off the critical line.

Proof. We prove the implications in the order

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \{\text{(iii)}, \text{(iv)}\}, \qquad \text{(ii)} \Leftrightarrow \text{(v)}, \qquad \text{(iii)} \Leftrightarrow \text{(vi)} \Rightarrow \text{(i)},$$
$$\text{(iv)} \Rightarrow \text{(i)}, \qquad \text{(i)} \Leftrightarrow \text{(vii)} \Leftrightarrow \text{(viii)}.$$

Under RH, $\lambda_\rho = \gamma^2 > 0$, so H is a sum of positive real exponentials and is completely monotone. Products of completely monotone functions are completely monotone, and $t^{-\alpha}$ is completely monotone for every $\alpha > 0$; hence (i) implies (ii)–(iv). Complete monotonicity immediately gives all-order Hankel positive semidefiniteness by the Gram representation. Conversely, the representation theorem for exponentially convex functions gives $H(t) = \int_{\mathbb{R}} e^{-tx} \nu(\mathrm{d}x)$ with $\nu \geq 0$; see Widder (1934). Since $H(t) \to 0$ as $t \to \infty$, ν cannot charge $(-\infty, 0]$. This makes H completely monotone.

If H/t is completely monotone, the Lévy-density characterization of complete Bernstein functions (Schilling et al., 2012, Chapter 6) makes B complete Bernstein. Conversely a complete Bernstein function has a completely monotone Lévy density, so uniqueness of the Lévy representation gives complete monotonicity of H/t.

If B is complete Bernstein, its Pick extension is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. The identity theorem extends $h = e^B$ from the positive axis to that domain, where the exponential cannot vanish. Every zero of the entire function h must therefore lie on the removed axis. Since $h(0) = 1$, its zeros are

negative real. For $\rho = \beta + i\gamma \in \mathcal{Z}_+$ the corresponding zero is

$$(\rho - \tfrac{1}{2})^2 = \delta_\rho^2 - \gamma^2 + 2i\gamma\delta_\rho.$$

As $\gamma > 0$, this is negative real exactly when $\delta_\rho = 0$. Thus completeness forces RH by combining slit-plane holomorphy with the pre-existing entireness of h.

More generally, suppose $\ell_\alpha(t) = H(t)t^{-\alpha}$ is completely monotone for one finite $\alpha > 0$. Write $\ell_\alpha(t) = \int_0^\infty e^{-rt}\mu(\mathrm{d}r)$ with $\mu \geq 0$. Then for $s > 0$, Tonelli gives

$$B'(s) = \int_0^\infty e^{-st}H(t)\,\mathrm{d}t = \Gamma(\alpha+1)\int_0^\infty (s+r)^{-\alpha-1}\mu(\mathrm{d}r).$$

At $s = 1$ this gives $\int(1+r)^{-\alpha-1}\mu(\mathrm{d}r) < \infty$. For every compact $K \subset \mathbb{C}\setminus(-\infty, 0]$, $|z+r| \geq c_K(1+r)$ for $z \in K$, $r \geq 0$, with $c_K > 0$. Thus the principal-branch integral defines a holomorphic function J on the slit plane, as in the proof of Polson (2026a, v9, Theorem 26). The identity theorem gives $h' = Jh$ there; a zero of h of multiplicity m would give orders $m-1$ and at least m on the two sides. Hence all zeros of h are negative real, proving (iv)⇒(i).

Under RH the genus-zero product (2.4) becomes

$$h(s) = \prod_{\gamma>0}\left(1 + \frac{s}{\gamma^2}\right), \qquad \Xi(z) = c_0\prod_{\gamma>0}\left(1 - \frac{z^2}{\gamma^2}\right),$$

where ordinates are counted with multiplicity and $\sum_{\gamma>0}\gamma^{-2} < \infty$. These are respectively the restricted and even Laguerre–Pólya products. Schoenberg's characterization applied to $1/(c_0 h)$ and $1/\Xi$ gives the PF densities; conversely, either PF representation forces the corresponding product and hence RH (Roynette and Yor, 2005, Théorème V.3.2; Gröchenig, 2020, Theorems 3–5). Uniqueness of the Laplace transforms identifies those densities with (2.8). Under RH the representing measures of H and of H/t^α have infinite support, so every nonzero polynomial square has strictly positive integral; Proposition C.2 then gives strict total positivity in every finite order. □

Appendix B. Estimates for H and the reciprocal laws

The reciprocal-law construction needs control of H at both ends of the time axis. Near zero, many high zeros contribute and zero counting determines the leading scale; at large times, the first zero determines the decay. These estimates justify the Lévy integrals and moment bounds used in the paper. We first check the counting constants and theta normalization, then derive the asymptotics of H at both ends, including their fixed-order derivative versions; the last subsection completes the regularity and moment statements of Lemma 2.4. Table 1 lists, for each result of the paper, the inputs beyond elementary calculus and linear algebra.

B.1. **The counting constant.** For $Y \geq e$, Bellotti and Wong (2025, Theorem 1.1) give the upper bound

$$0.11200\log Y + 0.12567\log\log Y + 3.77417.$$

Put $u = \log Y \geq 1$. The excess over $(\log Y)/8$ is $0.12567\log u - 0.013u + 3.77417$, whose maximum at $u = 0.12567/0.013$ is less than $3.934 < 4$. At $Y = e$ the same counting estimate gives $N(e) < 4$; thus $N(Y) \leq 3$ and $-1 < M(Y) \leq 0$ on $[0, e]$, which covers the remaining range. One-sided limits at zero ordinates give the same inequality for either endpoint convention. The counting theorem, including its analytic inputs and low-height checks, is used as published.

B.2. **The theta normalization.** The classical theta formula, in our normalization, is

$$\Xi(z) = \int_{\mathbb{R}} e^{izx}\Phi(x)\,\mathrm{d}x, \qquad \Phi(x) = \sum_{n\geq 1}(4\pi^2 n^4 e^{9x/2} - 6\pi n^2 e^{5x/2})e^{-\pi n^2 e^{2x}}. \tag{B.1}$$

This normalization follows by differentiating the Jacobi theta identity and integrating by parts; see Titchmarsh (1986, Section 2.16) and Konstantopoulos et al. (2024, eq. (2.7)). The kernel is even

Table 1: Numerical inputs by result.

Assertion	Inputs beyond elementary calculus and linear algebra
Positive H and $-H'$; the Bernstein and self-decomposable laws	Critical strip, zero counting, the verified first zero, and all zeros through height 100
Global log-concavity of Θ and Λ	Reality through height 100, the real zero enclosed near 21.022 (Appendix C.19), the shifted-exponent estimates, and Yamazato's shape theorems (Yamazato, 1978; Watanabe, 1992)
PF convolution factors below U	The estimate in Appendix C.14 and verified critical-line zeros below U
PF approximation in total variation	A retained zero in $(U/4, U/3]$, the reciprocal-square tail bound, the first exponential or Laplace factor, and the scalar check $225\, b_{T_*} < 4.5 \cdot 10^{-9}$ in Corollary C.21
$\Theta, \Lambda \in \mathrm{PF}_{k_*}$ with the stated strictness	Verified height U_{PT}, zero counting, signed factorization, the orbital integral, tilted rank comparison, moving-basis bound, support-boundary and Gaussian transfers, and directed scalar checks
Direct $\Theta \in \mathrm{PF}_4$ certificate	Verified height T_*, 64 rigorous real-zero enclosures, uniform endpoint estimates, directed Fourier and residue bounds on closed intervals
Direct $\Lambda \in \mathrm{PF}_{256}$ certificate	Verified height T_*, residual mean and shared-cumulant bounds, 270 rigorous real-zero enclosures, a closed interval cover, and an analytic tail bound
All $L_n[\Xi] \geq 0$ on the interval determined by verified zeros	Canonical product, critical-strip width, and finite verification through T_*

by the theta transformation, and each summand is positive for $x \geq 0$, since $2\pi n^2 e^{2x} > 3$. It has faster-than-exponential decay at both ends. Thus F is the characteristic function of the positive density Φ/c_0.

B.3. **Asymptotics of H at both ends.**

Proposition B.1 (Asymptotics)**.** *As $t \downarrow 0$,*

$$H(t) = \frac{\log(1/t) - \gamma_E - 2\log(4\pi)}{8\sqrt{\pi t}} + O(\log(1/t)). \tag{B.2}$$

For each fixed nonnegative integer j,

$$(-1)^j H^{(j)}(t) \sim \frac{\Gamma(j+\frac{1}{2})}{8\pi} t^{-j-1/2} \log(1/t). \tag{B.3}$$

At infinity $H(t) = m_1 e^{-qt} + O(e^{-(q+\varepsilon)t})$ for some $\varepsilon > 0$, where m_1 is the multiplicity of the first zero. All fixed derivatives have analogous bounds.

The leading term in (B.2) agrees with Polson (2026b, v24, Corollary 5.5), who also obtains the constant term $7/8$ from the explicit formula.

Proof. First replace every λ_ρ by γ^2. Stieltjes integration against (2.1) gives the main integral

$$\frac{1}{2\pi} \int_0^\infty e^{-ty^2} \log(y/(2\pi))\, \mathrm{d}y.$$

The error is bounded by $\int_0^\infty 2ty e^{-ty^2} [\frac{1}{8}\log(y+2) + 4]\, \mathrm{d}y = O(1 + \log(1/t))$. Set $x = \sqrt{t}\, y$ and use $\int_0^\infty e^{-x^2}\, \mathrm{d}x = \sqrt{\pi}/2$ and $\int_0^\infty e^{-x^2} \log x\, \mathrm{d}x = \sqrt{\pi}(-\gamma_E - 2\log 2)/4$ to obtain the stated main term. For $0 < t \leq 1$,

$$|e^{t\delta^2} \cos(2t\gamma\delta) - 1| \leq C(t + t^2\gamma^2).$$

Zero counting then bounds the total replacement error by $O(\sqrt{t}\,[1 + \log(1/t)])$. For (B.3), repeat the argument with $y^{2j} e^{-ty^2}$. Its counting error is $O(t^{-j} \log(1/t))$; second-order Taylor expansion

of $z^{2j}e^{-tz^2}$ on the short vertical segments from γ to $\gamma - i\delta$ gives an error $O(t^{1/2-j}\log(1/t))$. More explicitly, its real first-order term is zero, and the second derivative is bounded by a constant depending on j times $(y^{2j-2} + ty^{2j} + t^2y^{2j+2})e^{-ty^2/2}$, with the negative power omitted at $j = 0$. Integration of these bounds against N proves the claimed order. The Gaussian main integral has leading constant $\Gamma(j+1/2)/(8\pi)$. For the large-time statement, low zeros are discrete and real, and every tail real part exceeds $100^2 - 1/4 > q$. There is therefore a positive gap above the minimum real part q. Local absolute convergence at $t = 1$ controls the remaining exponentially decaying sum, also after any fixed number of differentiations. □

B.4. **Regularity and moments of the reciprocal laws.** The Bernstein representation gives a probability measure, but the later shape arguments need a smooth density that is positive in the interior of its support. We complete that step by extracting gamma convolution factors of arbitrarily high smoothness; the asymptotics of H then control the exponential moments.

Proof of Lemma 2.4. For each integer $N \geq 2$, (B.2) and the lower bound in Lemma 2.2 allow $b_N > 0$ such that $\alpha H(t) \geq Ne^{-b_N t}$ for every $t > 0$. To see this explicitly, first choose $t_N > 0$ so that $\alpha H \geq N$ on $(0, t_N]$, and then choose $b_N \geq q$ large enough that $Ne^{-(b_N-q)t_N} \leq \alpha(1 - \varepsilon_{100})$. Subtracting this Lévy density gives independent variables with

$$S_\alpha \stackrel{d}{=} \mathrm{Gamma}(N, b_N) + R_N. \tag{B.4}$$

The gamma density, extended by zero to the negative half-line, has bounded continuous derivatives through order $N - 2$. Convolution with the law of R_N preserves them. As N is arbitrary, θ_α is smooth. Every R_N has zero drift: its Laplace exponent divided by s tends to zero, by dominated convergence with the finite first Lévy moment. For a nonnegative variable Y, the elementary Laplace bounds show $\lim_{s\to\infty} -s^{-1}\log \mathbb{E}e^{-sY} = \inf \operatorname{supp} Y$. Hence $\inf \operatorname{supp} R_N = 0$, and convolution in (B.4) makes $\theta_\alpha(x) > 0$ for every $x > 0$.

The large-time expansion of Proposition B.1 shows $\int_1^\infty (e^{rt} - 1)H(t)\,\mathrm{d}t/t < \infty$ exactly for $r < q$. The Poisson exponential formula, first with truncations and then by monotone convergence, proves the exponential-moment assertions. Holomorphy follows by domination on compact subsets. Finally Stirling's formula in (1.1) gives

$$B(s) = \tfrac{1}{4}\sqrt{s}\log s + O(\sqrt{s}) \quad (s \to \infty). \tag{B.5}$$

In particular $h(s)^{-\alpha} \to 0$, so there is no atom at zero. For $p > 0$, Tonelli's theorem yields

$$\mathbb{E}S_\alpha^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty s^{p-1} h(s)^{-\alpha}\,\mathrm{d}s < \infty,$$

using (B.5). □

Appendix C. Higher Pólya frequency orders

We first give the complete analytic proof of the scalar criterion in Theorem 3.4: the signed factorization, the finite-exponential reference, its exact orbital identity, comparison of tilted corner arrays and rank increments, a global moving-basis bound, and the two transfers to the actual reciprocal densities. The numerical specialization is in Appendix C.10. We then prove the necessary first-pole obstruction and the exact scope of finite spatial stitching in Appendices C.11 and C.12. Subsequent subsections give fixed-split bounds, positive-residual factorizations, approximation results, and the direct certificates as independent results. All accepted numerical inequalities use directed interval arithmetic. The verified height and first-zero isolation are published inputs.

C.1. **Wronskian transfers.** The following is the classical Wronskian criterion for an extended complete Chebyshev system (Karlin and Studden, 1966). We include the short proof to fix the signs used in the PF arguments of this appendix and in the last assertion of Proposition A.1.

Lemma C.1. *Suppose real C^{k-1} functions $f_1, \dots, f_k$ on an interval have positive initial Wronskians, where $W(f_1, \dots, f_j) = \det[f_b^{(a-1)}]_{a,b=1}^{j}$: $W(f_1, \dots, f_j) > 0$ for $1 \le j \le k$. Then $\det[f_j(s_i)]_{i,j=1}^{k} > 0$ whenever $s_1 < \cdots < s_k$.*

Proof. Induct on k. Since $f_1 > 0$, divide the evaluation row at s_i by $f_1(s_i)$. Set $g_j = (f_{j+1}/f_1)'$. Direct differentiation gives

$$W(g_1, \dots, g_j) = \frac{W(f_1, \dots, f_{j+1})}{f_1^{j+1}} > 0.$$

Subtract consecutive normalized rows, starting at the bottom, and expand in the first column. The remaining determinant is the integral, over $[s_1, s_2] \times \cdots \times [s_{k-1}, s_k]$, of the evaluation determinant of $g_1, \dots, g_{k-1}$. It is strictly positive in the interior by the induction hypothesis. □

Positive differential Hankel determinants pass to distinct arguments by the same principle.

Proposition C.2. *If a smooth real f satisfies $[(-1)^{i+j} f^{(i+j)}(t)]_{i,j=0}^{k-1} \succ 0$ for all $t > 0$, then $f(x+y)$ is strictly totally positive through order k.*

Proof. Every initial differential Hankel determinant $\det[f^{(i+j)}(t)]_{i,j=0}^{r-1}$ is positive: the inserted row and column signs have product one. Fixing y, these are the initial Wronskians of $x \mapsto f(x+y), f'(x+y), \dots, f^{(r-1)}(x+y)$. Lemma C.1 first proves positivity of $\det[f^{(j-1)}(x_i + y)]_{i,j=1}^{r}$. Fixing the x_i, these determinants are the transposes of the initial Wronskians of the functions $y \mapsto f(x_i + y)$. A second application of the lemma proves the assertion. □

C.2. **Differential signs at the support boundary.** For a function with the required derivatives write

$$\mathscr{D}_k(f; x) = (-1)^{k(k-1)/2} \det[f^{(i+j)}(x)]_{i,j=0}^{k-1}.$$

Lemma C.3 (Differential criterion at a support boundary)**.** *Let $m \ge 1$ and let $f \in C^{2m-2}(\mathbb{R})$ vanish on $(-\infty, 0]$. Suppose that $\mathscr{D}_k(f; x) > 0$ for every $x > 0$ and $1 \le k \le m$. Then, for strictly increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$, with $k \le m$,*

$$\det[f(x_i - y_j)]_{i,j=1}^{k} \ge 0, \qquad \det[f(x_i - y_j)]_{i,j=1}^{k} > 0 \quad \Longleftrightarrow \quad x_i > y_i \quad (1 \le i \le k).$$

Proof. We first extend the evaluation criterion used in Lemma C.1. Suppose $a_1 < \cdots < a_n$, each $g_j \in C^{n-1}(\mathbb{R})$ vanishes for $t \le a_j$, and $W(g_1, \dots, g_r)(t) > 0$ for $t > a_r$, $r \le n$. Then $\det[g_j(t_i)] \ge 0$ at increasing nodes, with strict inequality exactly when $t_i > a_i$ for all i. The case $n = 1$ is immediate. If $t_i \le a_i$, the first i rows have nonzero entries in at most $i - 1$ columns. Otherwise $g_1 > 0$ on $[t_1, t_n]$. Put $h_j = (g_{j+1}/g_1)'$ for $t > a_1$. These functions extend by zero to the left of a_{j+1}, belong to C^{n-2}, and satisfy

$$W(h_1, \dots, h_r) = \frac{W(g_1, \dots, g_{r+1})}{g_1^{r+1}} > 0 \qquad (t > a_{r+1}).$$

Dividing rows by $g_1(t_i)$ and taking consecutive row differences gives

$$\frac{\det[g_j(t_i)]_{i,j=1}^{n}}{\prod_i g_1(t_i)} = \int_{t_1}^{t_2} \cdots \int_{t_{n-1}}^{t_n} \det[h_j(v_i)]_{i,j=1}^{n-1} \, \mathrm{d}v_{n-1} \cdots \mathrm{d}v_1.$$

Induction makes the integrand nonnegative. It is positive on the nonempty open rectangle $\max(t_i, a_{i+1}) < v_i < t_{i+1}$, proving the assertion.

Fix $n \le m$ and set $g_j(t) = f(t + x_{n+1-j})$, with $a_j = -x_{n+1-j}$. For $t > a_r$, all arguments in the initial Wronskian of order r are positive. The functions $f, -f', \dots, (-1)^{r-1} f^{(r-1)}$ have positive initial

Wronskians $\mathscr{D}_j(f;\cdot)$ on $(0,\infty)$. Lemma C.1 therefore gives their positive evaluation determinants. Reversing the arguments and removing the column signs cancel the two factors $(-1)^{r(r-1)/2}$, so $W(g_1,\dots,g_r)(t)>0$. Apply the preceding assertion at $t_i=-y_{n+1-i}$. Its evaluation matrix is the transpose of $[f(x_i-y_j)]$, with both orders reversed, and has the same determinant. Its strictness condition is precisely $x_i>y_i$ for every i. □

C.3. **A signed correction to an all-real product.** Throughout the next subsections, total variation means the full variation mass of a signed measure. Fix a verified height $100\le U\le U_{\mathrm{PT}}$, and retain every occurrence of each zero, including multiplicities. Define

$$K_0(u)=\sum_{\rho\in\mathcal{Z}_+}e^{-\gamma^2u},$$

$$\Delta K(u)=\sum_{\gamma>U}e^{-\gamma^2u}\{e^{\delta_\rho^2u}\cos(2\gamma\delta_\rho u)-1\},\qquad \Delta\nu(\,\mathrm{d}u)=\frac{\Delta K(u)}{u}\,\mathrm{d}u. \tag{C.1}$$

Conjugate occurrences are counted separately, so $H=K_0+\Delta K$ exactly. The real rates γ^2 above U are auxiliary; replacing λ_ρ by these rates does not assert that $\delta_\rho=0$.

Lemma C.4 (Finite signed factor)**.** *Let $\mathcal{R}_U,c_U(a)$ and $\epsilon_U(a)$ be as in (3.2). The signed measure $\Delta\nu$ is finite, with mass $d\in[-\mathcal{R}_U/4,0]$. The series*

$$\mathcal{M}_U=e^{-d}\sum_{j=0}^{\infty}\frac{\Delta\nu^{*j}}{j!},\qquad \Delta\nu^{*0}=\delta_0, \tag{C.2}$$

defines a real signed measure of mass one, supported on $[0,\infty)$, and

$$\int e^{au}|\,\mathrm{d}(\mathcal{M}_U-\delta_0)(u)|\le\epsilon_U(a)\qquad(0\le a<U^2-1/4). \tag{C.3}$$

If V_0 is the convergent sum of independent $\mathrm{Exp}(\gamma^2)$ variables over all positive zero occurrences, then

$$\mathrm{Law}(S_1)=\mathrm{Law}(V_0)*\mathcal{M}_U. \tag{C.4}$$

Proof. Stieltjes integration of (2.1) gives

$$\sum_{\gamma>U}\gamma^{-2}\le\mathcal{R}_U=\frac{\log(U/(2\pi))+1}{2\pi U}+\frac{\log(U+2)/4+8+1/16}{U^2}. \tag{C.5}$$

Indeed, writing $E(t)=\log(t+2)/8+4$, the error in integrating the counting main term is at most $E(U)U^{-2}+2\int_U^\infty E(t)t^{-3}\,\mathrm{d}t$. The inequality $\log(t+2)\le\log(U+2)+\log(t/U)$ evaluates this to the second term in (C.5); integrating M' gives the first term. The endpoint convention for N does not enlarge the bound.

For one occurrence, put $b=\gamma^2-\delta_\rho^2$. The inequalities $e^v-1\le ve^v$ and $1-\cos v\le v^2/2$ give

$$\frac{|e^{-bu}\cos(2\gamma\delta_\rho u)-e^{-\gamma^2u}|}{u}\le\delta_\rho^2e^{-bu}+2\gamma^2\delta_\rho^2ue^{-bu}.$$

Since $b-a\ge c_U(a)\gamma^2$ and $\delta_\rho^2\le1/4$, integration and (C.5) yield

$$\int e^{au}|\,\mathrm{d}\Delta\nu(u)|\le\frac{c_U(a)+2}{4c_U(a)^2}\mathcal{R}_U=:V_U(a). \tag{C.6}$$

This justifies absolute summation. Frullani's integral then gives

$$d=-\sum_{\gamma>U}\log(1+\delta_\rho^2/\gamma^2),\qquad-\mathcal{R}_U/4\le d\le0.$$

The series (C.2) converges in every weighted variation norm in (C.6). Its Laplace transform is

$$\int e^{-su}\,\mathrm{d}\mathcal{M}_U(u)=\exp\Big(\int(e^{-su}-1)\,\mathrm{d}\Delta\nu(u)\Big).$$

Separating the atom at zero, and using $e^{-d} \geq 1$, bounds its difference from δ_0 in weighted variation by $e^{-d} - 1 + e^{-d}(e^{V_U(a)} - 1) \leq e^{\mathcal{R}_U/4 + V_U(a)} - 1$. This proves (C.3). The measure need not be positive: if $d < 0$, its atom has mass greater than one, so it is not a probability law; if $d = 0$, the correction is δ_0.

The sum defining V_0 converges almost surely and in L^1, since $\sum \gamma^{-2} < \infty$. Its Laplace exponent has kernel K_0. Multiplying its Laplace transform by that of $\mathcal{M}_U$ gives $\exp[-\int (1 - e^{-su}) H(u)\, \mathrm{d}u/u] = h(s)^{-1}$. Uniqueness for finite signed measures on $[0, \infty)$ proves (C.4). □

C.4. **The full finite-exponential reference.** Here every norm is the Euclidean operator norm. The full reference has dimension n, although only derivatives through $2m - 2$ will be applied to an actual zero-extended convolution.

Lemma C.5 (Finite-exponential Jacobi bound)**.** *Let $f = *_{\ell=1}^{n} \mathrm{Exp}(q_\ell)$ with $0 < q_\ell \leq Q$. For each $x > 0$, its entire positive-argument branch $\widetilde{f}$ has a lower triangular polynomial normalization C_x with Gram $J_n = \mathrm{diag}((-1)^i)_{i=0}^{n-1}$. In that basis, multiplication by the derivative variable is a signed Jacobi matrix B_x satisfying*

$$\|B_x\|_2 \leq \frac{2n}{x} + nQ, \qquad C_x[\widetilde{f}^{(i+j)}(x - u)]_{i,j<n} C_x^T = e^{-uB_x} J_n. \tag{C.7}$$

Repeated rates are allowed.

Proof. Write $D_k(x) = \mathscr{D}_k(f; x)$, $D_0 = 1$. We first prove a positive Laplace representation, valid for $k \leq n$:

$$D_k(x) = d_k x^{k(n-k)} \int e^{-xA}\, \mathrm{d}\nu_k(A), \qquad d_k > 0, \quad \operatorname{supp} \nu_k \subset [kq_{\min}, kq_{\max}], \tag{C.8}$$

where ν_k is a probability measure independent of x.

For one exponential kernel its ordered minor is $q^k e^{-q\sum(u_i - v_i)} \det[\mathbf{1}_{u_i > v_j}]$. The step determinant is zero or one, and is one on the interlacing region $v_1 < u_1 < v_2 < u_2 < \cdots < v_k < u_k$. Iterated continuous Cauchy–Binet therefore represents a minor of f as a positive integral over a bounded interlacing network. Take its endpoint tuples to be $x\delta\xi_i$ and $x(1 + \delta\xi_i)$, with the same increasing ξ_i at both ends, and scale every intermediate coordinate by x. The domain is independent of x. Every increment is nonnegative and their sum is k, so the exponent is $-xA$ with $kq_{\min} \leq A \leq kq_{\max}$.

Normalize Lebesgue measure on this scaled domain and push it forward by A. Its volume is nonzero for sufficiently small δ: replacing every exponential by the step kernel gives $h(x) = x_+^{n-1}/(n-1)!$, and direct differentiation gives

$$\mathscr{D}_k(h; x) = \frac{x^{k(n-k)}}{((n-1)!)^k} \prod_{j=0}^{k-1} j!(n-1)_{\underline{j}} > 0.$$

The ratio of the two minors is the Laplace transform of the normalized pushforward, times $(\prod q_\ell)^k$. Choose one weakly convergent subsequence as $\delta \downarrow 0$; the measures have common compact support and are independent of x. Confluence of the endpoint minors yields (C.8), with the same limiting measure for every $x > 0$ and $d_k = (\prod q_\ell)^k \prod_{j<k} j!(n-1)_{\underline{j}}/((n-1)!)^k$. This proves strict positivity.

Compact support permits differentiating the representation. With expectation and variance under its tilt by e^{-xA},

$$\begin{aligned} (\log D_k)' &= \frac{k(n-k)}{x} - \mathbb{E}_{k,x} A, \\ (\log D_k)'' &= -\frac{k(n-k)}{x^2} + \mathrm{Var}_{k,x}(A). \end{aligned} \tag{C.9}$$

All leading Hankel determinants through size n are nonzero, so exact orthogonalization of $P \mapsto P(D)f(x)$ gives normalized polynomials of degrees $0, \ldots, n-1$ with squared norms $(-1)^i$. Multiplication has diagonal entries $\alpha_i = (\log(D_{i+1}/D_i))'$ and off-diagonal entries $B_{i,i+1} = b_i$, $B_{i+1,i} = -b_i$.

The last row is well defined: $\widetilde{f}$ solves $\prod_\ell (D+q_\ell)\widetilde{f}=0$, so multiplication takes place modulo $\prod_\ell (z+q_\ell)$. This also covers repeated rates.

Equation (C.9) and the supports of the probability measures give

$$|\alpha_i| \le \frac{n-1}{x} + nQ.$$

Here both expectations are nonnegative, so the absolute difference between the ones of orders $i+1$ and i is at most $(i+1)Q$. The signed Hankel Sylvester identity gives, for $1 \le k < n$,

$$b_{k-1}^2 = \frac{D_{k+1}D_{k-1}}{D_k^2} = -(\log D_k)'' \le \frac{k(n-k)}{x^2} \le \frac{n^2}{4x^2}.$$

The entrywise absolute matrix is symmetric, and its row sums are at most $(2n-1)/x + nQ$. This proves the norm bound.

Freeze the normalized polynomial basis at x. Quotient multiplication shows that the derivatives of the two expressions in (C.7) agree at $u=0$ at every order. Both sides are entire exponential polynomials, so the translation identity follows. □

C.5. **The Grassmannian law of a finite-exponential minor.** The compactly supported measure in (C.8) has an exact orbital interpretation. Identifying the measure, rather than only its support, will allow comparison of its separately tilted laws at successive ranks.

Lemma C.6 (Exact Grassmannian identity)**.** *Let $f = *_{\ell=1}^n \operatorname{Exp}(q_\ell)$, where $q_\ell > 0$ and repeated rates are allowed. Put $D_k(x) = \mathscr{D}_k(f;x)$, $D_0 = 1$, and*

$$\mathcal{C}_0 = 1, \qquad \mathcal{C}_j = \prod_{\ell=0}^{j-1} \ell!, \qquad d_k = \Big(\prod_{\ell=1}^n q_\ell\Big)^k \frac{\mathcal{C}_k \mathcal{C}_{n-k}}{\mathcal{C}_n}.$$

Let V have normalized Haar law on $U(n)$, and let P_k be the diagonal rank-k projection with eigenvalues $0^{n-k}, 1^k$. For $x>0$ and $0 \le k \le n$,

$$\frac{D_k(x)}{d_k x^{k(n-k)}} = \mathbb{E}\exp\left[-x \operatorname{tr}\big(\operatorname{diag}(q_1,\dots,q_n) V P_k V^*\big)\right]. \tag{C.10}$$

The constant d_k agrees with that in (C.8), and ν_k is exactly the law of

$$A_k = \operatorname{tr}\big(\operatorname{diag}(q_1,\dots,q_n) V P_k V^*\big).$$

Proof. First assume the rates are pairwise distinct and write $a_j = -q_j$, relabeling them so that $a_1 < \cdots < a_n$. Partial fractions give

$$f^{(r)}(x) = \Big(\prod_{\ell=1}^n q_\ell\Big) \sum_{j=1}^n \frac{a_j^r e^{x a_j}}{\prod_{\ell \ne j}(a_j - a_\ell)} \qquad (x>0,\ r \ge 0).$$

Cauchy–Binet applied to the resulting finite moment sum expresses the Hankel determinant as a sum over k-element subsets I of $\{1,\dots,n\}$. In each term,

$$\prod_{\substack{i,j\in I\\ i\ne j}} (a_i - a_j) = (-1)^{k(k-1)/2} \Delta(a_I)^2.$$

Thus the internal Vandermonde factors cancel, including the differential sign, and

$$\begin{aligned} D_k(x) &= \Big(\prod_{\ell=1}^n q_\ell\Big)^k S_k(a;x), \\ S_k(a;x) &= \sum_{\substack{I\subset\{1,\dots,n\}\\ |I|=k}} \frac{\exp(x\sum_{i\in I} a_i)}{\prod_{i\in I,\, j\notin I}(a_i - a_j)}. \end{aligned} \tag{C.11}$$

Empty products give $S_0 = 1$ and $S_n = e^{x\sum_j a_j}$. Individual terms of this subset sum need not be positive.

For distinct real vectors a, b, the normalized Harish–Chandra integral is

$$\int_{U(n)} e^{x\,\mathrm{tr}(\mathrm{diag}(a)V\,\mathrm{diag}(b)V^*)}\,\mathrm{d}V = \mathcal{C}_n x^{-n(n-1)/2}\frac{\det[e^{xa_ib_j}]_{i,j=1}^n}{\Delta(a)\Delta(b)}, \tag{C.12}$$

where $\Delta(a) = \prod_{i<j}(a_j - a_i)$; see Olshanski (2013, Eq. (12)). All unitary integrals here use probability Haar measure. One may equivalently obtain the real-exponential form from the Fourier form by analytic continuation, since the compact integral is entire in the spectral arguments.

Conflate the first $n-k$ entries of b to zero and the last k entries to one. Within each group, Taylor confluence replaces its columns by derivatives of orders $0, 1, \ldots$, divided by the corresponding factorials. Cross-group Vandermonde factors tend to one. Define the $n \times n$ matrix

$$M_{ij}^{(k)} = \begin{cases} a_i^{\,j-1}, & 1 \le j \le n-k, \\ e^{xa_i}a_i^{\,j-(n-k)-1}, & n-k < j \le n. \end{cases}$$

The determinant quotient in (C.12) consequently tends to

$$\frac{x^{\binom{n-k}{2}+\binom{k}{2}}}{\mathcal{C}_{n-k}\mathcal{C}_k}\frac{\det M^{(k)}}{\Delta(a)}.$$

Laplace expansion in the last k columns gives $\det M^{(k)}/\Delta(a) = S_k(a;x)$: its cross-Vandermonde signs are precisely the signs in (C.11). Since $\binom{n}{2} - \binom{n-k}{2} - \binom{k}{2} = k(n-k)$, we obtain

$$\mathbb{E}e^{x\,\mathrm{tr}(\mathrm{diag}(a)VP_kV^*)} = \frac{\mathcal{C}_n}{\mathcal{C}_k\mathcal{C}_{n-k}}\,x^{-k(n-k)}S_k(a;x).$$

Together with (C.11), this proves (C.10). The constants coincide because

$$\frac{\mathcal{C}_k\mathcal{C}_{n-k}}{\mathcal{C}_n} = \prod_{j=0}^{k-1}\frac{j!}{(n-1-j)!} = \frac{\prod_{j=0}^{k-1} j!(n-1)_{\underline{j}}}{((n-1)!)^k}.$$

To allow repeated rates, approximate a positive rate vector by pairwise distinct positive vectors. The entire positive branch of f solves $\prod_\ell(D+q_\ell)\widetilde{f} = 0$, with initial derivatives zero through degree $n-2$ and derivative $\prod_\ell q_\ell$ at degree $n-1$. Its companion-matrix exponential therefore depends continuously on the rates, as do all fixed derivatives on the positive axis, even when roots coalesce. The orbital integral is continuous by dominated convergence on $U(n)$. Hence (C.10) survives the limit; no simplicity assumption on the selected zeros is needed.

Finally $A_k \in [kq_{\min}, kq_{\max}]$. The identity just proved and (C.8) give equality of the Laplace transforms of its law and ν_k for every $x > 0$. These measures have compact support, so they are equal by Laplace-transform uniqueness. Equivalently, their entire transforms give equality of all moments, and polynomials are dense among continuous functions on their common compact support. Thus their tilted expectations are also identical:

$$\mathbb{E}_{k,x}A = \frac{\mathbb{E}[A_k e^{-xA_k}]}{\mathbb{E}[e^{-xA_k}]}. \tag{C.13}$$

□

C.6. **Order comparison for tilted corner arrays.** We need a comparison between separately normalized exponential tilts, not merely a coupling of the untilted projection matrices. The comparison below uses the full array of corner eigenvalues.

For a weakly increasing vector $\lambda = (\lambda_1, \dots, \lambda_n)$, put $X_\lambda = V \operatorname{diag}(\lambda) V^*$, with V Haar distributed on $U(n)$. Denote the increasingly ordered eigenvalues of its leading $r \times r$ corner by $z_1^{(r)}, \dots, z_r^{(r)}$, and write $\mathbf{z}(X_\lambda)$ for the array of proper corners. They interlace:

$$z_j^{(r+1)} \le z_j^{(r)} \le z_{j+1}^{(r+1)}, \qquad z^{(n)} = \lambda.$$

Put $S_r(X_\lambda) = \sum_{j=1}^r z_j^{(r)}$ and $S_0 = 0$. For fixed real fields $c_1, \dots, c_{n-1}$, let $\mathbb{E}_{\lambda,c}$ denote the probability law obtained from this orbital law by the positive weight $\exp(\sum_{r<n} c_r S_r)$, divided by its expectation. This definition applies also to repeated entries of λ.

Lemma C.7 (Comparison of tilted corner arrays)**.** *Fix the same real fields $c_1, \dots, c_{n-1}$ for two weakly increasing top rows λ, μ, and suppose $\lambda_j \le \mu_j$ for every j. For every continuous coordinatewise nondecreasing function Φ of the proper-corner array,*

$$\mathbb{E}_{\lambda,c} \Phi(\mathbf{z}) \le \mathbb{E}_{\mu,c} \Phi(\mathbf{z}).$$

In particular, $\mathbb{E}_{\lambda,c} S_r \le \mathbb{E}_{\mu,c} S_r$ for $r < n$. The fields may have either sign.

Proof. For $n = 1$ there are no proper-corner coordinates, so there is nothing to prove. Assume $n \ge 2$.

First let both top rows be strictly increasing and rational. For a strictly increasing boundary, the full proper-corner array of Haar orbital measure has constant density on the interlacing polytope $\mathrm{GT}(\lambda)$; the density is $\prod_{j=0}^{n-1} j! / \prod_{i<j} (\lambda_j - \lambda_i)$ by Olshanski (2013, Corollary 3.2). The tilted law therefore has density proportional to

$$\exp\left(\sum_{r<n} c_r \sum_{j=1}^r z_j^{(r)} \right) \mathbf{1}_{\mathrm{GT}(\lambda)}(\mathbf{z}). \tag{C.14}$$

The proportionality constant depends on the boundary, but the fields in the exponent do not.

We prove the comparison on a finite mesh before taking a continuous limit. Choose $h > 0$ such that both top rows lie in $h\mathbb{Z}^n$, and restrict every proper-corner coordinate to $h\mathbb{Z}$. The admissible set is finite and nonempty. Give each array the positive weight in (C.14). Conditional on all coordinates except $z_j^{(r)}$, its possible values form a nonempty lattice interval $[L, U] \cap h\mathbb{Z}$. Here L is the maximum of its lower interlacing neighbors and U is the minimum of its upper neighbors, including fixed boundary coordinates when appropriate. Both endpoints are nondecreasing functions of the other coordinates and of the top row. Its conditional weight at t is $e^{c_r t}$.

For any fixed positive weights $w(t)$, their normalized restrictions to two lattice intervals $[L, U]$ and $[L', U']$ are stochastically ordered when $L \le L'$ and $U \le U'$. Indeed, outside the common overlap the CDF comparison is immediate. For a test point in the overlap, write A for the weight below L', C for the weight from L' through the test point, D for the weight above it through U, and E for the weight above U through U'. The first CDF is $(A+C)/(A+C+D)$ and the second is $C/(C+D+E)$. Cross multiplication leaves $AD + (A+C)E \ge 0$. Thus common-uniform inverse-CDF sampling gives ordered conditional updates. In particular, c_r may have either sign.

For completeness, the single-site update chain is irreducible. The coordinatewise minimum admissible array for boundary λ is $z_j^{(r)} = \lambda_j$. From any other array there is a legal decrease of one coordinate by h towards this minimum. If a coordinate above its target cannot decrease by h, some lower neighbor has the same current value, because all coordinates lie on the same mesh. That lower neighbor is itself above its target: the target array satisfies the ordering constraint. A fixed boundary value cannot block such a decrease, since it would force the target of the original coordinate to be at least its current value.

Follow a blocking lower neighbor until reaching a coordinate that can decrease. This terminates: the directed interlacing edges are

$$(r+1, j) \longrightarrow (r, j) \longrightarrow (r+1, j+1),$$

and the integer $2j - r$ increases by one along each edge. Following lower neighbors therefore follows an acyclic graph in the opposite direction. Performing the resulting decrease reduces the finite integer sum of all mesh distances to the minimum by one. Iteration reaches the minimum. Every such move, and its reversal, has positive conditional probability. Hence random-scan conditional resampling is irreducible. It has positive self-loops, so is aperiodic. The weighted law is stationary by conditional resampling; the finite chain consequently converges to that unique stationary law.

Start the two boundary chains at their respective minimum arrays, which are ordered because $\lambda \le \mu$. Use the same random coordinate and the same inverse-CDF uniform variable at each update. The monotonicity of the interval endpoints and of the conditional laws preserves the order of the two arrays at every step. Taking the finite-chain stationary limit proves the desired expectation comparison for the two lattice laws.

Now let h tend to zero through common divisors of the rational boundaries. For a strictly increasing top row the interlacing polytope is bounded, has nonempty interior in the full space of $n(n-1)/2$ proper-corner coordinates, and has boundary of Lebesgue measure zero. Riemann sums therefore converge for both the positive normalizing integral in (C.14) and the same integral with the continuous factor Φ. The arrays lie in a common compact set, so this also covers functions Φ not globally bounded. Dividing the two convergent sums proves the comparison for strictly increasing rational top rows under the continuous orbital laws.

It remains to justify arbitrary real and repeated top entries. The tilted orbital law is weakly continuous in the top row. Use one Haar matrix V for a sequence of top rows tending to λ. Then $V \operatorname{diag}(\lambda) V^*$ and the ordered eigenvalues of each Hermitian corner vary continuously. On a bounded set of top rows the corner arrays have common compact support, and their exponential weights are bounded. Dominated convergence therefore applies to the weighted test integrals and their normalizing integrals; the limiting normalizing integral is strictly positive.

To approximate ordered weak boundaries while preserving their order, take $h \downarrow 0$ through positive rational values and put

$$\lambda_j^{(h)} = h\lfloor \lambda_j/h \rfloor + jh, \qquad \mu_j^{(h)} = h\lfloor \mu_j/h \rfloor + jh.$$

Both rows are rational and strictly increasing, $\lambda^{(h)} \le \mu^{(h)}$, and they converge to the respective original boundaries. The comparison already proved passes to the limit by the orbital continuity just established. In particular the conclusion applies to top rows consisting of zeros and ones. No ambient Lebesgue density on a collapsed polytope has been used. □

Theorem C.8 (Rank increments under exponential tilting)**.** *Let $D = \operatorname{diag}(q_1, \dots, q_n)$ be real, and put $q_- = \min_j q_j$, $Q = \max_j q_j$. Let P_k be an orthogonal projection of rank k, and define, with normalized Haar measure,*

$$Z_k(x) = \int_{U(n)} e^{-x \operatorname{tr}(DVP_kV^*)} \, \mathrm{d}V, \qquad \Gamma_k(x) = -\frac{\mathrm{d}}{\mathrm{d}x} \log Z_k(x).$$

For every real x and $0 \le k < n$,

$$q_- \le \Gamma_{k+1}(x) - \Gamma_k(x) \le Q. \tag{C.15}$$

Repeated rates are allowed. The assertion compares the separately normalized tilted laws at ranks k and $k+1$.

Proof. Compactness of the unitary group justifies differentiation under the integral for every real x. Thus $\Gamma_k(x)$ is the expected value of $\operatorname{tr}(DVP_kV^*)$ under the indicated normalized tilt. In particular $\Gamma_0 = 0$ and $\Gamma_n = \operatorname{tr} D$.

For any Hermitian matrix X, let $S_r(X)$ be the trace of its leading $r \times r$ corner. Since $X_{rr} = S_r - S_{r-1}$,

$$\operatorname{tr}(DX) = q_n S_n + \sum_{r=1}^{n-1} (q_r - q_{r+1}) S_r. \tag{C.16}$$

On an orbit with fixed top eigenvalues, S_n is constant and its factor cancels in the normalization of the exponential tilt. The fields of the proper-corner array are therefore

$$c_r = x(q_{r+1} - q_r),$$

the same at every top boundary.

The increasingly ordered top row $(0^{n-k}, 1^k)$ is coordinatewise at most $(0^{n-k-1}, 1^{k+1})$. Apply Lemma C.7 with these boundaries and the same fields. It gives

$$\Delta_r := \mathbb{E}_{k+1,x} S_r - \mathbb{E}_{k,x} S_r \geq 0 \quad (r < n), \qquad \Delta_n = 1,$$

where each expectation uses its own normalized tilt. The repeated top eigenvalues, including ranks zero and n, are covered by the limiting part of that lemma.

By Haar invariance, Z_k and Γ_k are unchanged when the q_j are permuted. First list them increasingly. Then $q_n = Q$ and $q_r - q_{r+1} \leq 0$, so (C.16) gives

$$\Gamma_{k+1} - \Gamma_k = Q + \sum_{r<n} (q_r - q_{r+1}) \Delta_r \leq Q.$$

For the lower bound, list the same rates decreasingly. The comparison lemma still applies, because it allows arbitrary real fields. Now $q_n = q_-$ and $q_r - q_{r+1} \geq 0$, giving $\Gamma_{k+1} - \Gamma_k \geq q_-$. These two arguments can use different couplings: their scalar left-hand sides agree by unitary invariance. This proves (C.15). □

Theorem C.8 applies to the *same* tilted means that occur in the finite-exponential reference. Indeed Lemma C.6 identifies the normalized determinant in (C.8) with Z_k, including its constant. Differentiation of this positive compact integral, together with (C.9), therefore identifies the expectations in Lemma C.5 with Γ_k. Its diagonal coefficients satisfy the sharper bound

$$\alpha_i(y) = \frac{n - 2i - 1}{y} - (\Gamma_{i+1}(y) - \Gamma_i(y)) \geq \frac{n - 2i - 1}{y} - Q, \qquad y > 0. \tag{C.17}$$

This conclusion uses the exact orbital identity and the comparison *after* tilting. A nested coupling before tilting alone would not imply (C.15).

C.7. **Uniform signed smoothing in a moving basis.** We now use the rank-increment bound to control actual positive shifts uniformly up to the support boundary. Write

$$\mathsf{H}_m[g](x) = [g^{(i+j)}(x)]_{i,j=0}^{m-1}, \qquad J_m = \operatorname{diag}((-1)^i)_{i=0}^{m-1}.$$

Here $\mathsf{H}_m[g]$ denotes a derivative Hankel matrix and is unrelated to the scalar heat trace H defined earlier. Every matrix norm in this subsection is the Euclidean operator norm.

Theorem C.9 (Moving-frame signed smoothing)**.** *Let m, n be positive integers such that*

$$n \geq 2m, \qquad \frac{n - 2m + 1}{2} \geq \sqrt{(m-1)(n-m+1)}. \tag{C.18}$$

In particular, $n \geq 8m$ suffices. Let $f = *_{\ell=1}^{n} \operatorname{Exp}(q_\ell)$, where $0 < q_\ell \leq Q$, with repeated rates allowed. For each $x > 0$ there is a lower triangular matrix C_x with positive diagonal such that

$$C_x \mathsf{H}_m[f](x) C_x^T = J_m, \qquad \|C_x \mathsf{H}_m[f](x-u) C_x^T\|_2 \leq e^{Qu} \quad (u \geq 0). \tag{C.19}$$

The shifted matrix in this formula uses the actual zero extension of f, including at $u = x$.

Consequently, if η is a finite real signed measure supported on $[0,\infty)$ and

$$\varepsilon_\eta := \int_{[0,\infty)} e^{Qu}\,\mathrm{d}|\eta|(u) < 1, \tag{C.20}$$

then $p = f * (\delta_0 + \eta)$ satisfies $\mathscr{D}_k(p;x) > 0$ for every $x > 0$ and $1 \le k \le m$. Its zero extension is PF_m, and its ordered minors through order m are strictly positive exactly when $x_i > y_i$ for every diagonal pair.

Proof. For $y > 0$, let $\mathcal{L}_y(P) = P(D)f(y)$. Lemma C.5 gives normalized polynomials $Q_i(y;z)$ of degree i, with positive leading coefficient, such that

$$\mathcal{L}_y(Q_iQ_j) = s_i\delta_{ij}, \qquad s_i = (-1)^i.$$

Their coefficients are smooth in y: exact orthogonalization expresses the monic coefficients in the analytic moments with nonvanishing leading determinants, and each normalization is the positive square root of a nonzero absolute pivot. Let C_y be the lower triangular coefficient matrix of the first m polynomials. Then $C_y\mathsf{H}_m[f](y)C_y^T = J_m$.

We first differentiate this basis. This differentiation is with respect to its evaluation parameter y, not its polynomial variable z. Put $K_y = C_y'C_y^{-1}$. Since a parameter derivative does not increase polynomial degree, K_y is lower triangular. Differentiating orthogonality and using $\partial_y\mathcal{L}_y(P) = \mathcal{L}_y(zP)$ for a frozen polynomial gives

$$0 = \mathcal{L}_y((\partial_yQ_i)Q_j) + \mathcal{L}_y(Q_i(\partial_yQ_j)) + \mathcal{L}_y(zQ_iQ_j).$$

In the signed multiplication matrix of Lemma C.5, the entries are $B_{ii} = \alpha_i$, $B_{i,i+1} = b_i > 0$, and $B_{i+1,i} = -b_i$. For $j < i$ the second term in the preceding display vanishes, so $(K_y)_{ij} = -B_{ij}$. For $j = i$ the two basis-derivative terms are equal. We have therefore obtained the exact closed system

$$C_y' = K_yC_y, \qquad (K_y)_{ii} = -\alpha_i(y)/2, \qquad (K_y)_{i,i-1} = b_{i-1}(y), \tag{C.21}$$

with every other entry zero. The first m rows are closed under this parameter derivative.

Let $\Gamma_k(y) = \mathbb{E}_{k,y}A$, with $\Gamma_0 = 0$. Lemma C.6 identifies these means with the tilted orbital means, so Theorem C.8 gives $\Gamma_{i+1}(y) - \Gamma_i(y) \le Q$. Equation (C.9) and the Jacobi identities consequently yield

$$\begin{aligned}
\alpha_i(y) &= \frac{n-2i-1}{y} - (\Gamma_{i+1}(y) - \Gamma_i(y)) \ge \frac{n-2i-1}{y} - Q,\\
b_{i-1}(y)^2 &= \frac{D_{i+1}(y)D_{i-1}(y)}{D_i(y)^2} = -(\log D_i(y))'' \le \frac{i(n-i)}{y^2} \qquad (1 \le i < n).
\end{aligned} \tag{C.22}$$

Write S_y for the strictly lower part of K_y. A lower bidiagonal shift has norm equal to its largest subdiagonal magnitude. Since $n \ge 2m$,

$$\|S_y\|_2 \le \frac{\sqrt{(m-1)(n-m+1)}}{y}.$$

The diagonal entries of K_y are at most $Q/2 - (n-2m+1)/(2y)$. Thus

$$\begin{aligned}
\lambda_{\max}\left(\frac{K_y + K_y^T}{2}\right) &\le \frac{Q}{2} - \frac{n-2m+1}{2y} + \left\|\frac{S_y + S_y^T}{2}\right\|_2\\
&\le \frac{Q}{2} - \frac{n-2m+1}{2y} + \frac{\sqrt{(m-1)(n-m+1)}}{y} \le \frac{Q}{2}.
\end{aligned} \tag{C.23}$$

The last step is precisely (C.18). For $n \ge 8m$, its first term satisfies $(n-2m+1)/2 \ge 3n/8$, while $\sqrt{(m-1)(n-m+1)} \le \sqrt{mn} \le n/\sqrt{8} < 3n/8$, which proves the asserted sufficient factor budget.

Fix $x > 0$. For $0 \le u < x$ define $R_x(u) = C_xC_{x-u}^{-1}$. The order of its factors matters: differentiation of (C.21) gives

$$R_x(0) = I, \qquad R_x'(u) = R_x(u)K_{x-u}.$$

For an arbitrary row vector v_0, its evolved row $v(u) = v_0 R_x(u)$ therefore satisfies

$$\frac{\mathrm{d}}{\mathrm{d}u}\|v(u)\|_2^2 = v(u)(K_{x-u} + K_{x-u}^T)v(u)^T \le Q\|v(u)\|_2^2.$$

Gronwall yields $\|R_x(u)\|_2 \le e^{Qu/2}$ for the actual nonautonomous right evolution. Since

$$C_x \mathsf{H}_m[f](x-u)C_x^T = R_x(u) J_m R_x(u)^T,$$

the required norm bound follows for $u < x$.

The finite exponential convolution has a C^{n-2} zero extension, and all its derivatives through degree $n-2$ vanish at zero. They are bounded on the real line, being exponential polynomials on the positive axis and zero on the negative axis. As $2m-2 \le n-2$, the actual matrix $\mathsf{H}_m[f](x-u)$ is zero for $u \ge x$. This proves (C.19) at and beyond the support boundary.

The same bounded derivatives justify differentiation of $p = f * (\delta_0 + \eta)$ under integration against the finite variation measure $|\eta|$. Thus

$$C_x \mathsf{H}_m[p](x) C_x^T = J_m + E_x, \qquad \|E_x\|_2 \le \int_{[0,\infty)} e^{Qu}\,\mathrm{d}|\eta|(u) = \varepsilon_\eta < 1.$$

Each leading compression $J_k + (E_x)_{[k]}$ has the same strict error bound. The symmetric path $J_k + t(E_x)_{[k]}$, $0 \le t \le 1$, remains nonsingular because the least singular value of J_k is one. Its inertia, and hence its determinant sign, therefore equals that of J_k. The lower triangularity of C_x makes this leading compression the congruence of $\mathsf{H}_k[p](x)$ by the leading k block of C_x; its determinant multiplier is positive. It follows that $\mathscr{D}_k(p;x) > 0$ for all $k \le m$.

Finally the zero extension of p belongs to $C^{2m-2}(\mathbb{R})$ and vanishes on $(-\infty, 0]$. Also $p \in L^1(\mathbb{R})$ because $f \in L^1(\mathbb{R})$ and η has finite variation. Lemma C.3 therefore proves the global PF assertion and the exact strictness pattern. If $\delta_0 + \eta$ has mass one, the $k = 1$ conclusion also shows that p is a positive probability density. □

C.8. **Closure for Θ and the infinite complement.** We now prove the Θ assertion of Theorem 3.4. The counting inequality selects the first n verified zero occurrences at most T, with $q_\ell \le Q = T^2$. Let f be their exponential convolution and put $p = f * \mathcal{M}_U$. By Lemma C.4, $\eta = \mathcal{M}_U - \delta_0$ satisfies $\int e^{Qu}\,\mathrm{d}|\eta|(u) \le \epsilon_U(Q) < 1$. Theorem C.9 therefore gives the required differential signs for this single selected core at every $x > 0$.

The actual p is C^{n-2} on the line, vanishes on the nonpositive half-line, and has all required strict differential signs. Lemma C.3 makes it PF_m, with strict ordered minors exactly when $x_i > y_i$ for every i. In particular $p > 0$ on the positive half-line. It integrates to one.

Let ν_0 be the infinite convolution of all remaining real $\mathrm{Exp}(\gamma^2)$ factors, including the auxiliary ones above U. To justify its PF property, retain two of these factors, whose convolution g is bounded, uniformly continuous and positive on $(0,\infty)$. Write $v(x) = \mathbb{E}g(x-W)$ for the remaining sum $W \ge 0$. Finite partial sums W_j converge in L^1, so uniform continuity gives $\mathbb{E}g(x-W_j) \to v(x)$ uniformly in x. Each finite convolution is PF_∞, and passing to the limit in each fixed determinant proves this property for v. Moreover $\Pr(W < \varepsilon) > 0$ for every $\varepsilon > 0$: choose a tail of mean less than $\varepsilon/4$, use Markov's inequality on that tail, and use independence and the positive small-value probabilities for the finite initial sum. Thus $v(x) > 0$ exactly for $x > 0$.

Equation (C.4) gives $c_0\Theta = p * v$. Continuous Cauchy–Binet shows that its ordered minor is the integral over increasing z_i of $\det[p(x_i - z_j)]\det[v(z_i - y_j)]$, a nonnegative integrand. Absolute convergence follows by expanding the determinants into products of finite one-dimensional convolution integrals. If $x_i > y_i$ for all i, take

$$y_i < z_i < \min(x_i, y_{i+1})\ (i < k), \qquad y_k < z_k < x_k.$$

On this nonempty open box the first determinant is positive, and the second is lower triangular with positive diagonal. The integral is strictly positive. If some $x_i \le y_i$, the support gives a zero

block with i rows and $k-i+1$ columns, forcing the determinant to vanish. This proves the full support-dependent assertion for Θ.

C.9. **Gaussian variance mixtures preserve the finite order.** The forward implication used for Λ is analytic and preserves the same order.

Theorem C.10 (Forward Gaussian transfer)**.** *Let θ be a Schwartz probability density on $\mathbb{R}$, zero on $(-\infty, 0]$, whose positive random variable V has every negative moment. Define*

$$\lambda(x) = \int_0^\infty \frac{e^{-x^2/(4u)}}{\sqrt{4\pi u}} \theta(u)\,\mathrm{d}u.$$

If $\theta \in \mathrm{PF}_m$, then λ has strictly positive ordered translation minors through order m on $\mathbb{R}$. Also $\mathscr{D}_r(\lambda; x) > 0$ for all $x \in \mathbb{R}$, $1 \le r \le m$.

Proof. We use the Fourier convention $\widehat{f}(t) = \int e^{itx} f(x)\,\mathrm{d}x$ in this proof. Fix $r \le m$, write $\Delta(a) = \prod_{i<j}(a_j - a_i)$ and $Z_r = r! \prod_{j=0}^{r-1} j!$. The derivative-ensemble density

$$P_\theta(a) = \frac{\Delta(a)}{Z_r} \det[(-D)^{j-1}\theta(a_i)]_{i,j=1}^r \tag{C.24}$$

is nonnegative by confluence of the second nodes of a PF minor. It integrates to one: Andreief's identity and integration by parts give the triangular moment matrix

$$\int a^{i-1}(-D)^{j-1}\theta(a)\,\mathrm{d}a = \begin{cases} (i-1)!\mathbb{E}V^{i-j}/(i-j)!, & i \ge j, \\ 0, & i < j. \end{cases}$$

All polynomial boundary terms vanish by the Schwartz hypothesis. The density in (C.24) vanishes if an $a_i < 0$, and its eigenvalues are strictly positive almost surely. Choose an independent Haar unitary U_r and set $S = U_r \operatorname{diag}(a) U_r^*$.

The Harish–Chandra integral followed by Andreief gives, for Hermitian T of eigenvalues t_j,

$$\mathbb{E}e^{i\operatorname{tr}(TS)} = \prod_{j=1}^r \widehat{\theta}(t_j). \tag{C.25}$$

These derivative-ensemble identities are recorded in Förster et al. (2021, Theorem 2.9(1), Corollary 3.4(1), Remark 3.5). To fix the constants, the unitary integral at distinct t_j is

$$\left(\prod_{j=0}^{r-1} j!\right) \frac{\det[e^{ia_i t_j}]}{\Delta(a)\Delta(it)}.$$

After integration, the resulting determinant has entries $(it_j)^{k-1}\widehat{\theta}(t_j)$, which cancel its Vandermonde and the normalizer Z_r. Repeated t_j follow by continuity. For diagonal T, (C.25) proves that the diagonal entries of S are independent with density θ. Unitary invariance consequently gives

$$\mathbb{E}e^{-\operatorname{tr}(AS)} = \prod_j L_\theta(a_j) \qquad (A \succeq 0), \tag{C.26}$$

where a_j are the eigenvalues of A and L_θ is the scalar Laplace transform.

Let G be independent complex Ginibre, with entry density $\pi^{-1}e^{-|z|^2}$, and define $X = \sqrt{S}\,G + G^*\sqrt{S}$. Gaussian conditioning gives

$$\mathbb{E}_G e^{i\operatorname{tr}(TX)} = e^{-\operatorname{tr}(ST^2)}, \qquad \mathbb{E}e^{i\operatorname{tr}(TX)} = \prod_j L_\theta(t_j^2) = \prod_j \widehat{\lambda}(t_j).$$

Here $\operatorname{tr}(TX) = 2\Re\operatorname{tr}(T\sqrt{S}G)$ and $\|T\sqrt{S}\|_F^2 = \operatorname{tr}(ST^2)$, which fix the Gaussian normalization. This law is unitarily invariant. Since $S > 0$, its conditional Gaussian is nondegenerate on the real space of Hermitian matrices and has an everywhere positive density.

All positive and negative moments of V imply that λ is Schwartz: weighted L^1 norms of its Gaussian derivatives are bounded by constants times $\mathbb{E}V^{(p-j)/2}$, and weighted suprema by $\mathbb{E}V^{(p-j-1)/2}$. Thus (C.24) with θ replaced by λ defines a finite signed spectral measure. The same transform calculation identifies its unitary lift with the law of X, by uniqueness of Fourier transforms of finite signed measures.

This identification is pointwise, not just almost everywhere. Negative moments give $\widehat{\lambda}(t) = \mathbb{E}e^{-Vt^2} \le C_N(1+t^2)^{-N}$ for every N. Weyl integration adds only $\Delta(t)^2$, so the matrix Fourier transform is integrable and its inverse p_X is continuous. Some event $\varepsilon I \preceq S \preceq MI$ has positive probability; on it the conditional Gaussian densities have a common positive lower bound on every compact set. Therefore p_X is strictly positive everywhere. Away from repeated eigenvalues, Weyl's formula and continuity imply

$$\frac{\det[(-D)^{j-1}\lambda(a_i)]}{\Delta(a)} = b_r p_X(\operatorname{diag} a) > 0, \qquad b_r > 0. \tag{C.27}$$

The exact positive Weyl constant is immaterial to the signs. For increasing a_i, the derivative evaluation determinant is positive. Fix increasing x_i and apply Lemma C.1 to $y \mapsto \lambda(x_i - y)$: each initial Wronskian is the transpose of such a positive determinant at $x_i - y$. This proves all strict translation minors.

Finally let all a_i tend to x in (C.27). Taylor confluence gives

$$\frac{\mathscr{D}_r(\lambda; x)}{\prod_{j=0}^{r-1} j!} = b_r p_X(xI) > 0.$$

The nonzero limit follows from the positive matrix density at xI. □

For the present laws take $\theta = c_0\Theta$ and $\lambda = c_0\Lambda$; their mixture relation is (2.8). The hypotheses follow from Lemma 2.4. More explicitly, its decompositions $S_1 = \mathrm{Gamma}(N, b_N) + R_N$ for arbitrarily large integers N give weighted derivative bounds for θ: choose $N > j+1$, convolve the jth derivative of the gamma density, and use $R_N \le S_1$ in that independent realization to bound all moments of R_N. The corresponding weighted suprema prove that θ is Schwartz. All negative moments were proved in Appendix B.4. Theorem C.10 now completes the Λ assertion of Theorem 3.4. It also gives the strict pointwise logarithmic curvature for Λ from $\Theta \in \mathrm{PF}_2$, using only the low-height inputs.

C.10. **Exact-budget scalar specialization and its scope.** Use the full published verified height

$$U = U_{\mathrm{PT}} = 3\,000\,175\,332\,800,$$

from Platt and Trudgian (2021, Theorem 1). The round value $3 \cdot 10^{12}$, used by the direct certificates, lies slightly below that theorem's endpoint. Take the exact profile

$$k_* = 1\,810\,540\,931\,075, \qquad n_* = 12\,363\,146\,804\,211,$$
$$T_{\mathrm{sel}} = \frac{6\,000\,347\,566\,081}{2}, \qquad Q_* = T_{\mathrm{sel}}^2. \tag{C.28}$$

Directed evaluation gives

$$C(T_{\mathrm{sel}}) - n_* > 0.889309597244,$$
$$\epsilon_U(Q_*) < 0.999999764856 < 1, \qquad Q_* < U^2 - \tfrac{1}{4}, \tag{C.29}$$
$$(n_* - 2k_* + 1)^2 - 4(k_* - 1)(n_* - k_* + 1) = 3\,780\,596\,175\,292 > 0.$$

Also $n_* \ge 2k_*$. These are all the scalar hypotheses of Theorem 3.4. The weighted error leaves a margin of about $2.4 \cdot 10^{-7}$; the replays below check it at higher precision. The uniform moving-frame estimate applies on the entire positive axis.

We give the exact capacity calculation for this sufficient criterion. For fixed $n \geq 2m$, squaring the nonnegative sides of (C.18) shows that its second condition is

$$8m^2 - (8n+12)m + n^2 + 6n + 5 \geq 0.$$

In the range $n \geq 2m$, this is equivalent to

$$m \leq \frac{2n + 3 - \sqrt{2n^2 - 1}}{4}. \tag{C.30}$$

Equivalently, for positive integer m, the least permissible factor count is

$$n_{\min}(m) = \max\left\{2m, \left\lceil 4m - 3 + 2\sqrt{2m^2 - 3m + 1} \right\rceil\right\}. \tag{C.31}$$

This is an exact integer calculation: the ceiling of the square root of $4(2m^2 - 3m + 1)$ is determined by an integer square root and one comparison. The required count increases with m.

Set $R = \mathcal{R}_U$, $r = R/4$ and $H = \log 2 - r > 0$, and define

$$c_* = \frac{r + \sqrt{r^2 + 8rH}}{2H}, \qquad A_* = U^2(1 - c_*) - \tfrac{1}{4}. \tag{C.32}$$

For $0 \leq a < U^2 - 1/4$, the function $\epsilon_U(a)$ is strictly increasing. Writing $c = c_U(a)$ shows that $\epsilon_U(a) < 1$ is equivalent to $Hc^2 - rc - 2r > 0$, and hence exactly to $a < A_*$. For the present height, $0 < A_* < U^2 - 1/4$. Since

$$C'(T) = \frac{1}{2\pi}\log\frac{T}{2\pi} - \frac{1}{8(T+2)} > 0 \qquad (T \geq 100),$$

every admissible selection height satisfies $C(T) < C(\sqrt{A_*})$. Outward bounds give

$$12\,363\,146\,804\,212.45185057 < C(\sqrt{A_*}) < 12\,363\,146\,804\,212.45185058. \tag{C.33}$$

Thus the gate $C(T) \geq n$ permits no integer factor count exceeding $12\,363\,146\,804\,212$. Equation (C.30) then gives at most k_*, and the accepted profile above attains that order. More directly,

$$\begin{aligned} n_{\min}(k_* + 1) &= 12\,363\,146\,804\,218, \\ C(\sqrt{A_*^{\mathrm{up}}}) - n_{\min}(k_* + 1) &< -5.54814942, \end{aligned} \tag{C.34}$$

where $A_*^{\mathrm{up}} \geq A_*$ is computed using an outward lower bound for R and outward upper endpoints thereafter. Decreasing R permits a larger weight, so this exclusion uses the optimistic endpoint. Larger orders require still more factors and are excluded as well. This proves exact maximality only for the displayed moving-frame budget, the fixed height U_{PT}, $Q = T^2$, the count gate $C(T) \geq n$, and the stated variation majorant. Allowing smaller verified heights cannot improve this maximum: $\mathcal{R}_U$ is strictly decreasing for $U \geq 100$, whereas $c_U(a)$ increases for fixed admissible a. Since $1 + (c+2)/c^2$ decreases for $c > 0$, $\epsilon_U(a)$ decreases with U. Thus every profile feasible at a smaller height remains feasible at U_{PT}. Even exploiting the integrality of $N(T)$ would not change the integer order here: it permits at most one additional factor, still fewer than $n_{\min}(k_* + 1)$.

For comparison, the simpler budget $n \geq 8m$ at the round height $U = 3 \cdot 10^{12}$ gives the maximum $1\,545\,299\,548\,368$, and the exact budget at that height gives $1\,810\,431\,035\,011$; the full published endpoint accounts for the remaining increase to (C.28). The round profile $m = 1\,500\,000\,000\,000$, $n = 8m$, $T = 2\,950\,000\,000\,000$, $U = 3 \cdot 10^{12}$ has the much wider error margin $\epsilon_U(T^2) < 6.886657 \cdot 10^{-10}$.

All evaluations are scalar and independent of the order; they check the hypotheses of Theorem 3.4. The accepted profile was checked at 100 and 160 decimal digits and replayed by an independent implementation at 180 digits; the capacity calculation was also repeated at 220 digits. All scalar bounds use outward interval arithmetic (Johansson, 2017); the verified height is the published input (Platt and Trudgian, 2021).

The value k_* is the largest order this criterion certifies; a different analytic estimate, for example a sharper matrix norm bound, could improve it with the same zero information. The asymptotic

scales are $\mathcal{R}_U = O(\log U/U)$ and $C(T) \asymp T \log T$. For fixed $0 < \beta < 1$, selection at $T = \beta U$ gives $\epsilon_U(T^2) = O_\beta(\log U/U)$ and permits $m \asymp U \log U$; the exact factor budget changes its constant.

C.11. **Necessary PF orders and the first nonreal poles.** The sufficient bounds above have a converse constraint: a high PF order forces a long prefix of real poles. We prove the constraint for each actual reciprocal density, with arbitrary multiplicities and coincident decay rates.

By (2.9), their transforms are

$$G_\Theta(s) := \frac{1}{c_0 h(s)} = \frac{1}{\xi(\frac{1}{2} + \sqrt{s})}, \qquad G_\Lambda(s) := \frac{1}{\Xi(s)}.$$

These identities hold on their respective initial convergence domains, including the imaginary axis. Corollary 2.5 gives the actual continuous densities; their positive probability normalization does not change the PF property. We continue to use the zero extension for Θ and the non-strict PF convention.

Write $\rho = \frac{1}{2} + \delta_\rho + i\gamma_\rho$ for a zeta zero with $\gamma_\rho > 0$, and let m_ρ denote its multiplicity. Thus $|\delta_\rho| < 1/2$. The rates of the poles in the left half-plane are

$$\lambda_\Theta(\rho) = (\gamma_\rho - i\delta_\rho)^2, \qquad \lambda_\Lambda(\rho) = \gamma_\rho - i\delta_\rho. \tag{C.35}$$

The corresponding pole is $s = -\lambda_f(\rho)$. Each pole has multiplicity m_ρ. We justify these facts, including the absence of a factor of two in the first multiplicity, in the contour argument below. Let $N_0(t-)$ and $N_0(t)$ count critical-line zero occurrences of positive ordinate strictly below, and at most, t, respectively.

Theorem C.11 (First nonreal pole, with multiplicities)**.** *Suppose an off-critical zero exists. Set*

$$\Gamma = \min_{\delta_\rho \neq 0} \gamma_\rho, \qquad A_\Theta = \min_{\delta_\rho \neq 0} (\gamma_\rho^2 - \delta_\rho^2), \qquad \tau = \sqrt{A_\Theta}.$$

These minima are attained, and $\tau < \Gamma$. For Θ, let r_Θ be the critical-line multiplicity at ordinate τ (zero if absent), and let μ_Θ be the largest multiplicity among off-critical zeros whose squared rate has real part A_Θ. For Λ, let r_Λ be the critical-line multiplicity at ordinate Γ (zero if absent), and let μ_Λ be the largest off-critical multiplicity at that ordinate. Put

$$d_\Theta = N_0(\tau-) + \max(0, r_\Theta - \mu_\Theta + 1), \tag{C.36}$$

$$d_\Lambda = N_0(\Gamma-) + \max(0, r_\Lambda - \mu_\Lambda + 1). \tag{C.37}$$

Then $\Theta \notin \mathrm{PF}_{d_\Theta+1}$ and $\Lambda \notin \mathrm{PF}_{d_\Lambda+1}$. In particular,

$$\Theta \in \mathrm{PF}_k \implies k \le d_\Theta \le N_0(\Gamma-), \qquad \Lambda \in \mathrm{PF}_k \implies k \le d_\Lambda \le N_0(\Gamma).$$

The last count for Λ includes a possible critical-line zero tied at the first off-critical ordinate. When that tied zero is absent, $d_\Lambda = N_0(\Gamma-)$.

This is a necessary condition. If RH holds, there is no first nonreal pole and the conditional upper bounds do not apply.

Corollary C.12 (PF order forces zero verification)**.** *Let $N(Y)$ count all nontrivial zeta zero occurrences with $0 < \gamma \le Y$, including multiplicity. For each $f \in \{\Theta, \Lambda\}$,*

$$f \in \mathrm{PF}_k \quad \text{and} \quad N(Y) \le k+1 \quad \Longrightarrow \quad \beta = \tfrac{1}{2} \text{ for every zero with } 0 < \gamma \le Y. \tag{C.38}$$

It suffices to have a rigorous real upper bound $N_{\mathrm{up}}(Y) < k + 2$.

Proof from Theorem C.11. If there were an off-critical zero through Y, its first ordinate $\Gamma \le Y$ would contain a reflected pair $\frac{1}{2} \pm \delta + i\Gamma$, of equal positive multiplicity. There are therefore at least two noncritical occurrences at that height. For either density, the theorem implies

$$k \le d_f \le N(\Gamma) - 2 \le N(Y) - 2 \le k - 1,$$

a contradiction. The strict upper-bound formulation follows because $N(Y)$ is an integer. This also covers Y equal to a zero ordinate. □

C.11.1. *Finite translates under weak total positivity.* We give the finite-order ingredient explicitly, including rank-deficient collocation matrices. A strict sign change means an alternation along an increasing finite sequence after zero values have been deleted.

Lemma C.13. *If $f \in \mathrm{PF}_k$ and $n \le k$, every real linear combination of n distinct translates of f has at most $n-1$ strict sign changes.*

Proof. For increasing real a_j, the initial Wronskians of $e^{a_j x}$ are positive exponentials times positive Vandermonde products. Thus Lemma C.1 makes every ordered evaluation determinant of these exponentials strictly positive.

Here is the finite matrix consequence needed below. Let A be a totally nonnegative $(n+1)\times n$ matrix. Suppose a vector y in its column space has $n+1$ nonzero alternating entries. The rank r of A is positive and at most n. Choose r independent columns A_C, with the original column order. All their maximal minors are nonnegative and at least one is positive. For $\varepsilon > 0$ set

$$(B_\varepsilon)_{ij} = e^{-(i-j)^2/\varepsilon}, \qquad 1 \le i, j \le n+1.$$

This matrix tends to the identity. It is strictly totally positive: positive row and column factors reduce its minors to those of $e^{2ij/\varepsilon}$, and the exponential evaluation fact applies. Cauchy–Binet now makes every r-row minor of $B_\varepsilon A_C$ strictly positive, since the same positive maximal minor of A_C contributes positively to each sum.

For sufficiently small ε, $B_\varepsilon y$ still has all $n+1$ alternating signs and belongs to the column space of $B_\varepsilon A_C$. Restrict to the first $r+1$ rows. The alternating maximal cofactors of this $(r+1)\times r$ matrix form a left null vector; all their magnitudes are positive. Its dot product with the first $r+1$ entries of $B_\varepsilon y$ has summands of one nonzero sign, so cannot vanish. This contradiction proves the matrix consequence. Rank zero is immediate.

Finally, n strict sign changes of $g(x) = \sum_{j=1}^{n} c_j f(x - y_j)$ would give $n+1$ increasing evaluation points with alternating nonzero values. Order the distinct y_j increasingly. The resulting $(n+1)\times n$ matrix $[f(x_i - y_j)]$ is totally nonnegative, since every possible minor has size at most $n \le k$. The preceding consequence excludes its column-space vector of alternating values. This proves the lemma for the non-strict PF convention, without assuming full rank or strictly positive minors. □

This is a special case of the classical finite matrix variation-diminishing theorem (Karlin, 1968). The proof above supplies exactly the version needed for finite translates.

C.11.2. *The pole obstruction with equal decay rates.*

Theorem C.14. *Suppose a real function has, as $x \to +\infty$, an expansion*

$$f(x) = \sum_{\lambda \in \mathcal{S}} e^{-\lambda x} P_\lambda(x) + O(e^{-Bx}), \tag{C.39}$$

where $\mathcal{S}$ is a finite set of distinct rates, closed under conjugation, each P_λ has exact degree $m_\lambda - 1$ with $m_\lambda \ge 1$, and conjugate rates have conjugate polynomials. Assume that $\mathcal{S}$ contains a nonreal rate, let A be the least real part of such a rate, and suppose every rate in $\mathcal{S}$ has real part at most A, with $B > A$. Thus all nonreal rates in the sum have real part A. Let

$$M_< = \sum_{\substack{q \in \mathcal{S} \cap \mathbb{R} \\ q < A}} m_q, \qquad r = \begin{cases} m_A, & A \in \mathcal{S}, \\ 0, & A \notin \mathcal{S}, \end{cases} \qquad \mu = \max_{\substack{\lambda \in \mathcal{S} \setminus \mathbb{R} \\ \operatorname{Re}\lambda = A}} m_\lambda.$$

Then, with $d = M_< + \max(0, r - \mu + 1)$, one has $f \notin \mathrm{PF}_{d+1}$.

Proof. Choose $h > 0$ small enough that $0 < h|\operatorname{Im}\lambda| < 2\pi$ for every nonreal rate in the group at A. Set $E_h f(x) = f(x+h)$, $\ell = \max(0, r-\mu+1)$, and

$$P(z) = \prod_{\substack{q\in\mathcal{S}\cap\mathbb{R}\\ q<A}} (z - e^{-qh})^{m_q}\,(z-e^{-Ah})^{\ell}. \tag{C.40}$$

It has degree d, so $P(E_h)f$ is a combination of $d+1$ translates. For a polynomial p,

$$(E_h - e^{-qh})[e^{-qx}p(x)] = e^{-q(x+h)}(p(x+h)-p(x)).$$

The factor lowers polynomial degree by one and kills constants. It follows that all real-rate terms below A are killed. The tied real term, when present, has remaining degree at most $\mu-2$, or is killed: if $r \ge \mu$, its degree drops from $r-1$ by $r-\mu+1$; if $r < \mu$, its original degree already is at most $\mu-2$. This includes $r=0$ and $\mu=1$.

For a nonreal rate λ at A, the operator multiplies the leading coefficient by $P(e^{-\lambda h}) \neq 0$. The roots associated with $q < A$ have different moduli, and the possible root at A has a different phase by the choice of h. Hence each nonreal term of maximal degree $\mu-1$ survives, with nonzero leading coefficient. Distinct such rates have distinct nonzero frequencies. Fixed finite translates preserve the exponential gap in the remainder, and therefore

$$P(E_h)f(x) = e^{-Ax}x^{\mu-1}(T(x)+o(1)), \tag{C.41}$$

where $T(x) = \sum_\nu a_\nu e^{i\nu x}$ is a nonzero real trigonometric polynomial, with distinct nonzero frequencies and $a_{-\nu} = \overline{a_\nu}$.

Direct integration gives its mean zero and positive mean square:

$$\lim_{L\to\infty}\frac{1}{L}\int_0^L T(x)\,\mathrm{d}x = 0, \qquad \lim_{L\to\infty}\frac{1}{L}\int_0^L T(x)^2\,\mathrm{d}x = \sum_\nu |a_\nu|^2 =: s^2 > 0.$$

Let $K \ge \max(1, \sup|T|)$ and choose $0 < \epsilon < s^2/K$. If $T \ge -\epsilon$ eventually, then $(T+\epsilon)(T-K) \le 0$ eventually. Taking long-time means gives $s^2 \le K\epsilon$, a contradiction. Apply the same argument to $-T$. Thus values above ϵ and below $-\epsilon$ both occur arbitrarily far out. They can be selected successively in an alternating increasing sequence. The $o(1)$ term in (C.41) is eventually smaller than $\epsilon/2$, so $P(E_h)f$ has arbitrarily many strict sign changes. Lemma C.13 contradicts this if $f \in \mathrm{PF}_{d+1}$. No rational independence of frequencies is needed. □

C.11.3. *Finite contour shifts for the actual transforms.* The rates in (C.35) are discrete, have positive real parts, and have only finitely many occurrences with real part below any fixed bound. For Θ, use $\operatorname{Re}\lambda_\Theta = \gamma^2 - \delta^2$ and $|\delta| < 1/2$; for Λ, use $\operatorname{Re}\lambda_\Lambda = \gamma$. Positive real parts follow also from the known first-zero isolation. All repeated occurrences at a rate are collected as its multiplicity.

Evenness of $\xi(\frac{1}{2}+z)$ makes $\xi(\frac{1}{2}+\sqrt{s})$ entire and single-valued. Near a nontrivial zero, the map $z \mapsto z^2$ has nonzero derivative. Consequently its zero multiplicity is unchanged by this map. The two zeros corresponding to opposite values of z give the same squared zero; choosing positive zeta ordinate selects one of them, without doubling its multiplicity. There is no branch-cut contribution or pole at zero. For Λ, the map defining Ξ is linear; evenness supplies the negative-half-plane pole $s = -\lambda_\Lambda$ with the same multiplicity. Reflection of zeta zeros in the critical line gives the conjugate rate, with equal multiplicity.

Fix $B > 0$ so that no pole lies on $\operatorname{Re} s = -B$. We require uniform large-$|t|$ estimates for $s = \sigma + it$, $-B \le \sigma \le 0$. For Θ, choose the principal square root solely for this estimate. Uniformly on the fixed strip,

$$\sqrt{\sigma+it} = \sqrt{|t|}\,e^{i\operatorname{sgn}(t)\pi/4} + O_B(|t|^{-1/2}).$$

Thus $w = \frac{1}{2} + \sqrt{s}$ has real part tending to infinity, and $w/2$ stays in a closed sector within $|\arg z| < \pi$. The Dirichlet series gives

$$|\zeta(w) - 1| \le 2^{-\operatorname{Re} w} + \frac{2^{1-\operatorname{Re} w}}{\operatorname{Re} w - 1} = o(1).$$

Apply sector Stirling (NIST, 2026, Section 5.11) to $\xi(w) = \frac{1}{2} w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)$. It gives

$$\log |G_\Theta(\sigma + it)| = -\frac{\sqrt{|t|}}{4\sqrt{2}} \log |t| + O_B(\sqrt{|t|}). \tag{C.42}$$

Indeed $\operatorname{Re}(w/2) = \sqrt{|t|}/(2\sqrt{2}) + O(1)$ and $\log |w/2| = \frac{1}{2} \log |t| - \log 2 + o(1)$; all other real parts in Stirling's formula contribute $O_B(\sqrt{|t|})$.

For Λ, the functional equation gives

$$\Xi(\sigma + it) = \xi(\tfrac{1}{2} + |t| \mp i\sigma),$$

using the minus sign when $t > 0$ and the plus sign when $t < 0$. The same Dirichlet-series estimate and Stirling formula, now with bounded imaginary part, give uniformly on the strip

$$\log |G_\Lambda(\sigma + it)| = -\tfrac{1}{2}|t| \log |t| + \tfrac{1}{2}(1 + \log(2\pi))|t| + O_B(\log |t|). \tag{C.43}$$

In particular, for either transform and every fixed integer $j \ge 0$,

$$\int_{\mathbb{R}} |(-B + it)^j G_f(-B + it)| \, \mathrm{d}t < \infty.$$

The same polynomial multiples tend to zero on horizontal edges uniformly in $\sigma \in [-B, 0]$. All constants here may depend on the fixed cutoff and derivative order.

Lemma C.15 (Actual finite residue expansion)**.** *For either $f = \Theta$ or $f = \Lambda$, every such B gives*

$$f^{(j)}(x) = \sum_{\operatorname{Re} \lambda < B} \left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^j [e^{-\lambda x} P_\lambda(x)] + R_{B,j}(x), \qquad x \ge 0, \tag{C.44}$$

where P_λ has exact degree $m_\lambda - 1$ and $|R_{B,j}(x)| \le C_{B,j} e^{-Bx}$ for a finite constant $C_{B,j}$. The sum is finite and conjugate rates have conjugate polynomials.

Proof. Fourier inversion on the imaginary axis is absolutely convergent by (C.42)–(C.43). It recovers the actual continuous density, since its transform there is already known. The same bounds justify every fixed number of derivatives. Shift this upward Bromwich contour to the upward line $\operatorname{Re} s = -B$. The bounding rectangle has positive orientation when its right edge is traversed upward; the two horizontal integrals vanish. If

$$G_f(s) = \sum_{a=1}^{m_\lambda} \frac{c_{\lambda,-a}}{(s + \lambda)^a} + \text{holomorphic}, \qquad c_{\lambda,-m_\lambda} \ne 0,$$

the residue of $e^{sx} G_f(s)$ is

$$e^{-\lambda x} P_\lambda(x), \qquad P_\lambda(x) = \sum_{a=1}^{m_\lambda} c_{\lambda,-a} \frac{x^{a-1}}{(a-1)!}.$$

This gives its exact degree and nonzero leading coefficient. Multiplying the integrand by s^j differentiates each finite residue term. The remaining vertical integral has absolute value at most

$$\frac{e^{-Bx}}{2\pi} \int_{\mathbb{R}} |(-B + it)^j G_f(-B + it)| \, \mathrm{d}t.$$

It supplies the asserted uniform remainder. Only finitely many residues are crossed. □

Completion of Theorem C.11. For either density choose B strictly above the least nonreal decay rate and strictly below the next larger rate real part. Local finiteness makes this possible. Lemma C.15 then supplies precisely the hypotheses of Theorem C.14. For Θ, the real rates are γ^2 with critical-line multiplicities, giving (C.36). For Λ, they are the critical-line ordinates themselves, giving (C.37).

An off-critical zero at the first ordinate Γ has $\delta \neq 0$, so $A_\Theta \leq \Gamma^2 - \delta^2 < \Gamma^2$. This proves $\tau < \Gamma$, even if the squared rate attaining A_Θ belongs to a different, slightly higher ordinate. Also $\max(0, r - \mu + 1) \leq r$ since $\mu \geq 1$. Consequently $d_\Theta \leq N_0(\tau) \leq N_0(\Gamma-)$ and $d_\Lambda \leq N_0(\Gamma)$. The theorem follows. □

C.11.4. *What a quadratic order would imply.* The counting inequality (2.1), derived from Bellotti and Wong (2025, Theorem 1.1), gives

$$N(Y) \leq N_{\mathrm{up}}(Y) := \frac{Y}{2\pi}\left(\log\frac{Y}{2\pi} - 1\right) + \frac{1}{8}\log(Y+2) + 4. \tag{C.45}$$

Directed scalar evaluation gives

$$N_{\mathrm{up}}(10^{23}) < 8 \cdot 10^{23}, \qquad N_{\mathrm{up}}(10^{24}) < 8.4 \cdot 10^{24}. \tag{C.46}$$

Hence Corollary C.12 proves the following implications for each density separately:

$$f \in \mathrm{PF}_{10^{24}} \implies \text{RH through height } 10^{23},$$
$$f \in \mathrm{PF}_{9\cdot 10^{24}} \implies \text{RH through height } 10^{24}.$$

The checks in (C.46) are repeated at 100 and 180 decimal digits with outward Arb arithmetic.

More generally $N_{\mathrm{up}}(Y) \sim Y\log Y/(2\pi)$. The inverse counting scale is $Y \sim 2\pi k/\log k$: for any fixed $0 < c < 2\pi$, taking $Y = ck/\log k$ satisfies the sufficient count inequality for all sufficiently large k. Thus $k \asymp T^2$ forces reality of zeros through heights of order $T^2/\log T$, far beyond T. A new analytic proof of that PF claim would simultaneously prove the stronger zero information.

The present sufficient construction, by comparison, uses the factor budget (C.18), whose minimal admissible factor count satisfies $n \sim (4 + 2\sqrt{2})k$, and $n \leq N(T)$, giving order of size $T\log T$. The pole obstruction explains why replacing this count scale by the squared-rate scale T^2 cannot be justified solely as a numerical optimization of the existing certificate.

C.12. **What a finite spatial sweep proves.** The variable in a spatial sweep of the reciprocal density is distinct from the ordinate of a zeta zero. We write L for a spatial cutoff and Y for a zero-height cutoff. There is a valid local-to-global passage on a fixed spatial interval, but it preserves the certified order and the domain of differences.

Lemma C.16 (Finite-window differential criterion)**.** *Let $m \geq 1$ and let $I = (a, b)$ be an interval. Suppose $f \in C^{2m-2}(I)$ and $\mathscr{D}_j(f;t) > 0$ for $t \in I$ and $1 \leq j \leq m$. Then, for increasing tuples of size $k \leq m$,*

$$\det[f(x_i - y_j)]_{i,j=1}^k > 0 \quad \text{if} \quad a < x_1 - y_k \quad \text{and} \quad x_k - y_1 < b. \tag{C.47}$$

There is also a support-boundary version. Let $L > 0$, suppose $f \in C^{2m-2}((-\infty, L))$ vanishes on $(-\infty, 0]$, and assume $\mathscr{D}_j(f;t) > 0$ for $0 < t < L$, $j \leq m$. Then

$$\det[f(x_i - y_j)]_{i,j=1}^k \geq 0 \qquad (x_k - y_1 < L), \tag{C.48}$$

and this determinant is strictly positive exactly when $x_i > y_i$ for every i.

Proof. Fix $k \leq m$. The functions $f, -f', \ldots, (-1)^{r-1}f^{(r-1)}$, $r \leq k$, have positive initial Wronskians on I, precisely the signs $\mathscr{D}_j$. Lemma C.1 therefore gives their positive evaluation determinants. Put $g_j(t) = f(t + x_{k+1-j})$. On the common interval $(a - x_1, b - x_k)$ the initial Wronskians of the g_j are positive: reversing the evaluation arguments and removing the derivative column signs cancels two factors $(-1)^{r(r-1)/2}$. Under the conditions in (C.47), all nodes $t_i = -y_{k+1-i}$ lie in this common

interval. A second use of Lemma C.1 proves positivity. The resulting evaluation matrix is the transpose of the desired matrix with both orders reversed, so its determinant is unchanged.

For the support version use the same functions on $t < L - x_k$, with support thresholds $a_j = -x_{k+1-j}$. Their initial Wronskians are positive for $a_r < t < L - x_k$: the arguments used are positive and less than L. The support-boundary evaluation argument in the proof of Lemma C.3 is local on the interval containing its nodes. Explicitly, if some $t_i \le a_i$, support gives the zero block forcing the determinant to vanish. Otherwise divide by g_1, subtract consecutive rows, and integrate the derived evaluation determinant over $[t_1, t_2] \times \cdots \times [t_{k-1}, t_k]$. Every integration variable stays below $t_k < L - x_k$. The induction for the derived functions therefore remains inside this same domain, and is strictly positive on $\max(t_i, a_{i+1}) < v_i < t_{i+1}$. No argument of f at or above L is used. The strictness condition $t_i > a_i$ is exactly $x_i > y_i$ after reversing the orders. □

Consequently, charts covering $(0, L)$ can be combined when every chart certifies all the strict differential signs through the *same* order m for the *same actual function* f. Their union supplies the hypotheses of Lemma C.16; its conclusion includes minors whose differences cross several charts. The bases used to bound those same signs may differ from chart to chart. Exact coverage, including chart endpoints, is necessary.

A practical sweep at a fixed order m can therefore combine an analytic endpoint bound on $(0, a]$ with rigorous bounds on closed intervals covering $[a, L]$. Every interval must prove all signs through m for the actual kernel, with certified joins. Positive scalar normalizations, or lower triangular congruences with positive squared determinant multipliers, can be chosen separately in each chart: they preserve the signs of the same actual leading determinants. For Λ, the first part of the lemma likewise applies to a cover of any interval of differences. The Gaussian-transfer theorem used in this paper assumes global PF order of Θ; a finite Θ window does not meet that premise, since its variance integration ranges over all $u > 0$. A finite-window transfer would require additional control of that unbounded integration domain.

For example, the finite-translate argument used in the pole obstruction can then be applied only where every matrix entry is in the certified domain. For the support version, a combination $g(x) = \sum_{j=0}^{d} c_j f(x + jh)$, $d < m$, has at most d sign changes on $x < L - dh$. Indeed every collocation matrix there has maximum cross difference less than L, and all its minors are covered by (C.48); the same finite-matrix argument applies. For the interval version with shifts $y_1 < \cdots < y_n$, the corresponding domain is $a + y_n < x < b + y_1$. These statements provide no control of the arbitrarily late values used by the pole obstruction.

Lemma C.17 (Finite-window positivity can precede a pole obstruction)**.** *For every integer $m \ge 2$ and every finite $L > 0$, there is a positive probability density f on $(0, \infty)$ whose zero extension is C^{2m-2}, which satisfies all conclusions of (C.48) through order m, but is not PF_m globally. Its Laplace transform has real poles, followed by a nonreal conjugate pair, with arbitrarily small nonzero imaginary parts, and then further real poles.*

Proof. Choose distinct positive rates

$$0 < q_1 < \cdots < q_{m-1} < A < r_1 < \cdots < r_{m-1}.$$

For $b > 0$ define the normalized transform

$$F_b(s) = \prod_{j=1}^{m-1} \frac{q_j}{s + q_j} \, \frac{A^2 + b^2}{(s + A)^2 + b^2} \prod_{j=1}^{m-1} \frac{r_j}{s + r_j}. \tag{C.49}$$

Its inverse is positive. To see this before convolving the remaining real exponentials, put $a = A - q_1 > 0$. The inverse of the first real factor and the conjugate pair is

$$\frac{q_1(A^2 + b^2)}{a^2 + b^2} e^{-q_1 x} \left[1 - e^{-ax} \left(\cos bx + \frac{a}{b} \sin bx \right) \right] \quad (x > 0).$$

The bracket starts at zero and has derivative $(a^2+b^2)e^{-ax}\sin(bx)/b$. Its positive-time local minima occur at $2\pi\ell/b$, where its value is $1-e^{-2\pi\ell a/b}>0$. It is therefore positive for every $x>0$. Convolution with the remaining positive exponential densities preserves positivity, and $F_b(0)=1$ gives unit mass.

There are $N=2m$ factors, counting the conjugate pair twice. The positive branch f_b solves the corresponding monic constant- coefficient ODE of order N, whose coefficients depend polynomially on b^2. Its initial derivatives vanish through degree $N-2$, and

$$f_b^{(N-1)}(0)=K_b:=(A^2+b^2)\prod_{j=1}^{m-1}q_jr_j>0.$$

These facts follow also by expanding (C.49) at infinity. The companion-matrix solution is analytic jointly in x and b^2. Thus the actual zero extension is $C^{N-2}=C^{2m-2}$.

For $1\le j\le m$, define for $x>0$

$$E_j(x,b)=x^{-j(N-j)}\mathscr{D}_j(f_b;x).$$

The power series of f_b at zero shows that this expression has a continuous, indeed analytic, extension in x and b^2 at $x=0$. To check the cancellation explicitly, factor $x^{N-1-i-\ell}$ from the (i,ℓ) derivative entry; the total power is $j(N-j)$, and the remaining entries are analytic power series. Their constant determinant is the signed determinant of the derivatives of $K_bx^{N-1}/(N-1)!$. Consequently

$$E_j(0,b)=\frac{K_b^j}{((N-1)!)^j}\prod_{a=0}^{j-1}a!(N-1)_{\underline{a}}>0.$$

At $b=0$, all N rates are real, with the rate A repeated twice. The positive representation (C.8) therefore gives $\mathscr{D}_j(f_0;x)>0$ for every $x>0$ and $j\le m$. Each $E_j(x,0)$ is positive on the compact interval $[0,L]$, including at zero by the displayed formula. Joint continuity and the finite number of indices imply $E_j(x,b)>0$ on this entire interval for all sufficiently small $b>0$. Lemma C.16 now gives the asserted finite-window minor signs and their support-dependent strictness.

Nevertheless, the first nonreal rates of F_b are $A\pm ib$. There are exactly $m-1$ preceding real poles, no tied real pole, and all these poles are simple. Theorem C.14 therefore gives $f_b\notin\mathrm{PF}_m$. The extra real rates above A supply the required boundary regularity without changing that obstruction. □

This example shows why a finite spatial certificate alone cannot replace the global PF premise in the zero-height implication. The conjugate rates $A\pm ib$ can be written $(\gamma_b\pm i\delta_b)^2$, with $\gamma_b\to\sqrt{A}$ and $\delta_b\to0$ as $b\downarrow0$. Thus the obstruction can lie at a bounded zero-height scale while its failure of PF escapes every prescribed finite spatial window. A finite-window zero-exclusion argument for the actual densities would need additional quantitative estimates placing enough alternating values of the annihilated tail inside the certified window. The qualitative residue expansion supplies no such uniform window: nonreal frequencies can approach zero, and the constants and gaps in the remainder also require control.

C.13. **Fixed-split bounds.** The following local and bulk estimates give a fixed-split criterion, which we keep as an independent bound.

Lemma C.18 (Local signed smoothing)**.** *Let $n\ge8m$, $0<q_\ell\le Q$, and $f=*_{\ell=1}^n\mathrm{Exp}(q_\ell)$. Let η be a finite real signed measure supported on $[0,\infty)$, put $p=f*(\delta_0+\eta)$, and set $\varepsilon=\|\eta\|_{\mathrm{TV}}$. Define*

$$r_m=\sqrt{\frac{m-1}{n-m+1}},\qquad c_m=\frac{r_m}{1-r_m}.$$

Then $\mathscr{D}_k(p;x)>0$ for every $k\le m$ and every $x>0$ satisfying

$$e^{(1+2c_m)Qx}-1+e^{2c_mQx}\varepsilon<1. \tag{C.50}$$

In particular, on $0 < x \le 1/(16Q)$ *it suffices that* $e^{7/48} - 1 + e^{1/12}\varepsilon < 1$.

Proof. Put $N = n-1$, $C = \prod q_\ell$, and $h(x) = x_+^N/N!$. At a fixed $x > 0$, the normalized polynomial functional $\mathcal{L}(z^j) = x^j h^{(j)}(x)/h(x) = N_{\underline{j}}$ has signed orthonormal polynomials

$$Q_i(z) = \frac{L_i^{(-n)}(-z)}{\sqrt{\binom{N}{i}}}, \qquad \mathcal{L}(Q_iQ_j) = (-1)^i\delta_{ij}, \quad 0 \le i,j < m. \tag{C.51}$$

These are finite polynomial identities. For completeness, the Laguerre generating function and $\mathcal{L}(e^{tz}) = (1+t)^N$ give

$$\mathcal{L}\Big[(1-u)^N e^{zu/(1-u)}(1-v)^N e^{zv/(1-v)}\Big] = (1-uv)^N;$$

coefficient comparison proves (C.51). All polynomial operator coefficients below are frozen at the chosen x; $z = xD$ is not an Euler operator acting on those coefficients.

Let L represent polynomial differentiation, $Q'(z) = LQ(z)$. Its strictly lower triangular entries are

$$L_{ij} = \sqrt{\frac{i!\,N_{\underline{j}}}{j!\,N_{\underline{i}}}} \le r_m^{i-j} \quad (j < i).$$

Both row and column sums are at most c_m, so $\|L\|_2 \le c_m$. Hence, for $v \ge 0$,

$$Q(z-v) = T(v)Q(z), \qquad T(v) = e^{-vL}, \qquad \|T(v)\|_2 \le e^{c_m v}. \tag{C.52}$$

The polynomial Euler operator $zQ'(z) = AQ(z)$ is instead bidiagonal: $A_{ii} = i$, $A_{i,i-1} = \sqrt{i(n-i)}$. Thus

$$\lambda_{\max}((A + A^T)/2) \le (m-1) + \sqrt{(m-1)N} < N/2,$$

because $N \ge 8(m-1)$ and $1/8 + 1/\sqrt{8} < 1/2$; for $m = 1$ the assertion is immediate. Differentiating squared Euclidean norms gives

$$\|e^{t(A - NI/2)}\|_2 \le 1 \quad (t \ge 0). \tag{C.53}$$

Let $J_m = \operatorname{diag}((-1)^i)_{i=0}^{m-1}$. For a shift $0 \le u < x$, put $r = (x-u)/x$ and $t = -\log r$. The normalized Gram matrix of the shifted monomial is exactly

$$G(r) = \left[\frac{Q_i(xD)Q_j(xD)h(x-u)}{h(x)}\right]_{ij} = r^N e^{tA} J_m e^{tA^T},$$

$$\|G(r)\|_2 \le 1. \tag{C.54}$$

Indeed its moments are $N_{\underline{j}} r^{N-j}$, which replaces each polynomial argument by z/r and multiplies by r^N. For $u \ge x$ its actual zero-extended derivative matrix is zero, including $u = x$, since $2m - 2 \le n - 2$.

The convolution-simplex formula reads

$$f(y) = Ch(y)\,\mathbb{E}e^{-\mathcal{A}y}, \qquad \mathcal{A} = \sum q_\ell V_\ell \in [0, Q],$$

where V is uniform on the standard simplex and its law is independent of y. Normalize derivative Grams by the fixed basis $Q_i(xD)$ and the positive scalar $Ch(x)$. For a fixed value $a \in [0, Q]$, the unshifted Gram of $Ch(y)e^{-ay}$ is $e^{-ax}T(ax)J_mT(ax)^T$. The matrix operator $X \mapsto -X - LX - XL^T$ has norm at most $1 + 2c_m$, so its difference from J_m has norm at most $e^{(1+2c_m)ax} - 1$. Averaging gives the first term of (C.50).

For $0 \le u < x$, the corresponding shifted Gram is

$$e^{-a(x-u)}T(ax)G((x-u)/x)T(ax)^T.$$

Its norm is at most $e^{2c_m Qx}$ by (C.52)–(C.54). The order of these matrices matters. For $u \geq x$ the Gram is zero. Integrating against η proves the second term of (C.50). Differentiation under that integral is legitimate: the zero-extended f has bounded continuous derivatives through order $n-2$.

An error of norm less than one from J_m preserves the inertia of every leading block along the straight-line homotopy. The basis change is lower triangular with positive diagonal, hence preserves all leading determinant signs. Finally $n \geq 8m$ gives $r_m < 2/5$ and $c_m < 2/3$, which proves the stated uniform specialization. □

Lemma C.19 (Bulk signed smoothing)**.** *Let $n \geq 2m$, $f = *_{\ell=1}^{n} \operatorname{Exp}(q_\ell)$ with $0 < q_\ell \leq Q$, and $p = f * (\delta_0 + \eta)$ for a finite real signed measure supported on $[0,\infty)$. If $x_0 > 0$ and*

$$a_0 = 2n/x_0 + nQ, \qquad \int e^{a_0 u} |\,\mathrm{d}\eta(u)| < 1,$$

then $\mathscr{D}_k(p;x) > 0$ for every $x \geq x_0$ and $k \leq m$.

Proof. At a fixed $x \geq x_0$, use the full normalization of Lemma C.5. In its leading m-block, the difference between the actual Grams of p and f is the integral of the actual shifted Gram of f against η. For $u > x$ this is zero. For $0 \leq u \leq x$ it is the leading block of $e^{-uB_x} J_n$. At $u = x$ the leading block is zero as well, because only derivatives through $2m-2 \leq n-2$ are involved. Thus the error norm is at most

$$\int_{[0,x]} e^{u\|B_x\|_2} |\,\mathrm{d}\eta(u)| \leq \int e^{a_0 u} |\,\mathrm{d}\eta(u)| < 1.$$

Compression and the signed-inertia argument in Lemma C.18 prove every leading determinant sign. There is no crossing error: the comparison is directly with the unit atom, and the actual shifted kernel is zero beyond its support. □

C.14. **Explicit PF convolution factors.** For this factorization we use a variant of Lemma 2.2. Its bounds hold with q replaced by any real q_* satisfying $2 \leq q_* \leq U^2/4 - 1/4$, and with $K_U(t) = e^{-q_* t} + H_{>U}(t)$: the same proof applies because $a_\rho - q_* \geq 3\gamma^2/4$ and $|\lambda_\rho| \leq q_* \gamma^2$.

For $100 \leq U \leq U_{\mathrm{PT}}$, choose a verified ordinate γ_* with $2 \leq \gamma_*^2 \leq U^2/4 - 1/4$; the first ordinate is always a possible choice. Let $\mathcal{A}_U$ be the finite multiset of ordinates $\gamma \leq U$, with *one occurrence* of γ_* deleted, and define

$$P_U(s) = \prod_{\gamma \in \mathcal{A}_U} (1 + s/\gamma^2).$$

Repeated zeros are retained with their multiplicities.

Theorem C.20 (Extraction of verified-zero PF factors)**.** *There is a positive self-decomposable random variable R_U such that*

$$S_1 \overset{d}{=} \sum_{\gamma \in \mathcal{A}_U} E_\gamma + R_U, \qquad E_\gamma \sim \operatorname{Exp}(\gamma^2), \tag{C.55}$$

$$X_{1,1} \overset{d}{=} \sum_{\gamma \in \mathcal{A}_U} L_\gamma + Y_U, \qquad L_\gamma \sim \operatorname{Lap}(\gamma), \tag{C.56}$$

where $\operatorname{Lap}(\gamma)$ has density $\gamma e^{-\gamma|x|}/2$ and $Y_U \overset{d}{=} \sqrt{2R_U}\, G_1$ with G_1 independent of R_U. The summands on each right-hand side are independent. The density of either finite sum is a PF_∞ function. The residual R_U has a smooth positive density on $(0,\infty)$; Y_U is self-decomposable and has an entire positive density.

Under RH, $1/h(s) = \prod_\gamma (1 + s/\gamma^2)^{-1}$ is the Laplace transform of a convergent sum of independent exponential variables, a generalized gamma convolution in Thorin's sense (Bondesson, 1992, Chapter 3), and correspondingly $X_{1,1} = \sum_\gamma L_\gamma$; for xi these factorizations are written in Polson

(2018, Section 2.1) and Polson (2026a, v9, Theorem 12(i)). The point here is the unconditional extraction of finitely many verified factors with a self-decomposable residual.

Proof. Set $C_U(s) = B(s) - \log P_U(s)$. By the product and integral formulas,

$$C_U(s) = \int_0^\infty (1 - e^{-st}) \frac{K_U(t)}{t}\, \mathrm{d}t, \qquad \mathbb{E}e^{-sR_U} = e^{-C_U(s)} = \frac{P_U(s)}{h(s)}. \tag{C.57}$$

Here $K_U(t) = e^{-\gamma_*^2 t} + H_{>U}(t)$, which is positive and decreasing by the variant of Lemma 2.2 stated above. Thus the Poisson and self-decomposability constructions from Section 2.3 apply. Multiplication of Laplace transforms proves (C.55). On the Fourier axis each factor becomes $(1 + t^2/\gamma^2)^{-1}$, the characteristic function of $\mathrm{Lap}(\gamma)$, proving (C.56). The finite exponential and centered Laplace densities are PF_∞ by Schoenberg's product characterization (Schoenberg, 1951). Equivalently, their transform reciprocals have only real zeros of the required sign. Removing finitely many exponentials leaves the small-time divergence of H unchanged and changes $B(s)$ by $O_U(\log(2+s))$. The density and entire-extension arguments in Section 2.3 therefore apply to both residuals. □

The densities in (2.8) therefore have explicit finite PF_∞ convolution factors. A higher retained zero makes the residual small and yields quantitative approximations.

Corollary C.21 (Approximation by verified-zero PF laws)**.** *For $1000 \le U \le T_*$, the retained ordinate can be chosen in $(U/4, U/3]$. Put*

$$b_U = \frac{16}{U^2} + \frac{2(\log U + 1)}{U}, \qquad A_U = \sum_{\gamma \in \mathcal{A}_U} E_\gamma, \qquad Z_U = \sum_{\gamma \in \mathcal{A}_U} L_\gamma.$$

Then $\mathbb{E}R_U \le b_U$. If d_{TV} denotes total variation distance between probability laws, with values in $[0,1]$, then

$$d_{\mathrm{TV}}(S_1, A_U) \le q b_U, \qquad d_{\mathrm{TV}}(X_{1,1}, Z_U) \le q b_U, \qquad q = \gamma_1^2 < 225. \tag{C.58}$$

Both bounds are less than $4.5 \cdot 10^{-9}$ at $U = T_*$. The approximating laws have PF_∞ densities.

Proof. The counting bound gives

$$N(U/3) - N(U/4) \ge \frac{U}{24\pi} \log \frac{U}{8\pi} - \frac{1}{4} \log(U+2) - 8 > 0 \qquad (U \ge 1000).$$

Indeed, the first term exceeds $U/32$; the resulting lower bound is positive at 1000 and increasing thereafter. This supplies γ_*, and its square satisfies the hypothesis of Lemma 2.2. Differentiating (C.57) at zero gives

$$\mathbb{E}R_U = \gamma_*^{-2} + \sum_{\gamma > U} \mathrm{Re}\, \frac{1}{\lambda_\rho} \le \frac{16}{U^2} + \sum_{\gamma > U} \gamma^{-2} \le b_U,$$

since $\mathrm{Re}(\lambda_\rho^{-1}) = (\gamma^2 - \delta_\rho^2)/(\gamma^2 + \delta_\rho^2)^2 \le \gamma^{-2}$.

The finite sums include the first-zero factors. Translation by $r \ge 0$ changes the law $\mathrm{Exp}(q)$ in total variation by $1 - e^{-qr} \le qr$. Convolution contracts total variation, so conditioning on R_U proves the first bound in (C.58). For the second, let $l(x) = \gamma_1 e^{-\gamma_1 |x|}/2$. Its second distributional derivative is $D^2 l = ql - q\delta_0$, of total variation $2q$. Consequently

$$\| l(\cdot - y) + l(\cdot + y) - 2l \|_1 \le 2qy^2.$$

This follows by integrating translates of $D^2 l$ against the triangular weight $(|y| - |v|)_+$. The residual Y_U is symmetric and $\mathbb{E}Y_U^2 = 2\mathbb{E}R_U$. Averaging the displayed inequality and dividing the resulting L^1 bound by two gives $d_{\mathrm{TV}}(L_{\gamma_1} + Y_U, L_{\gamma_1}) \le q\mathbb{E}R_U$. Convolving the other Laplace factors proves the claim. The final numerical bound follows from $q < 225$ and $U = T_*$. □

These estimates concern the normalized densities $c_0\Theta$ and $c_0\Lambda$.

C.15. **Residual moments for the higher-order certificate.** Put $U = T_*$ and retain $\gamma_* \in (U/4, U/3]$ as in Corollary C.21. Write $R = R_U$ and $Y = \sqrt{2R}G_1$. For order 256, we use sharper bounds that retain the common residual mean across all matrix entries. The counting estimate supplies the following enclosures. Set

$$T = \frac{\log(U/(2\pi)) + 1}{2\pi U}, \qquad C = \frac{\log(U+2)/4 + 129/16}{U^2}, \qquad D_0 = T + C,$$
$$B = D_0 + \frac{16}{U^2}, \qquad V_2 = \frac{D_0}{U^2} + \frac{256}{U^4}, \qquad A(s) = 1 - \frac{16(s+1/4)}{U^2}.$$

Lemma C.22. *The residual mean $\mu = \mathbb{E}R$ and variance satisfy*

$$\mu_- := T - C + \frac{9}{U^2} - \frac{3D_0}{4U^2} \le \mu \le \mu_+ := B, \qquad \mathrm{Var}(R) \le V_2. \tag{C.59}$$

For $s \ge 0$ with $A(s) > 0$,

$$\log \mathbb{E}e^{sR} \le \frac{Bs}{A(s)}, \qquad 0 \le \log \mathbb{E}e^{sR} - \mu s \le \frac{s^2 V_2}{2A(s)^2}. \tag{C.60}$$

In particular, with $\Omega = 3 \cdot 10^7$ and $\kappa = \exp(B\Omega^2/A(\Omega^2)) - 1$, $\mathbb{E}\cosh(\Omega Y) - 1 \le \kappa$.

Proof. Stieltjes integration of the counting error, including its boundary term at U, gives $\sum_{\gamma>U} \gamma^{-2} \in [T - C, T + C]$. For $\lambda = (\gamma + i\delta)^2$, $|\delta| \le 1/2$,

$$0 \le \gamma^{-2} - \mathrm{Re}\,\lambda^{-1} \le \frac{3}{4\gamma^4}.$$

Combine this with $9/U^2 \le \gamma_*^{-2} \le 16/U^2$ in (C.57) to obtain the mean interval. The second logarithmic derivative at zero is bounded by $\gamma_*^{-4} + \sum_{\gamma>U} |\lambda|^{-2} \le V_2$. For $0 \le v \le s$, each resolvent denominator in the first two logarithmic derivatives loses at most a factor $A(s)$ relative to its value at zero. Equivalently, integrate the positive residual kernel against e^{vt} and bound each zero term by its absolute value. The derivatives are bounded by $B/A(s)$ and $V_2/A(s)^2$, respectively. Integration and convexity prove (C.60). Gaussian conditioning gives the cosh bound. All domain inequalities are checked with directed arithmetic. □

Higher cumulants retain the same dependence. If $h_j = \kappa_j(R)/(j-1)!$, then

$$h_j = \gamma_*^{-2j} + \sum_{\gamma>U} \mathrm{Re}\,\lambda_\rho^{-j}.$$

Writing $q = 2j - 1$, put

$$T_j = \frac{\log(U/(2\pi)) + 1/q}{2\pi q U^q}, \qquad C_j = \frac{\log(U+2)/4 + 8 + 1/(16j)}{U^{2j}}.$$

The same integration and the inequality $0 \le \gamma^{-2j} - \mathrm{Re}\,\lambda^{-j} \le j(2j+1)\gamma^{-2j-2}/4$ yield

$$T_j - C_j + \frac{9^j}{U^{2j}} - \frac{j(2j+1)D_0}{4U^{2j}} \le h_j \le T_j + C_j + \frac{16^j}{U^{2j}}. \tag{C.61}$$

These bounds use only the published verified height and the counting estimate.

C.16. **Differential determinants and strict composition.** For a sufficiently smooth density p, define

$$\mathscr{D}_k(p; x) = (-1)^{k(k-1)/2} \det[p^{(i+j)}(x)]_{i,j=0}^{k-1}.$$

Lemma C.23 (Differential criterion for PF minors)**.** *If $p \in C^{2m-2}(\mathbb{R})$ and $\mathscr{D}_k(p; x) > 0$ for every $x \in \mathbb{R}$ and $1 \le k \le m$, then $p(x - y)$ has strictly positive ordered minors through order m.*

Proof. Fix y. The initial Wronskians of the functions $x \mapsto (-1)^j p^{(j)}(x-y)$, $0 \le j < m$, are precisely $\mathscr{D}_k(p; x-y)$. Lemma C.1 makes their ordered evaluation determinants positive. For fixed increasing x_i, these determinants are the transposes of the initial Wronskians in y of $y \mapsto p(x_i - y)$. A second application of that lemma proves the claim. □

Choose the 270 distinct verified ordinates $a_0 < \cdots < a_{269} < U/4$ specified in Appendix C.20, and let $f = \mathrm{Lap}(a_0) * \cdots * \mathrm{Lap}(a_{269})$ denote their convolution density. Theorem C.20 gives

$$c_0\Lambda = p * \nu, \qquad p = \mathbb{E}H_R f, \qquad H_t = e^{tD^2}, \tag{C.62}$$

where ν is the convolution of the remaining extracted Laplace laws. The positive Fourier transform $\widehat{f}(t) = \prod_l a_l^2/(a_l^2 + t^2)$ has integrable moments through degree 538, so f and p have bounded continuous derivatives through that degree. This exceeds the 510 derivatives needed for $m = 256$.

Lemma C.24 (Certified differential signs)**.** *For the density p in (C.62), $\mathscr{D}_k(p; x) > 0$ for every $x \in \mathbb{R}$ and $1 \le k \le 256$.*

Proof. The interval calculation in Appendix C.20 certifies all leading determinant signs uniformly on $[0, 3]$, using the residual bounds just proved. Its analytic tail estimate certifies the same signs on $[3, \infty)$. Evenness gives the negative half-line: the derivative matrices at $-x$ and x are congruent by $\mathrm{diag}((-1)^i)$. □

Lemma C.23 now gives strict ordered minors through size 256 for p. Convolution with each remaining Laplace density preserves strictness. The continuous Cauchy–Binet formula (Karlin, 1968) integrates the product of a strictly positive minor and a nonnegative Laplace minor over $u_1 < \cdots < u_k$. The latter is positive near $u_i = y_i$: the matrix $[e^{-a|y_i - y_j|}]$ is positive definite for distinct y_i, by its strictly positive Fourier density. Thus the integral is strictly positive. Iterating over the remaining finite set of factors in ν proves the assertion for $c_0\Lambda$, and hence for Λ.

C.17. **Fixed-order positivity near the one-sided boundary.** We use the exponential-tilting (saddlepoint) method of Daniels (1954) to obtain relative determinant estimates near the left endpoint of Θ. Under a sufficiently strong negative exponential tilt, the one-sided law becomes Gaussian after centering and scaling. To control determinants, weak convergence is insufficient: we prove convergence of every fixed number of density derivatives.

Proposition C.25 (Local PF positivity of Θ)**.** *For each fixed integer $m \ge 1$, there is $\varepsilon_m > 0$ such that*

$$\det[\Theta(x_i - y_j)]_{i,j=1}^{k} > 0 \qquad (1 \le k \le m)$$

whenever $x_1 < \cdots < x_k$, $y_1 < \cdots < y_k$, and $0 < x_i - y_j < \varepsilon_m$ for every i, j. More precisely, put $x_s = B'(s)$ and $v_s = -B''(s)$. As $s \to \infty$,

$$\mathscr{D}_k(\Theta; x_s) \sim \Theta(x_s)^k v_s^{-k(k-1)/2} \prod_{j=0}^{k-1} j! \qquad \text{for every fixed } k. \tag{C.63}$$

Proof. We use the branch of $B(z) = \log h(z)$ that is real on the positive axis. Stirling's expansion (NIST, 2026, Section 5.11), together with $\log\zeta(w) = O(2^{-\operatorname{Re} w})$ for $\operatorname{Re} w \to \infty$, gives

$$B(z) = \frac{\sqrt{z}}{2}\left(\log\frac{\sqrt{z}}{2\pi} - 1\right) + \frac{7}{8}\log z + C + O(|z|^{-1/2}) \tag{C.64}$$

uniformly in the right half-plane as $|z| \to \infty$. Indeed, $|\arg\sqrt{z}| \le \pi/4$ and $\operatorname{Re}\sqrt{z} \ge \sqrt{|z|/2}$ there, so the gamma expansion and Euler-product estimate are uniform. Cauchy's estimates permit termwise differentiation near the positive axis. In particular,

$$x_s \sim \frac{\log s}{8\sqrt{s}}, \qquad v_s \sim \frac{\log s}{16 s^{3/2}}, \qquad B'''(z) = O\left(\frac{\log s}{s^{5/2}}\right) \quad (|z - s| \le s/2). \tag{C.65}$$

For large s, $v_s > 0$ and x_s decreases continuously to zero.

The tilted density is $f_s(u) = e^{B(s)-su}\theta_1(u)$, with mean x_s and variance v_s. Write $\sigma_s = \sqrt{v_s}$ and let $g_s(z) = \sigma_s f_s(x_s + \sigma_s z)$ be its centered, standardized density. Its characteristic function is

$$\phi_s(t) = \exp\{B(s) - B(s - it/\sigma_s) - itx_s/\sigma_s\}.$$

Taylor expansion using (C.65) gives $\phi_s(t) \to e^{-t^2/2}$ for each fixed t.

To pass to density derivatives, we need an integrable bound for all frequencies. Write $\sqrt{1+iu} = a + ib$ with $a \geq 1$, and set $w = \sqrt{s}$, $L = \log(w/(2\pi)) - 1$. From (C.64),

$$\operatorname{Re} B(s + isu) - B(s) \geq \frac{wL}{2}(a-1) - \frac{C_1}{w}.$$

Here $\operatorname{Re}[(a+ib)\log(a+ib)] \geq 0$: for $\eta = |b|/a < 1$, its quotient by a is $\operatorname{arctanh}(\eta^2) - \eta\arctan\eta$, whose derivative is $\eta/(1-\eta^2) - \arctan\eta \geq 0$. The logarithmic term $\frac{7}{8}\log|1+iu|$ is also nonnegative. Finally,

$$a - 1 = \frac{u^2}{2(a+1)(\sqrt{1+u^2}+1)} \geq c_1 \min\{u^2, \sqrt{|u|}\}$$

for an absolute $c_1 > 0$. Put $A_s = s\sigma_s$, so $A_s^2 \sim wL/8$ and $A_s \to \infty$. Substitution of $u = t/A_s$ gives $|\phi_s(t)| \leq C_2 \exp\{-c\min(t^2, A_s^{3/2}\sqrt{|t|})\}$, with $c, C_2 > 0$ independent of large s. Since $A_s \geq 1$ eventually,

$$|\phi_s(t)| \leq C_2 \exp\{-c\min(t^2, \sqrt{|t|})\}.$$

The right side is integrable after multiplication by any fixed power of $|t|$. Dominated convergence in Fourier inversion therefore gives $g_s^{(r)}(0) \to \varphi^{(r)}(0)$ for every fixed r, where $\varphi(z) = (2\pi)^{-1/2}e^{-z^2/2}$.

The Gaussian moment determinant is

$$\mathscr{D}_k(\varphi; 0) = \varphi(0)^k \prod_{j=0}^{k-1} j! > 0.$$

For example, this follows by replacing monomials with monic Hermite polynomials in the Gaussian Gram matrix; their squared norms are $j!$. For $M_f = (f^{(i+j)}(x))_{i,j=0}^{k-1}$, Leibniz's rule gives $M_{e^{cx}f} = e^{cx}T_c M_f T_c^{\mathsf{T}}$, where T_c is lower triangular with $(T_c)_{ij} = \binom{i}{j}c^{i-j}$ for $j \leq i$. Since $\det T_c = 1$, tilting multiplies the differential determinant by e^{kcx}. Scaling then gives

$$\frac{\mathscr{D}_k(\Theta; x_s)}{\Theta(x_s)^k} = \sigma_s^{-k(k-1)}\frac{\mathscr{D}_k(g_s; 0)}{g_s(0)^k},$$

which proves (C.63). All initial signed determinants through size m are therefore positive on $(0, \varepsilon_m)$ for some $\varepsilon_m > 0$. The proof of Lemma C.23 applies locally: both Wronskian integrations remain in this interval when all the cross differences do. This proves the stated PF_k minor inequalities. □

C.18. **The global order-four certificate for Θ.** For global PF positivity, the differential criterion must also handle nonpositive cross differences, as in Lemma C.3.

For Θ, the required interior signs follow from uniform analytic bounds near zero and infinity, joined by closed interval computations.

Lemma C.26. *For every $x > 0$ and $1 \leq k \leq 4$, $\mathscr{D}_k(\Theta; x) > 0$.*

Proof. Write $p = c_0\Theta = \theta_1$. The positive constant c_0 does not affect the signs. The three domains used below are

$$(0, 0.00211005], \qquad [0.00211004, 0.01001], \qquad [0.008, \infty). \tag{C.66}$$

All terminating decimals in this proof denote exact rationals.

The uniform interval at zero. We make the bounds in the proof of Proposition C.25 explicit. With principal square roots and logarithms, put

$$\mathcal{F}(z) = \frac{\sqrt{z}}{2}\left(\log\frac{\sqrt{z}}{2\pi} - 1\right) + \frac{7}{8}\log z.$$

For $\operatorname{Re} z \geq 0$ and $|z| \geq 10^4$, the remainder in (C.64) satisfies

$$B(z) = \mathcal{F}(z) + C + E(z), \qquad |E(z)| \leq |z|^{-1/2}. \tag{C.67}$$

To check the constant, set $w = \sqrt{z}$ and $v = w/2 + 1/4$. Binet's formula (NIST, 2026, Eq. 5.9.10_2) gives

$$E(z) = \log\left(1 - \frac{1}{4w^2}\right) + \left(\frac{w}{2} - \frac{1}{4}\right)\log\left(1 + \frac{1}{2w}\right) - \frac{1}{4} + R(v) + \log\zeta(w + \tfrac{1}{2}),$$
$$R(v) = \int_0^\infty e^{-vt}\left(\frac{1}{2}\coth\frac{t}{2} - \frac{1}{t}\right)\frac{\mathrm{d}t}{t}.$$

The partial-fraction expansion of coth bounds the bracket by $t/12$. Thus $|R(v)| \leq \sqrt{2}/(6|w|)$. For $|w| \geq 100$, the first logarithm is at most $(4|w|^2 - 1)^{-1}$ in modulus, and the second term together with $-1/4$ is at most $3/[16|w|(1 - 1/(2|w|))]$. The Euler product and integral bound for the zeta tail give, with $\sigma = \operatorname{Re}(w + 1/2)$,

$$|\log\zeta(w + \tfrac{1}{2})| \leq \log\zeta(\sigma) \leq 2^{-\sigma}\left(1 + \frac{2}{\sigma - 1}\right) \leq \frac{1}{100|w|}.$$

Here $\sigma \geq |w|/\sqrt{2} + 1/2$; the last bound holds at $|w| = 100$ and improves thereafter. The sum of these bounds is less than $0.44/|w|$, proving (C.67).

Retain $x_s = B'(s)$, $v_s = -B''(s)$, $\sigma_s = \sqrt{v_s}$ and the standardized tilted density g_s from the preceding subsection. Put $w = \sqrt{s}$, $L = \log(w/(2\pi)) - 1$, $W = wL$, and $A_s = s\sigma_s$. Cauchy's estimate on the disk of radius $s/2$ gives $|E^{(r)}(s)| \leq r!2^r\sqrt{2}\,s^{-r-1/2}$ and hence

$$A_s^2 \in \left[\frac{W}{8} + \frac{7}{8} - \frac{8\sqrt{2}}{w}, \frac{W}{8} + \frac{7}{8} + \frac{8\sqrt{2}}{w}\right].$$

For $s \geq S := 275789$, these bounds imply $W/8 \leq A_s^2 \leq W/7$. Taylor's formula for the characteristic function ϕ_s gives, when $|t| \leq T := 3113/500 \leq A_s/2$,

$$\log\phi_s(t) = -t^2/2 + ib_s t^3 + R_s(t), \qquad |b_s| \leq a, \qquad |R_s(t)| \leq dt^4,$$

where the following expressions are evaluated at $s = S$:

$$a = \frac{1}{6}\left(\frac{8}{L}\right)^{3/2}\left(\frac{3L - 1}{16\sqrt{w}} + \frac{7}{4w^{3/2}} + \frac{48\sqrt{2}}{w^{5/2}}\right),$$
$$d = \frac{1}{24}\left(\frac{8}{L}\right)^{2}\left(\frac{15L + 167/4}{32w} + \frac{21}{4w^2} + \frac{384\sqrt{2}}{w^3}\right).$$

These expressions decrease for $s \geq S$. They follow by differentiating $\mathcal{F}$ three and four times and applying the same Cauchy bound; on $s + i[-s/2, s/2]$ one uses $|\log(z/s)| \leq 1/2$. With $c = 1/2 - dT^2 > 0$, the even part therefore satisfies

$$|\operatorname{Re}\phi_s(t) - e^{-t^2/2}| \leq \tfrac{1}{2}a^2t^6e^{-t^2/2} + dt^4e^{-ct^2} \qquad (|t| \leq T).$$

The sector estimate also bounds the entire Fourier tail. The calculation preceding (C.63) gives

$$|\phi_s(t)| \leq e^{2/\sqrt{s}}\begin{cases} e^{-7t^2/24}, & |t| \leq A_s, \\ \exp\{-\frac{W}{32}\sqrt{|t|/A_s}\}, & |t| \geq A_s. \end{cases}$$

Indeed, for $\alpha = \operatorname{Re}\sqrt{1+iu}$, the identity used there gives $\alpha - 1 \ge u^2/12$ when $|u| \le 1$ and $\alpha - 1 \ge \sqrt{|u|}/16$ when $|u| \ge 1$. Define

$$I_j(b,T) = \int_T^\infty t^j e^{-bt^2}\,\mathrm{d}t, \qquad J_j(W) = 2(W/7)^{(j+1)/2}\frac{\Gamma(2j+2, W/32)}{(W/32)^{2j+2}}.$$

The second part of the tail contributes at most $J_j(W)$. For $0 \le j \le 6$, its finite incomplete-gamma expansion is decreasing when $W \ge 80$; also $W(S) > \max(80, 32T^2)$. Fourier inversion thus proves, for even $j \le 6$ and every $s \ge S$,

$$\begin{aligned} |g_s^{(j)}(0) - \varphi^{(j)}(0)| \le e_j, \qquad \varphi(z) = (2\pi)^{-1/2}e^{-z^2/2}, \\ \pi e_j = \tfrac{1}{2}a^2 I_{j+6}(1/2,0) + dI_{j+4}(c,0) + I_j(1/2,T) \\ + e^{2/\sqrt{S}}\{I_j(7/24,T) + J_j(W(S))\}. \end{aligned} \tag{C.68}$$

The four Gaussian values are $\varphi^{(j)}(0) = (2\pi)^{-1/2}(1,-1,3,-15)$ for $j = 0,2,4,6$. Outward evaluation of (C.68) proves the four strict inequalities

$$g_s(0) > 0, \qquad -g_s''(0) > 0, \qquad g_s^{(4)}(0) - \frac{g_s''(0)^2}{g_s(0)} > 0, \qquad -g_s^{(6)}(0) - \frac{g_s^{(4)}(0)^2}{-g_s''(0)} > 0.$$

Their smallest certified lower bound exceeds $1.5585 \cdot 10^{-6}$. The even-index block of the derivative matrix is positive definite and the odd-index block is negative definite. Eliminating the even block leaves a negative definite Schur complement, whatever the mixed entries; the same argument applies to every leading block. Hence $\mathscr{D}_k(g_s;0) > 0$ for $k \le 4$. The scaling identity in the preceding proof transfers these signs to Θ at x_s. Finally,

$$x_S \ge \frac{L(S)+1}{4\sqrt{S}} + \frac{7}{8S} - \frac{2\sqrt{2}}{S^{3/2}} > 0.00211005.$$

Since $x_s' = -v_s < 0$ and $x_s \to 0$, this proves all signs on the first domain of (C.66).

The compact interval. For positive rational s, σ and rational x, use the exact density and characteristic function

$$g(z) = \sigma e^{B(s)-s(x+\sigma z)}p(x+\sigma z), \qquad \phi(t) = \frac{h(s)}{h(s-it/\sigma)}e^{-itx/\sigma}.$$

At $U = 3 \cdot 10^{12}$, Theorem C.20 and the retained anchor from Corollary C.21 extract the first 64 exponential rates $q_\ell = \gamma_\ell^2$ and leave a positive independent residual law. Tilting changes these rates to $s + q_\ell$. Consequently, with $r_\ell = (s+q_\ell)\sigma$,

$$|\phi(t)| \le P(t) := \prod_{\ell=1}^{64}(1 + t^2/r_\ell^2)^{-1/2}.$$

For $t \ge T_0 := 64$, logarithmic differentiation gives $P(t) \le P(T_0)(t/T_0)^{-\alpha}$, where $\alpha = \sum_\ell T_0^2/(T_0^2+r_\ell^2)$. Thus, whenever $\alpha > j+1$,

$$\int_{T_0}^\infty t^j|\phi(t)|\,\mathrm{d}t \le \frac{P(T_0)T_0^{j+1}}{\alpha - j - 1}. \tag{C.69}$$

Certified quadrature of $\pi^{-1}\operatorname{Re}\int_0^{T_0}(-it)^j\phi(t)\,\mathrm{d}t$, together with (C.69), encloses $v_j = g^{(j)}(0)$ for $0 \le j \le 9$. The transform is evaluated by the exact gamma–zeta formula (1.1).

Absolute Fourier integrals give bounds $N_j \ge \sup_z |g^{(j)}(z)|$ for $4 \le j \le 10$. On each interval $[u,v] \subset [0,64]$ of length $1/8$, use $t^jP(t) \le v^jP(u)$ and add (C.69). For $s \ge 10^4$, the smaller of this bound and the decreasing majorant

$$\exp\left\{\frac{2}{\sqrt{s}} - \frac{W}{2}\left(\sqrt{\frac{\sqrt{1+(t/(s\sigma))^2}+1}{2}} - 1\right)\right\}$$

is used on each rectangle; it follows from (C.67). All comparisons use directed upper endpoints. Taylor's theorem now encloses the whole derivative matrix $H(z) = [g^{(i+j)}(z)]_{i,j=0}^{3}$ on $|z| \le r$:

$$H_{ij}(z) = \sum_{n=0}^{3} v_{i+j+n} \frac{z^n}{n!} + R_{ij}(z), \qquad |R_{ij}(z)| \le N_{i+j+4} \frac{r^4}{4!}.$$

Each box carries an exact rational lower triangular matrix C with positive diagonal. Apply this congruence to the Taylor coefficients before enclosing z. Writing $CH(z)C^T = Q(z) + E(z)$, we have

$$|E_{ij}(z)| \le F_{ij} := \frac{r^4}{4!} \sum_{u \le i,\, v \le j} |C_{iu} C_{jv}| N_{u+v+4}.$$

Every leading block is transformed by the corresponding leading block of C, so its determinant changes by a positive square. One acceptance test bounds $\|CH(z)C^T - J\|_\infty < 1$ uniformly, where $J = \mathrm{diag}(1, -1, 1, -1)$; the homotopy from J preserves every leading determinant sign. The other test forms each $\det Q_k(z)$ as a polynomial in the same variable z and bounds its signed value below, using $z^n \in [-r^n, r^n]$ for odd n and $z^n \in [0, r^n]$ for positive even n. If $M_{ij} \ge \sup_{|z| \le r} |Q_{ij}(z)|$, the complete determinant error is bounded by

$$|\det(Q_k + E_k) - \det Q_k| \le \sum_{\pi \in S_k} \sum_{\varnothing \ne I \subset \{0,\dots,k-1\}} \prod_{i \in I} F_{i,\pi(i)} \prod_{i \notin I} M_{i,\pi(i)}.$$

The signed polynomial lower bound must exceed this error for every $k \le 4$. Both tests prove signs throughout the closed physical interval $[x - r\sigma, x + r\sigma]$, because the exponential tilt and positive scale preserve these signs. The records certify 89 boxes on $[0.00211004, 0.006]$ and 40 on $[0.006, 0.01001]$; 54 use the norm test and 75 use the polynomial test. Exact rational endpoint comparisons verify the closed cover.

The residue tail. Extract the same 64 factors and write $p = f_{64} * \mathrm{Law}(V)$, where V includes every omitted factor and the positive anchored high residual. The verified zeros satisfy $q_{64} < \tau := 29470 < q_{65}$, and the certified zero count below $\sqrt{\tau}$ is 64. Every remaining finite exponential factor therefore has a finite moment at τ. Lemma C.22 gives the same conclusion for the high residual, since $\tau < 40000$. Having established $\mathbb{E}e^{\tau V} < \infty$, analytic continuation of the cancelled transform gives

$$\kappa := \mathbb{E}e^{\tau V} - 1 = \frac{c_0}{\Xi(\sqrt{\tau})} \prod_{\ell=1}^{64} (1 - \tau/q_\ell) - 1.$$

The finite convolution and the actual residues are

$$f_{64}(x) = \mathbf{1}_{\{x>0\}} \sum_{\ell=1}^{64} c_\ell e^{-q_\ell x}, \quad c_\ell = \frac{\prod_h q_h}{\prod_{h \ne \ell}(q_h - q_\ell)}, \quad d_\ell = c_\ell \mathbb{E}e^{q_\ell V} = -\frac{2\gamma_\ell c_0}{\Xi'(\gamma_\ell)}.$$

The zero and derivative enclosures verify $(-1)^{\ell-1} d_\ell > 0$. Set $G(x) = \sum_{\ell=1}^{64} d_\ell e^{-q_\ell x}$. For polynomials P, Q of degree at most three, the part of the convolution with $V \ge x$ gives

$$|P(\partial_x)Q(\partial_x)(p - G)(x)| \le \frac{\kappa e^{-\tau x}}{1 - e^{-\tau x}} \sum_{\ell=1}^{64} |c_\ell P(-q_\ell) Q(-q_\ell)|. \tag{C.70}$$

Indeed, for $v \ge x$ and $q_\ell < \tau$, $e^{q_\ell v}/(e^{\tau v} - 1) \le e^{(q_\ell - \tau)x}/(1 - e^{-\tau x})$. Integrate against V and use $\mathbb{E}(e^{\tau V} - 1) = \kappa$. Differentiation through order six is valid because f_{64} and these derivatives match zero at the support boundary.

Use the normalized Newton polynomials

$$P_i(z) = \frac{\prod_{h=1}^{i}(z + q_h)}{\sqrt{|d_{i+1}|} \prod_{h=1}^{i}(q_{i+1} - q_h)}, \qquad 0 \le i \le 3.$$

Their coefficient matrix is lower triangular and invertible. After this congruence and a positive diagonal scaling with entries $e^{q_{i+1}x/2}$, the residue matrix has entries

$$A^0_{ij}(x) = \sum_{\ell>\max(i,j)} w_{\ell ij} e^{-\beta_{\ell ij}x}, \qquad w_{\ell ij} = d_\ell P_i(-q_\ell)P_j(-q_\ell), \quad \beta_{\ell ij} = q_\ell - \frac{q_{i+1}+q_{j+1}}{2}.$$

The leading diagonal term is $(-1)^i$; all other surviving exponents are positive. The actual transformed matrix $A(x)$ differs from $A^0(x)$ by at most

$$E_{ij}(L) = \frac{\kappa e^{-(\tau-(q_{i+1}+q_{j+1})/2)L}}{1-e^{-\tau L}} \sum_\ell |c_\ell P_i(-q_\ell)P_j(-q_\ell)| \qquad (x \ge L > 0).$$

At $L = 0.032$, the absolute row sums of $A - J$, bounded by the nonleading residue terms and these errors, are less than 0.880786. Every term decreases for larger x, so the same homotopy argument proves all four signs on $[0.032, \infty)$.

For a closed rational box $[L, H] \subset [0.008, 0.032]$, put $x_0 = (L+H)/2$ and $r = (H-L)/2$. The fourth-degree Taylor coefficients and the full remainder bound are

$$(C_n)_{ij} = \frac{1}{n!} \sum_\ell w_{\ell ij}(-\beta_{\ell ij})^n e^{-\beta_{\ell ij}x_0}, \qquad 0 \le n \le 4,$$

$$F_{ij} = \frac{r^5}{5!} \sum_\ell |w_{\ell ij}| \beta^5_{\ell ij} e^{-\beta_{\ell ij}L} + E_{ij}(L).$$

Each box supplies an invertible rational lower triangular matrix T. Transform the coefficients to TC_nT^T and the remainder to $|T|F|T|^T$ before inserting the parity intervals for $(x - x_0)^n$. Interval elimination on this enclosure verifies successive pivot signs $+, -, +, -$, checking each strict sign before division. Lower triangularity transfers these signs to the leading minors of A, and hence to those of p. The resulting 1041 closed rational boxes cover $[0.008, 0.032]$ exactly, completing the third domain of (C.66).

Certificate records. All numerical comparisons use outward Arb balls (Johansson, 2017); all interval joins and basis entries use exact rationals. The compact quadratures were produced at 40 decimal digits; the returned balls, including their radii, enter every later test. A global check replays the endpoint inequalities and all 129 compact boxes at 100 digits, independently regenerates the residue inputs and all 1041 tail boxes, and verifies the monotone tail and exact overlaps. Separate implementations replayed both finite covers at 120 digits, the compact one from the certified quadrature balls. The smallest acceptance margin of the compact replay exceeds 0.0147355, and the smallest signed pivot of the tail replay exceeds 0.0100751; these margins refer to the recorded congruences. The checks verify all three overlapping domains in (C.66), proving the lemma. □

The density Θ is smooth and flat at zero, so its zero extension belongs to $C^6(\mathbb{R})$. Lemmas C.26 and C.3 therefore prove the global order-four assertion for Θ, including its exact strictness condition.

C.19. **Low-zero enclosures for log-concavity.** For the low-height arguments, define $I_i = [\bar a_i - 10^{-40}, \bar a_i + 10^{-40}]$, $i = 1, 2$, by the following exact decimal centers:

i	$\bar a_i$
1	14.134725141734693790457251983562470270784257115699243 18
2	21.022039638771554992628479593896902777334340524902781 75

Interval evaluation gives opposite signs for Ξ at the two endpoints of every I_i, so continuity gives a real zero in each box. The boxes are positive, disjoint, and increasing. The numbering and simplicity of these zeros are not needed.

The evaluation uses Euler–Maclaurin with $N = 1000$, $p = 64$, and $s = 1/2 + it$:

$$\zeta(s) = \sum_{n=1}^{N-1} n^{-s} + \frac{N^{1-s}}{s-1} + \frac{1}{2}N^{-s} + \sum_{k=1}^{p} \frac{B_{2k}}{(2k)!}(s)_{2k-1}N^{-s-2k+1} + R_p(s),$$

$$|R_p(s)| \le \frac{4|(s)_{2p}|}{(2\pi)^{2p}(\operatorname{Re} s + 2p - 1)N^{\operatorname{Re} s+2p-1}}.$$

Here $(s)_j = s(s+1)\cdots(s+j-1)$ and B_{2k} are exact rational Bernoulli numbers. The remainder follows by integration of the periodic Bernoulli remainder and $\|B_{2p}(\{\cdot\})\|_\infty/(2p)! \le 4/(2\pi)^{2p}$. The resulting disc is enclosed by a complex rectangle and multiplied by $s(s-1)\pi^{-s/2}\Gamma(s/2)/2$, with the gamma factor also in interval arithmetic, to enclose $\Xi(t)$.

C.20. **The order-256 certificate.** We describe the finite inequalities used in Lemma C.24. All zero enclosures, signed sums, matrix operations, and remainders use outward ball arithmetic (Johansson, 2017). The 270 selected zeros are specified by their increasing indices in the certificate; the largest index is 19930, with ordinate approximately 17991.0263. Each component regenerates ordinate balls using the rigorous `acb.zeta_zero` routine and checks the positivity, ordering, and analytic cutoffs needed by its method. The final independent audit also checks the exact critical-line identity, disjointness, and the bounds $a_l < \min(U/4, \Omega)$ for every selected index.

Signed matrices and Fourier norms. Put $S = 100$, $r_l = a_l/S$, and, for $x \ge 0$,

$$f(x) = \sum_{l=0}^{269} c_l e^{-a_l x}, \qquad c_l = \frac{\prod_h a_h^2}{2a_l \prod_{h\ne l}(a_h^2 - a_l^2)}.$$

The signs of c_l alternate, starting positive. Let $\phi(t) = \widehat{f}(t) > 0$. For even $j \le 538$,

$$F_j = \frac{1}{2\pi}\int_{\mathbb{R}} (t/S)^j \phi(t)\,\mathrm{d}t = (-1)^{j/2}\sum_l c_l r_l^j > 0.$$

For a fixed lower triangular polynomial basis P_i of degree i, define, whenever $2i + q \le 538$ and q is even,

$$R_{i,q}^2 = \frac{1}{2\pi}\int_{\mathbb{R}} (t/S)^q |P_i(it/S)|^2 \phi(t)\,\mathrm{d}t. \tag{C.71}$$

These are finite signed combinations of the F_j; each is enclosed and proved positive before taking its square root. Fourier inversion and Cauchy–Schwarz bound the derivative of order q of the transformed entry by $S^q R_{i,q}R_{j,q}$. The same bound applies to p, since $|\widehat{p}(t)| = \phi(t)\mathbb{E}e^{-Rt^2} \le \phi(t)$.

The matrix to be enclosed is $[p^{(i+j)}(x)/S^{i+j}]_{i,j=0}^{255}$. A lower triangular congruence with nonzero diagonal preserves the signs of every leading determinant. The numerical preconditioners are selected from midpoint factorizations, but their coefficients are fixed dyadic numbers. Signed sums and congruences are completed before uncertain common coefficients are inserted. Every rounding used to store a completed coefficient is outward.

The central interval. Choose an exact dyadic center μ_0 of the mean enclosure and a directed bound $\Delta \ge |\mu - \mu_0|$. Set

$$g_L = \sum_{l=0}^{L} \frac{\mu_0^l}{l!} D^{2l} f, \qquad \eta = \mu - \mu_0.$$

Taylor's formula in the heat parameter and contraction of H_t give

$$p = g_L + \eta D^2 g_{L-1} + \mathcal{R}.$$

In the transformed (i, j) entry the remainder is bounded by

$$\frac{(V_2 + \Delta^2)S^4}{2} R_{i,4}R_{j,4} + \left(\frac{\mu_0^{L+1}}{(L+1)!} + \frac{\Delta\mu_0^L}{L!}\right) S^{2L+2} R_{i,2L+2}R_{j,2L+2}. \tag{C.72}$$

Indeed, expanding $H_R f$ about $H_{\mu_0} f$ leaves a second-order error bounded by $\mathbb{E}(R-\mu_0)^2/2 \le (V_2+\Delta^2)/2$; expanding $H_{\mu_0} f$ and $D^2 H_{\mu_0} f$ gives the other terms. The heat parameters along the first Taylor segment are nonnegative.

On a closed box, expand the two reference functions about a fixed center z through odd spatial degree K. If $h = \max_{x \text{ in the box}} |x-z|$, their spatial remainder is at most

$$\frac{h^{K+1}}{(K+1)!}\left\{\sum_{l=0}^{L}\frac{\mu_0^l S^{2l+K+1}}{l!}R_{i,2l+K+1}R_{j,2l+K+1} + \Delta\sum_{l=0}^{L-1}\frac{\mu_0^l S^{2l+K+3}}{l!}R_{i,2l+K+3}R_{j,2l+K+3}\right\}. \tag{C.73}$$

The common variable η and spatial powers are retained until after the signed congruences. Odd and even powers have their correct symmetric and nonnegative enclosures. Intersecting termwise and Horner enclosures can tighten a polynomial without changing its range guarantee. The factor condition

$$n \ge m + L + (K+1)/2$$

supplies every required Fourier moment. For $n = 270$, $m = 256$, the certificate uses $(L,K) = (10,7)$ on $[0, 1/16000]$ and $(8,11)$ on $[1/16000, 1/1000]$. Both use moments only through degree 538. These choices of (L,K) shift the derivative budget toward the spatial remainder.

Shared cumulants outside the center. For $x > 0$ write $g(x) = \sum_l c_l M_l e^{-a_l x}$, where $M_l = \mathbb{E}e^{a_l^2 R}$. Independent interval choices for the M_l lose the cancellation needed at high order. Choose dyadic centers h_j^0 of (C.61), for $1 \le j \le N$, and put

$$L_0(s) = \sum_{j=1}^{N} h_j^0 s^j/j, \qquad \mathbb{E}e^{sR}e^{-L_0(s)} = \sum_{n\ge 0} d_n s^n.$$

Here $d_0 = 1$ and $n d_n = \sum_{j=1}^{n}(h_j - h_j^0)d_{n-j}$ for $1 \le n \le N$. For $\rho < U^2/16$, define

$$H = 16^{N+1}/U^{2N+2} + D_0/U^{2N},$$
$$G = \sum_{j=1}^{N}|h_j - h_j^0|\rho^j/j + \frac{H\rho^{N+1}}{(N+1)(1-16\rho/U^2)}.$$

The reciprocal-zero series bounds the omitted logarithmic coefficients by this geometric majorant. Cauchy's estimate therefore gives, for $0 \le s < \rho$,

$$\left|\mathbb{E}e^{sR}e^{-L_0(s)} - \sum_{n=0}^{N} d_n s^n\right| \le e^G \frac{(s/\rho)^{N+1}}{1-s/\rho}. \tag{C.74}$$

For each common d_n, form the signed derivative matrix first, and then multiply by its interval enclosure. The radius and every convergence inequality are checked in the component record. Conservative far-region specializations retain only d_0, d_1, moving the higher terms into the absolute error, or use the independent enclosures supplied by (C.60):

$$M_l \in \exp\left(a_l^2[\mu_-,\mu_+] + \left[0, \frac{a_l^4 V_2}{2A(a_l^2)^2}\right]\right).$$

For any fixed basis P_i, the branch-crossing error between p and g is bounded entrywise by

$$\frac{2\kappa e^{-\Omega x}}{(1-e^{-\Omega x})^2}\sum_l |c_l|(|P_i(r_l)P_j(r_l)| + |P_i(-r_l)P_j(-r_l)|). \tag{C.75}$$

Only $Y > x$ changes the exponential branch. On that event, $\cosh(\Omega Y) - 1 \geq e^{\Omega Y}(1 - e^{-\Omega x})^2/2$; together with $a_l < \Omega$ this proves the bound. The finite convolution is smooth enough to differentiate before splitting the expectation.

The outer bases start from the Newton polynomials

$$Q_i(w) = \frac{\prod_{h<i}(w + r_h)}{\sqrt{|c_i|}\prod_{h<i}(r_i - r_h)}.$$

Their zeros $Q_i(-r_l) = 0$ for $l < i$ are structural identities. The components either freeze the basis and its positive normalization at the box center, keep the normalization $\mathrm{diag}(e^{a_i x/2})$ moving, or apply a fixed positive rescaling $\mathrm{diag}(e^{\lambda_i(x-z)})$ to a frozen basis. All are valid triangular congruences throughout the box. For the Jacobi components, an interval LDL correction gives the orthogonal polynomial basis for the signed discrete weights $c_l \exp(L_0(a_l^2) - a_l z)$; its verified three-term recurrence evaluates the signed matrices and Fourier norms. These rate-dependent bases are enclosed throughout; only the auxiliary midpoint preconditioners and chosen rescaling rates are fixed dyadics. For a term with exponent β, the degree-K spatial error is at most its absolute coefficient times

$$\frac{h^{K+1}|\beta|^{K+1}}{(K+1)!}\max\{e^{-\beta(z-h)}, e^{-\beta(z+h)}\}.$$

Both signs of β are covered; a structurally constant term has zero remainder. Absolute errors are propagated through a further congruence C by $|C|E|C|^{\mathsf{T}}$.

Fourier remainder in the near region. On $[1/1000, 3/1000]$, use $N = 8$ shared cumulants and $K = 25$. For a frozen basis followed by the positive rescaling above, put $q = K + 1$ and $u_{ij} = \lambda_i + \lambda_j$. The spatial remainder of the *true* transformed entry is at most

$$\frac{h^q e^{|u_{ij}|h}}{q!}\sum_{d=0}^{q/2}\binom{q/2}{d}|u_{ij}|^{q-2d}S^{2d}R_{i,2d}R_{j,2d}. \tag{C.76}$$

This follows from $|u + it|^q = (u^2 + t^2)^{q/2}$, $|\widehat{p}| \leq \phi$, and Cauchy–Schwarz. The largest moment index is $510 + 26 = 536 \leq 538$. Taylor coefficients are computed from the shared-cumulant reference; their Cauchy and branch errors are also included. Summing their absolute Taylor errors costs at most $e^{(a_l+|u_{ij}|)h}$. Evaluating the corresponding error bounds at $z - h$ and multiplying by $e^{|u_{ij}|h}$ covers this cost, since $a_l < \Omega$. Thus (C.76) holds for the actual density p. Every side interval left by a successful centered box is checked separately.

Acceptance and the unbounded tail. A finite box is accepted only after strict leading determinant signs have been proved. The implementations use alternating nonzero pivots, a strict Neumann bound relative to $J = \mathrm{diag}((-1)^i)$, or positive definiteness of the even-index block together with negative definiteness of the odd-index block. The first gives the signs by elimination. If $\|A - J\|_\infty < 1$, each leading-block homotopy from J is nonsingular, giving the same signs. For the parity test, a Schur complement gives $\lceil k/2 \rceil$ positive and $\lfloor k/2 \rfloor$ negative eigenvalues in every leading k-block, again the required signs. Every inequality excludes zero; a failed enclosure is inconclusive.

For $x \geq 3$, use the normalized Newton basis and compare the actual matrix with $J_M = \mathrm{diag}((-1)^i M_i)$. Since $M_i \geq 1$, $\|J_M^{-1}\|_\infty \leq 1$. Every nonleading reference term decays with a strictly positive exponent. The branch error is bounded by

$$\frac{2\kappa W_i W_j e^{-[\Omega-(a_i+a_j)/2]x}}{(1 - e^{-\Omega x})^2}, \qquad W_i^2 = \sum_l |c_l|\{Q_i(r_l)^2 + Q_i(-r_l)^2\}.$$

The certificate verifies all decay exponents and bounds every error row sum at $x = 3$ by less than 0.834. These bounds decrease on the whole ray. Each leading-block homotopy from J_M therefore stays nonsingular, proving all required determinant signs for $x \geq 3$.

Coverage. The certificate uses the following exact cover, where $x_* = 491697171693/1638400000000$:

Range	Selected closed pieces	Method
$[0, 1/1000]$	98	Central heat expansion
$[1/1000, 3/1000]$	81	Near Fourier remainder
$[3/1000, 3/100]$	71	Jacobi basis
$[3/100, 1/10]$	101	Jacobi basis
$[1/10, 1/5]$	160	Jacobi basis
$[1/5, x_*]$	133	Jacobi basis
$[x_*, 1/2]$	17	Newton basis
$[1/2, 1]$	13	Newton basis
$[1, 3]$	9	Newton basis
$[0, 3]$	683	Complete finite cover
$[3, \infty)$	—	Decreasing analytic bound

An assembly step checks exact rational coverage and endpoints, a common order and selected factor list, admissible parameters, and one strict tail record. The cover uses 683 of 684 accepted intervals.

Setup and the main checks use 2000 decimal digits, with 1000 digits for the far-region matrix checks. The Jacobi components use $N = 8$, $K = 49$; the far components use $K = 17$, retaining either the linear common correction or independent moment enclosures, with every omitted term included in the error bound. Central coefficients are stored in outward-rounded 256-bit balls only after their signed congruences. The largest accepted central Neumann bound is less than 0.990; the tail row-sum bound is approximately 0.833815816692. A separate completed audit regenerated all 270 selected zeros at 180 digits and checked their identities and cutoffs.